	2020	2021E	2022E
Profit Before Tax	30.11	34.47	39.21
Tax	9.03	10.34	11.76
Net Income	21.08	24.13	27.45
FCFE		10.13	5.45
Cost of Equity	18%		
Terminal Equity Value			
Equity Value	90.14		

Table 1 here shows the summary figures of Reshma's FCF calculations. Table 2 sh summary figures of the FCFE calculations. She assumed sales growth rate to be the next 3 years. Thereafter, she assumed a perpetual growth rate of 5% for the co She computed the free cash flow by subtracting the net investment figures from Similarly, she computed the free cash flow to equity by subtracting the net investme net income. As per the current agreement with the bank, the current debt of ₹ 200 will be repaid by 2023.

Required:

1. Has Reshma computed the free cash flow figures correctly? Re-compute the if you think she has made any error.
2. Has she computed the cost of capital correctly? If not, then make the ne corrections.
3. Why do you think, she is getting two different equity values? Make the ne adjustments to her workings and show that the two methods yield the same value.

3 CHAPTER

DISCOUNTED CASH FLOW METHOD: THE COST OF CAPITAL

I have listened to so many nonsensical cost-of-capital discussions. I have heard CFOs talk about it, but nobody knows what it is. The real test is whether the capital that we retain generates more in market value than is retained.

- Warren Buffet

In Chapter 1, we discussed the different approaches to valuation. I also made an observation that in the absence of real options, the DCF approach is the only scientific method of valuing the company. The three key parameters in DCF method are cash flow (its definition depends on how we are defining the cost of capital and which model we use), the cost of capital, and the growth rate in the cash flows. In this chapter, we will discuss the various issues relating the estimation of the cost of capital :

(1) We will first discuss about certain theoretical issues that one should keep in mind while estimating the cost of capital. We will also talk about the different consistencies one must maintain while computing the cost of capital.
(2) We will discuss, how one can estimate the cost of debt of a company (both private and public) and the types of data that one would require for this and the sources of such data.
(3) We will then discuss about the estimation of cost of equity for any company (both listed and unlisted).

3.1 Cost of Capital

Providers of fund to a company obviously expect to be compensated for the risk they are bearing while providing funds to a company. In order to compute the cost of capital, we need to know the minimum rate of return that each provider of fund expects. The cost of capital is simply the weighted average of the minimum returns expected by the providers of funds (weighted by the market value of the amount of capital they provide). It is defined as:

$$WACC = \sum_{j=1}^{J} K_j \times \frac{A_j}{V} \quad \text{... 3(1)}$$

Here, there are 'i' different sources of fund, K_j is the minimum return (to be adjuste for tax if necessary[1]) that the providers of j^{th} source of fund expect from the compan A_j is the total market value of funds provided by the j^{th} source of fund, and V is th market value of the total capital provided by all the investors. Thus for example, whe debt and equity constitute the only two sources of fund, then the cost of capital (fo the FCF method) will be defined as :

$$WACC = K_e \times \frac{E}{V} + K_d \times (1-t) \times \frac{D}{V} \quad \text{... 3(2)}$$

Here, K_e is the cost of equity, K_d is the pre-tax cost of debt, E is the market value o the equity, D is the market value of debt, and V is sum of the market value of deb and equity.

3.1.1 Four Technical Points about Cost of Capital

Before getting into the computation of the cost of capital, we must understand four important points about the estimation of cost of capital.

1. Pre-tax cost of debt is not the same thing as the coupon rate of the bond.

Pre-tax cost of debt is defined as the interest rate that is expected by the investors ***now***. It is that rate of interest that a company has to pay if it borrows money ***now***. It is not the coupon rate that the company is paying on its existing bonds. Thus for example, if a company has borrowed money last year at 10%, then its pre-tax cost of debt will be 10% only if it can borrow money ***now*** at 10%. If lenders lend to the company only at 12% now, then its pre-tax cost of debt is 12% and not 10%.[2]

Another point that needs to be emphasized is that the cost of debt also depends on how much the company wants to borrow. Thus for example, a company may be able to raise ₹100 million now at 12%, but may have to pay 13% if it wants to raise more than that. So when we say that pre-tax cost of debt of a company is the interest rate that is expected by the investors now, we actually refer to a *marginal* increase in borrowings by the company. Let's try to understand why pre-tax cost of debt is defined this way by using two examples.

Example 3.1: Let's assume the following numbers for a hypothetical company.

FIGURE 3.1: DATA FOR EXAMPLE 3.1

Cost of Debt = 10%
Cost of Equity = 20%
Debt-to-Equity ratio = 1:1
Tax Rate = 30%
Coupon Rate on Debt = 5%

1. We know from Chapter 2 that tax adjustment is necessary for debt instruments if we value a company using the FCF method. But no tax adjustment is required if we value the company using the CCF method.
2. Similarly, if it can borrow at 8% now, then its pre-tax cost of debt will be 8% and not 10%.

The true cost of capital is given by:

$$0.2\times\frac{1}{2}+0.1\times(1-0.3)\times\frac{1}{2}=0.135\,or\;\;13.5\%.$$

However if one insists on using 5% (the coupon rate) as the cost of debt, then one will wrongly compute the cost of capital as equal to

$$0.2\times\frac{1}{2}+0.05\times(1-0.3)\times\frac{1}{2}=0.1175\,or\;\;11.75\%$$

You may wonder as to what is wrong with the second equation above. The reason why 11.75% is the wrong cost of capital is that it may lead the management to take the wrong investment decisions. Thus for example, what will the management do if it gets a project that promises an internal rate of return equal to 12%? If the management uses the wrong cost of capital, then it will lead to value destroying decisions. Here the management will apparently accept the project since the IRR is greater than the cost of capital as computed by the firm. The investors however expect 13.5% return on their investment, not 11.75%.

The coupon rate can also be higher than the yield to maturity of the bond. Suppose, the coupon rate on the bonds is 12% and not 5%. However, the before tax cost of debt is 10% as given in the question. The wrong cost of capital will be 14.2%. You may wonder that if the company uses a higher cost of capital, then it will never invest in value destroying projects. True. But it faces a problem of a different sort.

To appreciate the nature of the problem, just imagine what the management will do if it gets a project with an IRR equal to 14%. Our company will reject the project even though it would have created shareholders' wealth. It is important to understand that both the decisions (rejecting a good project and accepting a bad project) are against the shareholders' interest. One destroys value. The other avoids value-maximizing decisions.

In the above example, I just mentioned that using a wrong cost of capital destroys (or avoids positive NPV projects) shareholders' value. In example 3.2, I will quantify the loss that arises when a company takes a value-destroying project.

Example 3.2: Let's assume the following figures for XYZ Limited.

FIGURE 3.2: DATA FOR EXAMPLE 3.2

Earnings before Interest and Tax	₹100 million
Book Value of Debt	₹100 million
Tax Rate	30%
Cost of Equity	20%
Yield to maturity	12%
Coupon rate	10%

Here, the YTM of the bond is 12%, and the coupon rate is 10%. This means that the market expects a higher coupon rate from the company (if it borrows now) than what it is currently paying on its existing debt. Another interpretation is that companies with similar risk profile are raising loans at 12% now. Therefore, the pre-tax cost of debt is 12% and not 10%.

Let's assume that XYZ Limited is a no-growth company. The market value of equity therefore is ₹315 million.[3] What about the market value of debt? The interest expenses come to ₹10

3. Since the actual interest expenses are ₹10 million, the profit after tax is ₹63 million. The free cash flow to equity is also equal to ₹63 million as the growth rate is 0. This gives us the equity value as ₹315 million.

million every year. Since the yield to maturity of the debt is given by 12%, the actual mar value of debt is ₹83.33 million. The enterprise value is therefore ₹398.33 million.

The cost of capital of XYZ Limited is given by:

$$wacc = 0.2 \times \frac{315}{398.33} + 0.12 \times \frac{83.33}{398.33} \times (1 - 0.3) = 17.57\%$$ [4]

To see why the pre-tax cost of debt is not 10%, let's see what problems one will encounte one decides to use 10% as the pre-tax cost of debt. What will the market value of debt be this case? If a company uses the coupon rate as the pre-tax cost of debt, then very likely i also using the face value of the debt as the market value of debt.[5] The company will wron compute the cost of capital as follows:

$$wacc = 0.2 \times \frac{315}{415} + 0.1 \times \frac{100}{415} \times (1 - 0.3) = 16.87\%$$

Suppose, XYZ Limited finds a project with an IRR equal to 17%. The initial investment requir is ₹100 million and every year, and the project will generate an additional yearly perpetu free cash flow of ₹17 million.

This implies that if the company accepts the project, then the additional EBIT that the proje will generate is ₹24.2857 million. This number has been derived by dividing ₹17 million wi (1- tax rate)[6]. Let's assume that the company decides to keep the debt-to-equity ratio consta and hence borrows ₹24.096 million, and obtains the remaining ₹75.904 million through iss of equity to finance the project. You can see that this is consistent with the wrong DE ra (that is 100/315 as wrongly estimated by the firm). But as soon as the project is adopted t market value of equity drops by ₹1.024 million.[7] I have shown the calculations in Figure 3

FIGURE 3.3: MARKET VALUE OF EQUITY IN EXAMPLE 3.2

EBIT (after the project)	₹124.2857 million
Interest	₹12.89 million (= ₹10m + ₹24.0963 * 0.12)
PBT	₹111.394 million
PAT	₹77.976 million
Market Value of Equity	₹389.88 million

Since the market value of equity before the project was accepted was ₹390.904 (₹315 milli + ₹75.904 million of additional equity issued) million, this exercise resulted in a reduction equity of ₹1.024 million. The market value of equity declined because we invested in a proje where the company should not have invested. It invested in a project where the NPV w negative. It invested in a project where the IRR was lower than the cost of capital. In sho the cost of capital was not 16.87%. It was higher than 17%. It was 17.57%. Cost of debt is n

4. Alternatively one can find the value of the company by dividing ₹70 million with 17.57%.
5. One can of course find the market value of the debt from the secondary market data if the bond trad in the market actively. We will ignore this aspect here. But you can verify that our basic conclusion w not change if we use this number as the market value of debt.
6. Do not jump to the conclusion that you can always derive the EBIT by dividing the free cash flow wi (1-tax rate). In this case we could do it because we assumed that the growth rate is zero and that the was no non-operating revenue or expenses for the company.
7. Here, we have made the simple assumption that the market did not know about the project at the ti of issue. This is a bit unrealistic assumption. But this helps us in simplifying the matter and helps us understanding the basic concepts.

the same thing as the coupon rate. Never use the coupon rate to estimate the cost of debt. It leads to wrong investment decisions.

In the above example, the yield to maturity was higher than the coupon rate. What if the YTM of the bond happens to be 9% rather than 12%? We can similarly show that if the company computes its cost of debt as 10% rather than 9%, it may miss opportunities of shareholders' wealth maximization. I leave it to you as an exercise.

Therefore, we do not use the coupon rate as the pre-tax cost of debt because it is not consistent with the objective of shareholders' wealth maximization. In finance, we use the terms ***over-investment*** and ***under-investment*** to explain the above phenomenon. When a company invests its money in negative NPV projects, we say that the company over-invests. Similarly, when a company skips positive NPV projects (for various reasons), we say that the company under-invests.

Therefore, if a company uses the coupon rate as the pre-tax cost of debt, then it may over-invest (if the coupon rate is less than the YTM of the bond) or under-invest (if the coupon rate is higher than the YTM of the bond). And neither option is in the best interest of the shareholder.

2. Cost of equity is not the same as dividend yield.

Cost of equity is defined as the minimum return the shareholders require from their equity investment in the company. This required return has two components namely, dividend and capital gains. One should not define the cost of equity as only the dividend yield because this definition completely ignores the return expected by the shareholders from capital gains.

An article published in *Management Accountant* journal once argued that cost of equity should be defined as dividend yield[8]. The author gives the example of Eicher Tractors Limited (ETL) to explain that cost of equity should be defined as dividend yield. Apparently ETL was earlier defining cost of equity as the total return the shareholders expect from the company. Following the advice of the author, ETL started defining its cost of equity as the dividend yield. Since the dividend yield was much lower compared to its cost of debt, the company decided to issue equity and repay back the debt. This apparently improved the profitability figure of the company. If one studies the case of ETL (now it is merged with Eicher Limited), then one will observe that it was heavily dependent on debt. The book debt-to-equity ratio for the company was 4.33:1 as on 30.6.1987. The leverage of the company was more than its optimum leverage. Hence, when the company repaid debt by the issue of equity (and internal accruals) the financial position of the company improved not because it redefined its cost of equity, but because it moved towards its optimum leverage.

There are a few problems with defining cost of equity as the dividend yield. The dividend yield for most companies in the world is extremely low. If a company defines its cost of equity as the dividend yield, then it will invest money in projects giving extremely lower rates of return. This is a sure recipe for over-investment by a company. Another unpalatable conclusion that one can derive is that a company

8. See Ghosh, T P, "Cost of Capital and Leverage" in Management Accountant, Vol 29, No.4, April 1994, pp 244-245.

can actually reduce its cost of equity to zero by just refusing to pay any dividend to its shareholders. In fact, if the company takes this advice to an extreme, then it will be fully equity financed. In that case, it will lose all the interest tax shields (that debt brings with it).

A quick survey on the dividend payments made by the Indian companies for the year ending 31st March 2011 revealed that as many as 19,450 (approximately 82%) companies out of a maximum of 23,659 companies have not paid any dividend in that year[9]. The dividend yield of another 311 companies is less than 5%. If these companies start defining cost of equity as dividend yield, then they will accept all projects with IRR greater than 5%.

The number of companies whose shares are listed is much lower. One may argue that these companies have a better dividing payment record as otherwise nobody will invest in their shares. In Table 3.1, I show the number of Indian companies (whose shares are listed and whose dividend yield is zero) during 2006-2012.

TABLE 3.1: ABOUT 58% OF THE LISTED COMPANIES IN INDIA DO NOT PAY ANY DIVIDEND AT ALL.

	2006	2007	2008	2009	2010	2011	2012
Total No of Companies	2984	3120	3195	3259	3447	3582	3523
Companies with Div. Yield = 0	1671	1701	1712	1930	1936	2068	2028
%	56.0%	54.5%	53.6%	59.2%	56.2%	57.7%	57.6%

You can see that more than half of the listed companies do not pay any dividend at all. Now, if we define cost of equity as the dividend yield, then these companies will estimate their cost of equity as 0. This will give them a justification for investing in practically any project.

Let's take a simple example to explain the point. Assume that the shareholders require 20% return on their equity investment from a company. The expected dividend yield is 5%, and the expected capital gain is 15% on their investment. Let's assume that our company is an unlevered company. The cost of capital is therefore also equal to 20%. Suppose, the company gets a project with an IRR equal to 10%. If the company decides to use dividend yield as the cut-off point for accepting the projects, then it will surely accept the project. However, the shareholders will not get their money's worth and the share prices will fall, as the project does not meet the minimum return they require. I leave it to you to quantify the value loss to the shareholders as an exercise. Refer to the chapter-end exercise for this.

3. The weights in the computation of cost of capital must be based on market values of debt and equity and not their book values.

When we compute cost of capital, we compute the debt equity ratio by dividing market value of debt with market value of equity. This is the only definition that is consistent

9. I collect data from the Prowess Database of CMIE on 27 November, 2012.

with shareholders' wealth maximization objective. Investors pay the market value of equity (and debt) to invest in the equity (debt) of a company. Therefore they expect return on the market value of their investment and not on the book value which the accountant reports in the balance sheet.

The shares of Infosys, for example, were trading at ₹2,500 during the 3rd week of April, 2012. The book value of Infosys stock was a little more than ₹ 575. The cost of equity for Infosys is about 14%. This is the minimum return that the shareholders of Infosys require on their investment. When someone invests in Infosys, he is paying ₹2,500, not ₹575. So he is expecting the 14% return on the ₹2,500 investment, not on ₹575. In fact, you can easily see that the ₹575 figure is an irrelevant figure here.

What about an existing investor? Imagine that somebody invested in Infosys from the day the company started its operations. Will he use ₹575 to estimate the cost of capital? The answer is NO again. The promoters (and all the other investors) have the option of selling their investments at ₹2,500 per share today. When they reinvest their money in some other similar risky asset, they would require 14% return on that. So why would they be happy in receiving 14% return on ₹575, when they can get 14% on ₹2,500 by investing their money somewhere else (with the same risk)?

Example 3.3: Let's illustrate this point by taking the example of ABC Limited. In Figure 3.4, I report some of the relevant figures for ABC Limited.

FIGURE 3.4: ASSUMPTIONS FOR EXAMPLE 3.3

Cost of Equity	20%
Cost of Debt	10%
Tax Rate	30%
Market Value of Equity	₹120 million
Book Value of Equity	₹16 million
Market (Book) value of Debt	₹60 million[10]

Here, one must compute the cost of capital by using the market value figures rather than the book value figures. Therefore, the true cost of capital is given by:

$$wacc = 0.2 \times \frac{120}{180} + 0.1 \times (1-0.3) \times \frac{60}{180} = 15.67\%.$$

Instead if one uses the book value of equity while computing the cost of capital, then one will wrongly compute the cost of capital as:

$$wrong\_wacc = 0.2 \times \frac{16}{76} + 0.1 \times (1-0.3) \times \frac{60}{76} = 9.736\%.$$

Suppose, ABC Limited gets a project that requires an initial investment of ₹100 million, and the project has an IRR equal to 12%. Should it accept the project? Let's assume that the operating risk of the project is exactly the same as that for the company. Let's also assume that ABC

10. Here I have assumed that the market value of debt is equal to the book value of debt. This assumption is not necessary, but it simplifies the calculations and at the same time explains the basic concepts.

Limited wants to raise debt and equity in proportion to the existing debt-to-equity ratio keep the financing risk of the project same as that for the company.[11]

Suppose, the company sticks to the book value method. Then it will raise ₹21.05 million fr equity and ₹78.95 million from debt. (We will see later that this actually increases our cost debt, cost of equity, and cost of capital further.) Since the IRR is 12%, we will make matt simpler by assuming that the project will generate an extra EBIT of ₹17.14286 million (=₹ million/(1-0.3)) every year.

Since the company has borrowed ₹78.95 million, the interest to be paid on that will be ₹7.8 million.[12] The profit available to be distributed to the shareholders is therefore ₹6.47 milli What will happen to the value of equity here?

To find that figure, we must find out the new cost of equity. The cost of equity is an increasi function of the debt-to-equity ratio and therefore when the market debt-equity ratio ris the cost of equity will similarly rise.[13] The new cost of equity will be around 22.895%. (This based on the assumption that the risk free rate of return is 10%, and the market risk premiu is 10%.[14])

The value of equity drops to ₹133.1 million from ₹141.05 million, a loss of ₹7.95 million f the shareholders. (Here in order to derive these numbers, I have assumed that the compa was a no growth company before accepting this project.)

Why do we get these findings? The reason is that when somebody buys the share of th company, he pays ₹120, and not ₹16[15]. He will therefore expect a 20% return on ₹120 and n on ₹16. Hence we should define cost of capital using market value of debt and equity an not book value of debt and equity.

What if the Market Value of Debt is not available?

The definition of cost of capital as given in Equation 3(2) often creates problems in re life. We need the market values of debt and equity to find the cost of capital of a compan However the market values of debt of most Indian companies are not available. There ar various reasons for that.

Most of the Indian companies are heavily dependent on the banks and financial institution for their debt capital. It is not possible to obtain the market values of the debt in such case Secondly, even when the companies have issued bonds and debentures, one cannot obtain th market values of debt because these bond instruments do not trade actively in the marke The debt market in India is not as active as in other developed countries. This makes the jo of finding the market values of debt difficult. We encounter similar problems while valuin private companies for which the market values of neither debt nor equity are available.

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11. This immediately raises the question as to which debt-equity ratio that ABC must take into account to decide how to finance the new project. We will assume that the same debt-equity that has been used while computing the cost of capital has been used here.
12. Actually the company may have to pay a higher interest because since the debt-to-equity (in marke value terms) is increasing, the cost of debt may also increase. I have ignored this possibility here.
13. Here the management may not realize that the company has become riskier because it is thinking in terms of book leverage. But since the market looks at market leverage, it will demand a higher risk premium because of the increase in market leverage.
14. The beta of the stock was 1 prior to the increase in debt. We use MM's second proposition to estimate the cost of equity.
15. For simplicity, we assume that there are 1 million shares outstanding in ABC Limited.

When we estimate the debt-to-equity ratio by dividing the market value of equity with the book value of debt we actually assume that the market value of debt is equal to the book value of debt. This is equivalent to the assumption that the coupon rate is equal to the pre-tax cost of debt. In quite a few cases, this assumption may vitiate the entire valuation process. Thus for example, when we are valuing a troubled company, we certainly cannot assume that the book value of debt is equal to the market value of debt. We have to estimate the market value of debt by discounting the coupons and the principal payment using the pre-tax cost of debt.

4. The discount rate must match the riskiness and the type of cash flows being discounted.

One of the first principles of DCF valuation is that the discount rate must match the riskiness of the cash flows being discounted. In fact, in this book, I constantly (and somewhat repeatedly) advocate maintaining four types of consistency in the valuation. One of these consistencies, what I call 'risk consistency' simply says that the cash flows and the risk must be consistent with each other. If the cash flows are risk-less cash flows, then the discount rate must be risk-free discount rate. Equity cash flows must be discounted at cost of equity. Debt cash flows (coupon and principal repayments) must be discounted at cost of debt. Firm cash flows must be discounted at cost of capital. We already discussed about this in Chapter 2.

Here, I will talk about the consistency between the type (and ***not*** riskiness) of cash flow and the discount rate. In particular, we must ensure that:

(*a*) Nominal (real) cash flows are discounted at the nominal (real) discount rate, and

(*b*) Rupee (US$) cash flows are discounted at rupee (US $) discount rate.

I will give two examples to explain these two points.

Example 3.4:

Assume that MSL Limited produces and sells 1,000 units of JLTs each year. Also assume that MSL Limited does not expect any growth in the number of JLTs, it will sell in future. The unit selling price of each JLT was ₹100 last year (year # 0). The cost of production of each JLT was ₹60 last year. Assume that all revenues are collected in cash and that all expenses are incurred in cash. Also assume that there are no other expenses (costs) other than the ₹60 per unit cost of production.

Assume that the real discount rate is 3% and the expected inflation rate is 4%. The nominal discount rate is, therefore given by:

Nominal rate (r_N) = (1+3%) × (1+4%) - 1 = 7.12%[16].

How much value will you put on MSL Limited? Last year (year ending today), MSL generated free cash flow of ₹40,000. Since MSL does not expect any growth in JLTs to be produced and sold in future years, the real cash flows of MSL will remain at ₹40,000 for all the future years. The real discount rate being 3%, the value of MSL will be given by:

16. The nominal rate can be found by using the equation: (1 + real rate) × (1 + inflation rate) – 1. One can also approximately estimate the nominal rate by simply adding the real rate and the inflation rate. This is the well-known Fischer Equation.

$$\text{Value of MSL} = \frac{40,000}{0.03} = ₹1,333,333.$$

If we project the nominal cash flows for MSL, then the projected free cash flow for the n year (year # 1) will be ₹41,600. That is because, next year the expected selling price of ea unit of JLT will be ₹104 and expected cost of production of each unit of JLT will be ₹62.40 next year, the total projected cash flows will equal 1000 × (104 - 62.4) = ₹41,600.

When we look at the nominal cash flows, we can see that the cash flows are growing at per annum. This is exactly the inflation rate that we projected in this example. The pres value of the nominal cash flows will therefore, be given by:

$$\text{Value of MSL} = \frac{41,600}{7.12\% - 4\%} = ₹1,333,333.$$

We can see that, we obtain the same present value in both the cases. So as long as we maint this consistency between the discount rate and the cash flow, we can rest assured that will get the same present value[17].

In this example, we assumed that MSL would sell the same 1000 units of JLT every year. results will not change if we assume a growth rate in the number of units of products t MSL sells. Assume for example, that MSL Limited expects 2% growth rate in the numbe JLTs it will sell every year in future. I leave it to you as an exercise to show that you get same present value no matter whether you use real or nominal cash flows as long as y maintain the above consistency. Refer to the chapter-end exercises for this.

We will find the above result useful whenever inflation bothers us. Imagine valuing compan in a high-inflation regime. All that this example suggests is that just ignore inflation and direc obtain the value of the enterprise by discounting real cash flows by using real discount ra

Example 3.5: Let's look at the second type of consistency (matching the currency of the ca flow and the discount rate) we mentioned earlier. Imagine that an American investor wa to invest in MSL Limited (refer to Example 3.4 above). Imagine that the US real discount r is also 3% and the expected inflation rate is 1%. So the nominal rate in the US is 4.03%. Assu that the spot exchange rate between rupee and US$ is ₹54/1 US$.

The expected exchange rate after year 1 is given by:

$$\text{Expected spot rate after 1 year} = ₹54 \times \frac{1.0712}{1.0403} = ₹55.60$$[18]

The US$ nominal cash flows of MSL Limited will be given by:

$$\text{Nominal Cash flows (\$)} = \frac{41,600}{55.60} = ₹748.15$$

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17. In real life, we face a few practical problems. The inflation rate in the economy may be different fr the rate at which the product prices of the company are increasing. The cost of production may experience a different growth rate. We ignored these possibilities here.

18. This follows from the Exchange Rate Parity Theorem. Under this theorem, we can estimate the expec exchange rate F (₹/$) by multiplying the current spot rate S (₹/$) with $\frac{1 + r_{IND}}{r_{US}}$. Here r_{IND}, is the inte rate prevailing in India, and r_{US} is the interest rate prevailing in the U.S. Instead of using the actual r free rate, I have used the discount rates for the company as a proxy here.

Since the current cash flows are 40,000/54, that is, ₹740.74, the nominal \$ cash flows are expected to grow at 1% every year[19]. So the present value of the nominal \$ cash flows are given by:

$$\text{PV of Nominal Cash Flows (\$)} = \frac{748.15}{4.03\% - 1\%} = ₹24{,}691.36.$$

Since the current spot rate is Rs.54/US \$, the equivalent rupee value of MSL is

Rupee Value of MSL (in ₹) = ₹24,691.36 × ₹54/\$ = ₹1,333,333.

This is exactly the same figure that we arrived at earlier.

So if currency bothers us while valuing any company, we can simply value the company using a different currency that we are comfortable with and in which we can get the relevant data. Let's assume that we are valuing a company based in Dubai. Since Dubai Government does not borrow money from the market, we do not directly get the risk-free rate. This also affects our cost of capital calculation. In such a case, however, we can attempt to value the company (say) in US dollars and finally get its UAE Dirham value by converting the \$ value to Dirham value by using the current \$-Dirham exchange rate.

3.2 Cost of Debt

In the previous section we defined the cost of capital as the weighted average of the costs of the different sources of fund. Traditionally, debt and equity constitute the two primary sources of fund for a company. With the rapid development of innovative instruments in the recent past, companies have started raising funds by issuing these innovative instruments. We will first study how to find the costs of the plain vanilla instruments. Subsequently we will discuss how one can find the costs of some of these innovative instruments.

3.2.1 Cost of Straight Bond

Pre-tax cost of debt is defined as the coupon that a company will pay if it borrows ***now***. In this definition, the key word is now. We are therefore not interested in the coupon rate that the company is paying on its existing borrowings. To arrive at the cost of debt, we need to find out the coupon that the bondholders are expecting from the company *now*. Here, following Ehrhardt (1994), I recommend four possible ways to find the cost of debt of a company[20]. I call it a pecking order in the sense use the second method only if you cannot use the first method. Similarly use the third method only if the first two methods cannot be used.

Method 1: Find the YTM of the Bond (Pecking Order 1)

If the debt instruments issued by the company are trading in the market, then one can easily find the cost of debt. Once we know the market price of the bond, we can

19. When \$740.74 grows to \$748.15, the implied growth rate is 1%. This is actually equal to the expected inflation rate in the U.S.

20. See Ehrhardt, Michael, C., 1994, The Search for Value: Measuring the Company's Cost of Capital, Harvard Business School Press, Boston.

use equation 3(3) to find the cost of debt of the company. Here, 'y' is the pre-tax c of debt.

$$Market\_Price = \frac{Coupon_1}{(1+y)^1} + \frac{Coupon_2}{(1+y)^2} + ... + \frac{Coupon_T}{(1+y)^T} + \frac{Face\ value}{(1+y)^T} \quad ...\ 3(3)$$

For a traded bond, we know the market price, the coupon payments and the matur period of the bond. We have to solve for 'y' to get the pre-tax cost of debt. A spreadsheet package can find the exact value of 'y' and hence one does not need use any approximate figure for 'y' here. One can also use the 'yield' function in Exc to find the yield of a traded bond.

Here I have assumed that the entire principal is repaid at the end of year T. T current market price is ex-interest that is last year's coupon has already been pa and the next coupon payment is due exactly after one year. We can always adju for the accrued interest component in the calculation.

Though I recommend this method as the most suitable method of estimating the co of debt, one technical point is in order. In Equation 3(3), we assume a flat yield curv Therefore we discount all the coupons and the principal (to be received at differe points of time) using the same discount rate. If this assumption is not correct, th the 'y' we obtain by solving Equation 3(3) can be looked as some sort of a weight average of costs of debt of different maturities.

Method 2: Ask someone who knows (Pecking Order 2)

In most cases however, it is difficult (and sometimes impossible) to obtain the mark value of debt. There is no active debt market in India. In an illiquid market, it is difficu to obtain the market value of the bond. Secondly, sometimes loans are obtaine from the banks and financial institutions and these loans do not trade in any marke

As on 31 March, 2012, 440 of the 500 companies (that are part of NSE's CNX 50 Index) have some debt in their balance sheet. Out of this, as many as 299 (68%) hav not borrowed anything from the public. The median public borrowing is 0% (averag being 8%). Indian companies prefer to borrow from banks and financial institutions In only about 8% of the cases, we find issue of bonds and debentures made by th companies. And most of them do not trade actively[21].

In such cases, we have to use alternative methods to obtain the cost of debt for a company. If a company has obtained funds from commercial banks and financia institutions, then we can ask the banks and FIs directly as to what interest they wil charge if they lend now? The philosophy behind this method is: ***"If you do not know something, ask someone who knows."***

If you are valuing the company as part of a consultancy assignment, then you can approach the banks and financial institutions directly and get this information. This is much simpler than using complicated equations to find the cost of debt.

21. I collected this data from the Prowess database. I accessed the database on 22nd April, 2012.

Method 3: Use the Credit Rating of the Bonds issued by the Company (Pecking Order 3)

If we know the credit rating of the company[22], then we can find the cost of debt of the company. Thus, for example, if a company has an AAA rating and another company which has a AAA rating is paying 10% now to issue new bonds, then we can use the 10% figure as the pre-tax cost of debt of the first company. Credit rating agencies regularly publish the updates of the ratings of different companies. Therefore, we can use these ratings to obtain the pre-tax cost of debt. The logic behind this method is that the market treats all bonds with similar ratings alike as far as fixing the interest rate is concerned and therefore, if one AAA rated company is paying 10% now, then all companies with the same rating will also pay 10% *if they borrow now.*

Example 3.6: Cost of Debt of Tata Steel

Let's take an example to understand the process. The current credit rating of the bonds issued by Tata Steel is AA (for its domestic borrowings). I show the relevant disclosure made by the company in its website in Figure 3.5.

FIGURE 3.5: CREDIT RATING OF LONG-TERM CORPORATE BONDS BY TATA STEEL

Credit Ratings

Standard & Poor's | Moody's | Fitch

Outlook (long-term rating)	Stable
Long-term Corporate Credit Rating (Foreign Currency)	BB+
Long-term Corporate Credit Rating (Domestic Currency)	AA
Short-term Corporate Credit Rating (Domestic Currency)	F1+

Source: http://www.tatasteel.com/investors/investor-information/credit-ratings.asp

In order to find the cost of debt of Tata Steel, we need to find out what AA rated bonds are paying now. From the website of www.debtonnet.com[23], I find the corporate bond spread on AA rated bonds is 143 basis points (as on 22nd April 2012). Since the G-sec yield on the same date was 8.59%, the required yield on a AA rated bond comes out to:

8.59% + 1.43% = 10.02% ≈ 10%.

So, while determining the cost of capital of Tata Steel, we can use 10% as its cost of debt.

Here, we used information on the credit rating of one of Tata Steel's existing debt. Brickwork has assigned BWR AA+ rating to Tata Steel's proposed ₹3,000 crores non-convertible debentures

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22. Rating agencies actually rate the bond instruments of companies and not the company itself. Here, I am using the term a bit loosely.
23. You can also obtain the credit default spread data from the website of FIMMDA: http://fimmda.org/modules/bonds/corporate-bonds.aspx?m=btd

issue[24]. For AA+ rated bonds, the spread is 116 basis points. One will obtain a lower co[st of] debt for Tata Steel, if one uses the Brickwork rating.

Method 4: Rate the Bonds Yourself (Pecking Order 4)

As per the Disclosure and Investor Protection Guidelines, 2000 of the SEBI, cr[edit] rating (from at least one rating agency) is mandatory while issuing any convert[ible] debt instrument. However, if the company borrows money directly from banks [and] financial institutions, then we do not get the rating figures. In such cases, we nee[d to] do the rating ourselves to find out what rating a particular company[25] would h[ave] got had it gone for a rating.

Using data for the fiscal year ending March 2011, I prepared a table that shows the median values of three accounting ratios for companies whose bonds have b[een] rated by CRISIL, ICRA, and CARE. I obtain their current rating and these account[ing] figures from the Prowess database. I provide the details in Table 3.2.

TABLE 3.2: MEDIAN ACCOUNTING RATIOS FOR DIFFERENT CREDIT RATINGS

Rating	Rating	Debt-Equity Ratio	EBIT/Interest
Highest safety	AAA	0.6	4.98
High Safety	AA-, AA, AA+	0.67	4.57
Adequate safety	A-, A, A+	0.84	3.07
Moderate Safety	BBB-, BBB, BBB+	1.215	2.79
Inadequate safety	BB-, BB, BB+	1.515	1.58
High Risk	B-, B, B+	1.77	1.63
Substantial Risk	C-, C, C+	1.85	1.16
Default	D	1.89	1.20

Source: Prowess database.

As you can see from Table 3.2, the median interest coverage ratio (EBIT/Interes[t]) for companies with AAA rated bonds is 4.98. So, if we have a company with intere[st] coverage ratio close to 5 (or much greater than 5), we can assign its bonds AA[A] rating. I do not argue that credit rating agencies look only at the interest coverag[e] ratio. But a higher coverage ratio definitely indicates a stronger company in term[s] of its ability to pay the interest and repay the principal.

Let's take the example of Tata Steel again. Suppose, one does not have access to i[ts] credit rating and hence wants to rate it by using its interest coverage ratio (as give[n] in Table 3.2). From the profit and loss account of Tata Steel for the year endin[g] 31 March 2011, we obtain the following information:

EBIT = ₹10,024 crores

Interest = ₹16,86.3 crores

24. Source: http://www.moneycontrol.com/news/brickwork-research/brickwork-assigns-bwr-aa+-ratin[g-] to-tata-steel%60s-ncd-issue_695680.html
25. I will again repeat here: credit rating agencies do not rate the companies. They rate the different bo[nd] instruments issued by the companies.

The interest coverage ratio comes to 5.95. As per Table 3.2, the implied rating (for this coverage ratio) is AAA. The corporate bond spread for AAA rating is 95 basis points. So the cost of debt for Tata Steel comes to:

8.59% +0.95% = 9.18%

This is almost 1% lower than what we obtained by using the Fitch rating earlier. Which number is correct? If you have followed the discussion till now, you will realize that the 10% figure (Method 3) is more accurate as it uses the actual credit rating of Tata Steel's bonds. Our imputed rating for Tata Steel is AAA. However, since we know the actual rating of Tata Steel, we should use that information (rather than trying to rate it ourselves) while estimating its cost of debt.

Which tax rate should we use?

One comes across three types of tax rates while analyzing the financial performance of a company. The first one is the **statutory tax rate**. The current corporate tax rate is around 33.445%[26]. The multinational companies effectively pay 42.02% tax in India. The second tax rate is known as the **average tax rate**. This is computed by dividing current year's tax expense with the profit before tax. There are many reasons why the average tax rate is different from the statutory tax rate. Companies use different depreciation methods (and figures) while reporting to the shareholders and the income tax authorities. Export earnings (from export of computer software) are tax-free under Section 80HHE of the Income Tax Act, 1961. So software companies usually report higher earnings to the shareholders but pay lower taxes to the tax authorities. The third type of tax rate is known as the **marginal tax rate**. This is defined as the additional tax that the company will pay for every additional rupee of taxable income that the company earns. For most profitable companies, this rate will equal the statutory rate or will lie somewhere between the statutory rate and the average rate. Which rate should we use?

The tax rate that one should use while computing the cost of capital actually equals:

$$t = \frac{Tax^U - Tax^L}{Interest} \quad \text{... 3(4)}$$

Here, Tax^U refers to the unlevered tax and Tax^L refers to the levered tax of the company[27]. For a highly profitable company, this will equal the statutory tax rate of the company.[28] If a country follows progressive tax system, then the tax rate will equal the marginal tax rate for most profit making companies. In particular, if the company uses the same tax rate (t_1) to compute both the levered and unlevered taxes, then you can use this tax rate (t_1) to compute the cost of capital. I explain this with the help of a simple example.

26. The 2013 Budget increased the effective tax rate for domestic companies to 33.99%. The effective tax rate for the multinational companies increased to 43.26%.
27. Source: Mohanty, P., 2013, Proper Use of Tax in Company Valuation, Unpublished Paper, XLRI, Jamshedpur.
28. This can be easily proved. $Tax^U - Tax^L = t_m \times EBIT - t_m \times (EBIT - Int) = t \times Interest$. So the tax rate given in Equation 3(4) will t_m. Here, t_m refers to the marginal tax rate.

Example 3.7: MJL Limited is a software company that gets ₹150 million of revenue each Its total operating expenses (excluding depreciation) are ₹50 million. Its book depreci figure is ₹10 million, while its tax depreciation figure is ₹20 million. MJL follows Straigh Method of depreciation while reporting to the shareholders, while it reports the Written D Value Method of depreciation while reporting to tax authorities. It has borrowed ₹100 m at 10% and hence each year, it pays interest of ₹10 million.

In Figure 3.6, I show these figures. I have assumed that MJL will not have to pay any ta the ₹50 million revenue it earns by selling software to the U.S. I show the tax figures u two different tax scenarios. In the first scenario, I assume a flat tax system, where all tax income is subject to a flat 30% tax rate. In the second scenario, I assume that the com faces a progressive tax system, where taxable income below ₹ 25 million is subject to 20 rate. However, taxable income exceeding ₹ 25 million is subject to 30% tax rate.

FIGURE 3.6: WHICH TAX RATE TO USE?

	Flat Tax Rate	Progressive Tax Syste
Revenue	**Amount (₹ Million)**	**Amount (₹ Million)**
Domestic	100	
Export	50	
Total	150	
Expenses		
Operating Expenses	50	
Depreciation (Book)	10	
Depreciation (Tax)	20	
Interest	10	
Earnings Before Tax	80	
Taxable Income	20	
Tax @ 30%	6	
Average Tax Rate	7.5%	5.0
Statutory Tax Rate	30%	20% for Income upto $ Million; 30% after that
Marginal Tax Rate	30%	20
Tax Rate to be used	30%	-40.0

In Scenario-1, since the taxable income is ₹20 million, MJL pays ₹20 million × 30% = ₹6 mill as tax. The average tax rate is 7.5% as shown below:

$$\text{Average tax rate} = \frac{6}{80} = 7.5\%$$

Though MJL will pay a tax of ₹6 million, it will show ₹24 million (30% of ₹80 million) as tax expense. The remaining ₹18 million figure will be shown as deferred tax expense. Some analysts divide the entire ₹24 million with ₹80 million to obtain the average tax rate. In most cases this will come closer to the statutory tax rate.

Before analyzing the tax rate in the second scenario, we will address the main question here: Which tax rate we should use to find the after-tax cost of debt - 7.5% or 30%?

If MJL does not have any debt in its balance sheet, its interest expenses would have been 0 and the taxable income would have been ₹30 million, with a tax bill of ₹30 million × 30% = ₹9 million. The additional tax bill comes to ₹9 - ₹6 = ₹3 million. And that is exactly 30% of the ₹10 million of interest expenses. So we should use the statutory tax rate while finding the after-tax cost of debt. Since for most profit making companies, the marginal tax rate equals the statutory tax rate, you can directly use the marginal tax rate to find the after-tax cost of debt.

There can be exceptions, of course. Look at the second scenario given in Figure 3.6. Here, the company actually pays ₹4 million of tax (20% of ₹20 million). However, as an unlevered company, it would paid ₹6.3 million as tax. As an unlevered company, it would face 30% as the marginal tax rate. However, as a levered company, since the taxable income is lower than ₹25 million, it is facing a marginal tax rate of 20%. In such case, its average tax rate is 5%. But while finding the after-tax cost of debt, following Equation 3.4, we should use 23% as the tax rate. This rate is neither the effective tax rate nor the marginal tax rate. Use of this rate alone will give us the correct free cash flow and the correct enterprise value. We will discuss more on this in Chapter 4 of this book.

What if there are different debt instruments with different costs of debt?

Understanding this section helps us maintaining what I call the ***'liability consistency'*** in valuation. A company may have got various types of liabilities in its balance sheet. We do not consider all of them while computing the cost of capital. Our computation of the cash flow is related to which liability is included and which liability is excluded from the valuation.

Quite often we observe that the same company can obtain debt from the market at different coupon rates. Companies issue different types of debt instruments to raise capital. The costs of the different instruments will not always be equal. The credit ratings of the different instruments may not be the same. Credit rating agencies rate the different instruments and not the company. The different instruments may not have equal backing of assets. Some of the bonds issued by a company may be unsecured. Therefore, different debt instruments issued by the same company may have different pre-tax costs of debt.

Example 3.8: Let's assume that XYZ Ltd. has two types of debt instruments with costs of debt equal to 12% and 14% respectively. Which number should we use while finding the cost of debt? The answer depends. It depends on which debt is included in the cost of capital calculation. If we include both the debt instruments in the cost of capital calculation, then we should use both the numbers (12% and 14%) while finding the cost of capital. Let's see why.

Let's assume that the face value of bond 1 (with 12% yield) is ₹100m and that of bond 2 (with 14% yield) is ₹200m. Let's also assume that it is a no-growth company. We further assume that the market value of debt and the book value of debt are equal.

XYZ Limited will report ₹100 million as EBIT every year in the coming years. Its profit after tax will be ₹42 million each year, as shown below:

EBIT = ₹100 m.

Interest = ₹40 m. (=12% × ₹100m + 14% ×₹ 200m)

Profit Before Tax = ₹60m.

Tax (@30%) = ₹18m.

Profit After Tax = ₹42m.

If we assume the cost of equity equal to be 20%, the value of equity comes to **₹210 m.** Th enterprise value is therefore ₹210 m + ₹100 m + ₹200 m = ₹510 m.

The free cash flow is given by 100 × (1-0.3) = ₹70 m.

The cost of capital is given by

$$wacc = 0.2 \times \frac{210}{510} + 0.14 \times (1-0.3) \times \frac{200}{510} + 0.12 \times (1-0.3) \times \frac{100}{510} = 13.72\%.$$

We can verify that 13.72% of the correct cost of capital. We can compute the enterprise valu of the company by dividing ₹70 million with 13.72%, and as can be verified, the value is exactl equal to ₹510 million. **Therefore, if a company has issued different types of bonds with differen pre-tax costs of debt, use all these different numbers while finding the cost of capital.**

One can alternatively, decide to ignore one of the two debt figures while computing the cos of capital[29]. Will one get the same value of equity if we follow this method? The answer i yes. If a company has issued two (or more) debt instruments, we can value the equity of th company by using any one of the following three methods:

- Method 1: We can directly value the equity using the FCFE method.
- Method 2: We can use only one (or more) of the two (or more) types of debt issued b the company while finding the cost of capital.
- Method 3: We can use all the debt instruments issued by the company while finding the cost of capital.

All these three methods will give us the same equity value. I explain this using the numbers used in Example 3.8.

In Example 3.8, we found the market value of equity by using the first (the FCFE method) and the third (consider all the liabilities while computing the cost of capital) method. The market value of equity was ₹210 million. We could obtain this figure by using the first method, where we discounted the constant free cash flow to equity of ₹42 million using 20% cost of equity as the discount rate. Using the third method, we obtained ₹510 million as the enterprise value. From this figure, when we subtract ₹300 million as the market value of debt, we also obtain ₹210 million as the market value of equity.

Let's look at the second method here. Suppose, I decide to consider only the ₹200 million of debt the company has issued with a yield of 14%. I decide to ignore the ₹100 million of debt the company has issued with a yield of 12%. In this case, we will obtain 15.02% as the cost of capital, as shown below:

$$\frac{210}{410} \times 20\% + \frac{200}{410} \times 14\% \times (1-30\%) = 15.02\%$$

29. You may wonder 'Why would one want to do that?' I will give you a few examples later to show that sometimes, it makes sense to ignore certain types of liabilities, while computing the cost of capital.

Since, I do not consider the ₹100 million of debt issued by the company, while computing its cost of capital, I am effectively treating it as an operating activity. So I must subtract the interest the company pays on this debt (₹12 million) before computing the EBIT. The revised EBIT will therefore equal ₹88 million. The free cash flow will equal ₹88 million × (1-0.3), that is, ₹61.6 million. The implied enterprise value comes to ₹61.6/15.02%, that is ₹410 million. From this figure, when we subtract the ₹200 million of debt (we considered while computing the cost of capital), we obtain ₹210 million as the market value of equity. If you ignore the ₹200 million of debt issued by the company while computing the cost of capital, you would also get the same equity value, provided you compute the cash flow correctly.

If you do not consider a particular liability while computing the cost of capital, you must consider all the expenses the company incurs to service the liability as operating expense and you must subtract this expense, while computing the EBIT.[30] This is ***liability consistency***. The way you compute the EBIT (for valuing the business) depends on which liability is (not) considered for cost of capital computation.

All the three methods yield the same market value of equity figure. So which one should we use? Use that method which is practically the most convenient. Sometimes, we encounter situations where the rate of interest is not explicitly stated. In such situations, we can simply ignore this liability instead of trying to guesstimate the cost of debt figure. I show an example of this in Section 3.2.2 (while discussing non-interest bearing liabilities). We sometimes encounter situations where the company has treated certain liabilities as operating activities. Again, we can either reclassify such activities as financing activities and follow the third method or simply ignore them and follow the second method while computing the cost of capital. I show an example of this approach in Section 3.2.2 (while discussing leasing). As long as we maintain liability consistency, we will obtain the same value of equity.

3.2.2 Cost of certain other types of debt instruments

In the previous section, we discussed the different ways in which one can find the cost of what is called, a straight bond or a plain vanilla bond. If you look at the capital structure of a company, you will also find certain other types of liabilities in the capital structure. While estimating the cost of capital, we must consider the costs of these other liabilities as well (or we must know why we are ignoring a particular type of liability). In this section, we will discuss how the costs of these debt instruments are found out. In particular, we will discuss about the following types of liabilities.

- Non- interest bearing liabilities.
- Leasing
- Convertible bond
- Foreign currency bonds.
- Floating interest bond.
- Preferred stocks.

We will discuss about each of the above briefly here. We will discuss about cost of convertible bonds later in this chapter after discussing cost of equity.

30. There is a second adjustment you need to make. We will discuss that in Chapter 4, while computing net investment.

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- Convertible bond
- Foreign currency bonds.
- Floating interest bond.
- Preferred stocks.

We will discuss about e[illegible] discuss about cost of convertible bonds lat[illegible] equity.

30. The[illegible] apter 4, while computing net inv[illegible]

Non- Interest Bearing Liabilities

If we look at the liability-side of the balance sheet, we will find certain types o liabilities, which do not carry any *explicit* coupon on them. These liabilities are called non- interest bearing liabilities (or NIBL).

Current Liability (with the exception of short-term debt and current portion of long term debt) is the most prominent among these liabilities. Let's take the example o accounts payable (known as sundry creditors in India). When a company buys good worth ₹10 m on a 3-months credit basis, it pays ₹10 m only to the supplier after months. It doesn't pay any interest on this amount. It is this feature of current liability which makes it different from other liabilities.

Sometimes, people wrongly believe that current liabilities are free, that is there is no cost associated with it. This is not correct however. Current liabilities do have costs attached to them. But they are not explicitly stated.

Take the previous example of a credit purchase of ₹10 m for 3 months. Does it come free? Certainly not. Instead of paying after 3 months, if the company decides to pay immediately, the supplier will obviously be happy to sell the same raw materials at a reduced price.

Let's assume that the supplier is willing to accept ₹ 9.7m if the company decides to pay in cash now. This difference of ₹ 0.3m is actually the cost associated with buying on a 3-months credit basis. One can therefore, look at this credit purchase as a loan from the supplier for a 3-months period. It is as if the supplier is giving the company a loan of ₹ 9.7 m today and he is charging an interest of ₹ 0.3m on that for the 3-months period for which he has extended the credit.

Therefore, current liabilities are not free. But the interest component is not mentioned explicitly. How does this implicit cost affect our cost of capital calculations? That depends on how we treat the implicit interest expenses of ₹0.3 million. We can treat the sundry creditors as a loan from the supplier, and use the implicit annualized interest as the pre-tax cost of debt. This is equivalent to method-3 in Example 3.8. Thus for example, if we treat the ₹0.3m as the interest on a loan of ₹9.7m for three month's period, the implicit interest cost would come to:

$$\left(1+\frac{0.3}{9.7}\right)^4 - 1 = 12.96\%$$

We can therefore treat the sundry creditors as a different source of liability and use 12.96% as the cost of debt. In that case, we must show ₹9.7m (and NOT ₹10m) as the debt figure. Secondly, as I explained in example 3.8, we must also restate the EBIT figure (for computation of the free cash flow). We can alternatively, just ignore this type of liabilities while finding the cost of debt (an example of the second method discussed in Example 3.8).

Ignoring a particular type of liability is not a problem at all if the interest on this liability (explicitly stated or otherwise) has already been subtracted when you compute the EBIT. When we ignore the ₹ 9.7 million of debt while computing the cost of capital, we must include the ₹ 0.3 million of (implicit) interest as an operating expense. Since the accountant treats this ₹0.3 million of implicit interest as an operating expense anyway (the COGS is shown as ₹10 million and not ₹9.7 million), this condition is automatically satisfied. So it is pretty convenient to ignore current liabilities, while computing the cost of capital.

So, remember that you must follow one of the following two methods while adjusting for non-interest bearing liabilities[31]:

- Consider the non-interest bearing liability as a source of fund while finding the cost of capital. The debt (current liability) must be shown as ₹ 9.7 million. The COGS figure will be restated as ₹ 9.7 million (and not ₹10 million). Both EBIT and FCF figures will be based on this revised COGS.
- Ignore non-interest bearing liabilities while computing the cost of capital. Subtract the implicit interest expenses from EBIT and based on this EBIT, compute the free cash flow. Effectively, the debt will not include ₹9.7 million and the implied interest of ₹0.3 million will be subtracted while computing the EBIT and FCF.

We face certain practical problems while using the first method. At any time, a company purchases raw materials from different suppliers. We need to ask each and every supplier as to how much implicit interest they are charging for the credit they are giving to the company. This is not a practical solution. The second solution is more practical. We can simply ignore the non-interest bearing liabilities while computing the cost of capital.

In certain other types of non-interest bearing liabilities, it is difficult to estimate the implicit costs. For example, in quite a few of the Indian companies' annual reports, one can find the presence of what is called deferred sales-tax loan. Here the companies collect the sales tax from the customers at the time of sales but do not pay it immediately to the state government. They pay it usually after certain number of years.

But why do certain state governments give this benefit to the companies? The reason is obvious. They want the companies to set up their factories in their states. This is done to encourage industrialization in the states.

What is the implicit cost associated with it? The companies do not set up the factory at the best possible location, where they can get raw materials or skilled labour easily. When a company sets up its factory at a place different from the most desirable location, it incurs certain expenses, e.g., transportation of raw material, payment of excess wages and salaries, developing township around the factory, etc. One can look at these expenses as the implicit interest that the company is paying on the deferred sales tax loan. Instead of finding out the implicit interest cost, one can adopt the second approach: Just ignore these liabilities while finding the cost of capital.

31. I ignore the first method discussed in the earlier section, namely, ignore all the debt and simply use the FCFE method.

Leasing

Sometimes, companies get fixed assets through finance leases. Instead of takin loan and then buying the fixed asset, the company can get the asset through finar lease. Therefore, a finance lease actually replaces debt for a company[32]. While find the cost of capital, we can treat finance leasing in one of the following two ways we can explicitly show leasing as a source of capital, ii) we can completely ignc leasing by not showing as a source of capital for the company. This second treatme is similar to our treatment of NIBL in the previous section.

Example 3.9: Let's take an example to understand the adjustment process. RLM Limited i small aluminum manufacturer based on New Delhi. It wants to acquire a new machine th costs ₹5,000,000 through leasing. It plans to enter into a lease agreement with SSM Limit a leasing company based in Delhi. As per the agreement, RLM has to pay yearly lease rent of ₹937,220 for the next 8 years to SSM Limited. The pre-tax cost of debt of RLM is 10%. T after-tax cost of debt is 7%.

As per Indian Accounting Standard 17 (Ind_AS 17)[33], on the date of commencement of t finance lease term the lessee will show the asset in its balance sheet at its fair market val or if lower, at the present value of the future lease rentals payable. Subsequently, the lea payments must be apportioned between the finance charges and the reduction in outstandi liability. Though the lessee is not the owner of the asset, Ind_AS 17 requires the lessee depreciate the asset over the life of the lease period[34]. Since the lease rental in a given ye need not be equal to the sum of finance charge and depreciation, the asset and liability figur for the given asset may be different during the lease period.

How should RLM estimate its cost of capital? RLM has two choices with it: i) it can consid finance leasing as a source of fund (Method-3 of Example 3.8), or ii) it can simply ignore (Method-2 of Example 3.8). If it adopts the first option, it should show ₹5,000,000 as a sour of fund. The cost of this debt should be 10% and it must be adjusted for tax, as we do for th cost of any debt instruments. Following Ind_AS 17, RLM must show finance charges an depreciation separately and hence we do not require any further adjustment in the cash flow

Alternatively, we can just ignore this finance lease while computing the cost of capital fo RLM. As I showed in Example 3.8, this will not affect the free cash flow to equity figure fo RLM and hence which treatment is adopted does not affect the equity value of the compan However, we need to restate the EBIT figure for RLM, if we decide to ignore this financ lease while computing the cost of capital. If the finance lease is ignored in cost of capit calculation, then all the expenses incurred by RLM in serving the finance lease must be treate as operating expenses and hence must be subtracted before computing the EBIT. Since th interest expenses (finance charges for the finance lease) are included along with other interes expenses, we need to separate out the interest expenses on finance lease and subtract thi figure from EBIT and compute the adjusted EBIT.

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32. The riskiness of the cash flows that the finance company gets from a manufacturing company in the form of interest and principal payments (debt cash flows) and in the form of lease rentals (lease cash flows) are the same.
33. Source: http://www.mca.gov.in/Ministry/pdf/Ind_AS17.pdf
34. If ownership of the asset is not likely to be transferred to the lessee at the end of the lease period, then the asset must be fully depreciated over the life of the lease.

As we can clearly see, it is better to treat the finance leases as a source of capital, while computing the cost of capital. The big advantage of doing this is that you do not to do any extra adjustment to compute the free cash flow. Let's take the example of Kingfisher Airlines. As on 31 March, 2011, finance leases constituted ₹610.41 crores (12%) of the total secured loans of ₹5184.53 crores. So, we should include this figure of ₹610.41 crores as a source of capital while estimating Kingfisher's cost of capital.

Not all leases are finance leases. Many companies lease assets for a short time period. Such leases are called **operating leases.**[35] Retail companies and restaurants normally use leased properties to locate their shopping malls (or restaurants). The ATMs (and in a number of cases, the bank branches) of most banks are located on leased properties. For other companies, the amount of operating lease (as % of total capital) will be very small. Many companies, for example, use leased flats to accommodate their employees. Operating lease is actually a debt for the company. However, how you want to treat it in your cost of capital calculation depends on convenience and its importance. Remember that, for operating leases, the lessee does not divide the lease rental into a finance charge and depreciation. Second, companies treat operating lease as an off-balance sheet item.

Example 3.10: Let's take the example of Pantaloon Retail (India) Limited[36]. Since operating lease is an off-balance sheet item, you need to read the notes to account to get details of operating lease agreements entered into by the company. As on 30 June, 2011, Pantaloon Retail has entered into operating lease agreements for its fixed assets (mostly its shopping outlets) with future lease rentals payable of ₹1,904.91 crores. The company gives further details of these lease rentals payable in its notes to accounts. I give the details here.

Time Period	Lease Rentals Payable
Within 1 Year	₹ 656.06 crores
Between 1 Year and 5 Years	₹ 1,112.52 crores
After 5 Years	₹ 136.33 crores

In order to find the market value of the operating lease, we need to find the present value of these future lease rentals payable. Since the risk of operating lease is similar to the normal debt of a company, we need to discount the above cash flows using the cost of debt of Pantaloon Retail.

Care has assigned a Care A rating to Pantaloon Retail[37]. This is the most recent rating that is available for the company (as on May 3, 2012). The credit spread for a A-rated bond is around 200 basis points. Since the risk-free rate is 9.12%, the cost of debt of Pantaloon Retail is around 11.12%.

We need to discount the lease rentals payable using 11.12% as the discount rate to find the market value of the operating lease. Since the company does not give year-wise breakup of the lease rentals payable between year 1 to year 5, I simply divide ₹1,112.52 crores with 4 to get the lease rentals payable from Year 2 to Year 5. I assume that the ₹68.165 crores (₹136.33 crores divided with 2) of lease rental is payable in year 6 and 7. The present value of the lease rentals comes to ₹1,433.75 crores. I show my workings in the nearby table.

35. This is an oversimplified definition. Read Ind-AS17 on how leases are classified into finance and operating leases.
36. In May 2012, the name of Pantaloon Retail India got changed to Future Retail India Limited.
37. Source: http://www.careratings.com/Portals/0/CareAdmin/CompanyFiles/PR/FUTURE%20AGROVET%20LIMITED-05032012.pdf

Year	Lease Rental Payable	PV of Lease Rental @ 11.12%
1	656.06	590.41
2	278.13	225.25
3	278.13	202.71
4	278.13	182.42
5	278.13	164.17
6	68.165	36.21
7	68.165	32.58
	Total	1,433.75

In its balance sheet, Pantaloon Retail shows ₹7,846.14 crores as interest bearing liabilities now need to add the capitalized value of operating lease rentals payable of ₹1,433.75 crore this figure to get the total debt of Pantaloon Retail. The total debt will come to ₹9,279.89 cro

The beta of Pantaloon Retail is 1.58. The implied cost of equity therefore comes to 20.3 The cost of debt of Pantaloon is 11.12%. The market cap of Pantaloon Retail was ₹4,483 crores in late June 2012. I show the capital structure of Pantaloon Retail here.

Sources of Capital	Mkt Value (₹ Crores)	% Weight	After-tax cost
Equity	4,483.39	32.58%	20.39%
Debt	7,846.14	57.01%	7.41%
Operating Lease	1,433.75	10.42%	7.41%
Total	13,763.28		

The cost of capital of Pantaloon comes to 11.64%. While valuing Pantaloon Retail using FCF method, we should use 11.64% as its cost of capital. Since we treat the capitalized val of operating lease rentals as a debt, we must adjust the reported EBIT figure. This step necessary as it assures **liability consistency**.

Pantaloon Retail has reported ₹895.77 crores as its Earnings before Interest and Taxes (EBI for the year ending 30 June, 2011. The company has included the operating lease ren of ₹617.86[34] crores as an operating expense. We must divide this figure into an equivale depreciation and interest figure and restate the EBIT by subtracting only the depreciatio One way of doing it is to find the implied interest expense on ₹1,433.75 crores. Since the co of debt is 11.12%, the implied interest expense comes to ₹159.43 crores:

Implied Interest Expense = ₹1,433.75 crores × 11.12% = ₹159.43 crores.

We can now find the implied depreciation figure by subtracting ₹159.43 crores from ₹617.8 crores:

Implied Depreciation Expense = ₹617.86 - ₹159.43 = ₹458.43 crores.

Schedule 17 (page 97) of the annual report, the company has shown ₹775.28 crores as rent includi ase rentals. However, in 2010, the company has shown ₹617.86 crores as lease rental payable in th ext year. I am just using this figure directly. I therefore assume here that the company has not enter into any new operating lease agreement during 2010-11 that can affect this figure of ₹617.86 crores

The adjusted EBIT of Pantaloon Retail can now be found out by adding[39] the interest components on the capitalized lease rentals to the reported EBIT figure. It is given by:

Adjusted EBIT = ₹895.77 + ₹159.43 = ₹1,055.20 crores.

While computing the free cash flow of Pantaloon Retail, we must use ₹1,055.20 crores as the EBIT of Pantaloon Retail.

In case operating lease is a small part of the total debt, you can follow the second method we discussed where we just ignore the operating lease. Since the entire operating lease rentals are subtracted anyway before computing the EBIT, we do not need to restate the EBIT figure in this case. Most of the time, you will find this method convenient (that of ignoring the operating leases while computing the cost of capital). This method assures that we do not need to do any additional adjustment while computing the free cash flow.

Foreign Currency Loans

The adjustments that need to be made in the case of foreign currency loans are quite straight forward. Let's take a hypothetical example to understand how this is done.

Example 3.11: Assume that a company has borrowed US$ 10,000 @6% per annum. Suppose the current rupee-dollar spot rate is ₹56/1$. The 1-year forward rate is ₹58/1$. We cannot use 6% as the pre-tax cost of debt because the cash flows of our company are expressed in rupee terms. We must find an equivalent rupee interest rate. (Actually interest rate is expressed in percentages. By rupee interest rate, I mean an interest rate that can be used to discount the rupee cash flows. Remember our earlier discussion that we must match the riskiness and type of the cash flow and the discount rate.)

Our company borrows ₹10,000 × 56 = ₹560,000 today. It has to repay US$10,000 × 1.06 after 1-year. If it wants to buy the US$ through a 1-year forward contract, then it requires ₹10,000 × 1.06 × 58 = ₹614,800. Therefore our company is actually borrowing ₹560,000 and repaying ₹614,800 after 1-year. The implicit rupee interest rate comes to

$$\frac{614800 - 560000}{560000} \times 100 = 9.79\%$$

We must therefore use 9.79% (and not 6%) as the pre-tax cost of debt in this case[40].

In this example we assumed that the company would purchase US$10,600 with a 1-year forward contract. Will our answer be any different if the company decides to speculate? The answer is no. Even if the company decides not to hedge by speculating in the forex market, the expected spot rate one year from today will equal the one-year forward rate prevailing today. So the best guess that anyone can have on the spot rate likely to prevail one year from today is ₹58 per 1$. So the rupee cost of debt will still be 9.79%.

In the above example, we assumed that our hypothetical company borrows the money and repays it within one year. What if, it borrows for 5 years and repays the entire principal in one bullet payment? Using the expected spot rates (you will not get forward rates data for 5-year period), first find out the expected rupee outflow and then find the IRR of this series of cash flow. In order to find out the expected spot rates, use the interest rate parity theorem (IRP). Since we used IRP in Example 3.5 earlier, I do not explain it here. In Table 3.3, I show the computations.

39. We need to add back the interest expense figure, because, the entire operating lease rental (including this interest) was subtracted while computing the EBIT.

40. Of course, if the free cash flow is denominated in US$, then we must use 6% as the cost of debt.

TABLE 3.3: COST OF DEBT COMPUTATION FOR A 5-YEAR $10,000 LOAN

Year	0	1	2	3	4	
Loan Amount	$10,000.00					
Interest Payments		$600.00	$600.00	$600.00	$600.00	$600.0
Principal Repayment						$10,000.0
Expected Spot Rate	56	58.17	60.43	62.78	65.22	67.7
Rupee Interest Rate	7%					
U.S. Interest Rate	3%					
Overall US$ cash flows	$10,000.00	($600.00)	($600.00)	($600.00)	($600.00)	($10,600.00
Rupee Cash Flows	560,000.00	(34,904.85)	(36,260.38)	(37,668.55)	(39,131.41)	(718,169.00
Implied Rupee Cost of Debt	10.12%					

The implied cost of debt comes to 10.12%. As you can see, this is very close to the figure w obtained earlier by assuming that the entire loan will be repaid within 1 year.

Preference Shares

Though preference shares are categorized under the broad heading 'Shareholder Funds' in the balance sheet of the Indian companies, they have a lot of feature common with the debt capital issued by the companies. The preference dividen rate is usually fixed and at the end of a pre-specified number of years, the preferenc shareholder either gets back his/her money or gets his/her preference share converte into common stock. It is therefore appropriate to treat it like debt, while valuing th company[41]. The cost of preference capital is 'K_p' in the following equation:

$$P = \sum_{t=1}^{T} \frac{Div_t}{(1+K_p)^t} + \frac{\text{Redemption Value}}{(1+K_p)^T} \quad \text{... 3(5)}$$

Here, P is the market value of the preference share that will mature in year T,

Div_t is the dividend received in year t[42], and

K_p is the cost of preference share.

If the market price is equal to the face value, and if redemption takes place at par then the cost of preference share will be equal to the dividend rate. One can use the above equation to find out the cost of a plain vanilla preference share. Preferenc shares can also have conversion options attached to them. We will discuss abou them in Section 3.4.

While estimating the cost of preference shares, we encounter more or less simila problems that we encounter while finding the cost of debt. Most of the preference shares do not trade in the market. Therefore, it is very difficult to find the marke

41. Of course, we will not do any tax adjustment, as the companies get no tax benefits by having preferenc capital. Preference dividends are not tax-deductible payments.
42. Since companies pay the same dividend every year, one can replace the subscript.

value. Sometimes people find the cost of preference shares by just taking the arithmetic average of the cost of equity and the pre-tax cost of debt. This approach is often justified by saying that preference shares have got features of both equity and debt.

We must be aware about two tax-related issues while finding the cost of preference shares.

1. Unlike in the case of debt, we don't need to multiply the cost of preference shares with (1-t). This is so because, preference dividends are not tax-deductible expenses. Therefore, the pre-tax and the post-tax costs of preference shares are the same.
2. Indian companies are required to pay a dividend tax on preferred dividends. As of 2013, the dividend distribution tax rate (including 10% surcharges and 3% cess) is 16.995%. We already discussed about dividend distribution tax on equity dividends in the appendix to Chapter 2. I therefore do not discuss this issue any further.

In this section we discussed how to find the cost of certain types of debt instruments usually issued by the companies. With the interesting innovation in the financial market, we expect companies to issue more complex instruments. Finding the cost and value of such instruments no doubt will be a daunting task. However, once one understands that cost of debt is actually the opportunity cost of using the debt capital, it will become easier to deal with any complex security. In the next section we will discuss the issues relating cost of equity.

3.3 Cost of Equity

Earlier in the chapter we define cost of equity as the minimum return that the shareholders require from their equity investments from the company. It is not defined as the dividend yield. It is not defined as the actual return that the stock has generated over a specific time period[43]. There are two components in the shareholders' expectations. Dividend yield is just one part of it. The other part, often the more important part is the capital gains that the shareholders expect from their investments. Therefore, while you define (or estimate) the cost of equity, make sure to account for both these components.

In this section, we will discuss the different approaches used by the analysts to estimate the cost of equity of companies. Let's just think about the magnitude of the problem ahead of us. There are thousands of shareholders in a company. It is possible that each one of them requires different returns from his stock investment. A non-diversified investor will require higher return from his investment than a diversified investor. An internationally diversified investor will require different returns than a domestic investor (who is not internationally diversified). In this case how will we estimate the cost of equity?

43. It is also not defined as the return expected by the shareholders. Thus for example, when the stock is undervalued, the shareholders expect a higher return from the stock. However, in an efficient market, the required return will equal the expected return. In this section, I use the two terms interchangeably.

Well, in real life, we make life easier by assuming that the investors have homogeneo expectations. That is each shareholder (be it an individual or a mutual fund) requir the same minimum return from the stocks. All we need to do therefore is to find out t return that any representative shareholder is looking for from his investments. Whi doing so, we make certain assumptions about the behaviour of this representati investor. The most crucial assumption that is of interest to us is that the representati investor holds a diversified portfolio. If all investors hold diversified portfolios, the the market will not price non-systematic risk. (Why price a risk that can be diversifie away?) Therefore, the investors expect to be compensated for bearing the systemat risk and not the total risk of the stock[44].

We can use any of the following three models[45] to find the price of the systemati risk (although analysts use CAPM much more frequently than the other models):

- Capital Assets Pricing Model
- Arbitrage Pricing Model
- Multi-Factor Model

The total risk than an investor faces is divided into two categories, namely **systemati** and **unsystematic risk.** That part of the risk, which can be diversified away b holding diversified portfolios, is known as the unsystematic (or idiosyncratic risk o diversifiable risk). The risk that cannot be diversified away is called systematic risk The investors need to worry about the systematic risk alone because this cannot b diversified away and the investor has to bear it if he wants to invest in risky assets

The asset pricing models predict that since the non-systematic risk can be diversifie away, only the systematic risk gets priced in the market. The required return will b related only to the systematic risk. These models also postulate that the required retur is linearly related to some measure of systematic risk. In this chapter, I will discus about CAPM at length. I briefly discuss about the other two models in Appendix (of this chapter[46].

Capital Assets Pricing Model

The central tenet of CAPM is that the required return on any capital asset[47] is linearl related only to beta (the measure of systematic risk), the ratio of its covariance wit

44. Cost of equity is also often defined as the minimum return required by a marginal investor. The margina investor is one who trades frequently and hence is more likely to be the next investor for a company It is usually assumed that the marginal investor is well-diversified (at least within the country). Sinc rupee is not fully convertible in the capital account in India as of now, Indian investors are not allowe to invest outside the country (with a few exceptions).
45. One can find varieties of models in academic literature to find the price of systematic risk. However, i this book, I will restrict our discussion to these three models.
46. A general idea about the three models can also be obtained from any standard book on corporate financ or investments. See for example, Brealey, Myers, Allen, and Mohanty (2014).
47. Though we generally use this model to find the required return on stocks, it can be used to find th required return on any capital asset.

the market portfolio and the variance of the market portfolio. The required return from stock investment can be written as:

$r_i = r_f + \beta_i(r_m - r_f)$, where

r_i is the required return from the stock,

r_f is the risk free rate of return,

β_i is the measure of systematic risk, and

$r_m - r_f$ is the market risk premium

If one uses CAPM to estimate the cost of equity of a company, then one needs to know how to estimate the risk free rate of return, the expected market risk premium, and the beta of the stock. Now, I briefly discuss about how one can estimate the above three parameters here.

Estimation of the risk- free rate of return

While estimating the risk free rate of return we must keep in mind that ***CAPM is a one- period model.*** While deriving the CAPM, we assume that the investor has a single period investment horizon[48]. The risk-free rate of return, therefore, depends on the investment horizon of the investor. If the investor has a one-year investment horizon, he should use the one-year risk-free rate. If the investor has a 5-year investment horizon, he should use the 5-year risk-free rate. When we assume the investor to have a 5-year horizon, we actually assume that the investors invests his money today and gets all the cash back at the end of 5 years. What investment horizon should we use when we value the company? There are two practical issues that we must keep in mind while answering this question.

We project free cash flows for each year for the next infinite number of years. We never encounter flat yield curves in real life and hence, each cash flow should be discounted using a discount rate that uses spot rate for that period.

Example 3.12: A risk-free asset will give the following cash flows in the next 3 years.

Year	*1*	*2*	*3*
Cash Flows	100	100	200

Let's also assume that the following spot rates[49] are prevailing today.

Year	*1*	*2*	*3*
Spot Rate	5%	6%	7%

We can now, discount the three cash flows using the corresponding spot rates and find the present value of the risk-free asset.

$$Value = \frac{100}{1.05} + \frac{100}{1.06^2} + \frac{200}{1.07^3} = 374.50$$

48. We derive the CAPM equation by maximizing the investor's end-of-period utility function.

49. These rates are used to discount cash flows on zero-coupon bonds. Thus for example, the two-year spot rate of 6% will be used to discount cash flows from an asset that gives a single cash flow at the end of year 2.

We can also find out, what is known as the yield of the asset, one single rate, that when appl to the three cash flows will also give us the same present value. The yield can be found solving the following equation:

$$374.50 = \frac{100}{1+r} + \frac{100}{(1+r)^2} + \frac{200}{(1+r)^3}$$

By solving the above equation for 'r', we obtain

$r = 6.52\%$

So, when we have a non-flat yield curve, we can find the present value of the cash flows one of the following two ways:

(*i*) Discount the t[th] year's cash flow by using the t-year spot rate prevailing today and th add up all the discounted values.

(*ii*) Discount each cash flow using a single discount rate, known as the yield, and add- all the discounted cash flows.

Though these two methods give us identical answer, the problem with the second approa is that the yield depends not only on the spot rates, it also depends on the cash flows (th amount and timing). Thus for example, in the above example, if the last cash flow of 20 comes in year 1 rather than in year 3[50], then the yield will be 6.09%. Remember that whi valuing a company, the terminal value will always influence the overall cash flow pattern. S the longest-term spot rate available will influence the risk-free rate that we should use fo valuation more than the intermediate term spot rates.

This implies that if we want to use one risk-free rate to estimate the cost of capital[51], then w must not only know the spot rates for the next infinite number of years, we must also kno the pattern of cash flows for each year. Theoretically, the risk-free rate for a high grow company (higher cash flows in the distant future) will be different from the rate that we ca use for a mature and stable company.

You can get the zero-coupon yield curve (ZCYC) from the website of the National Stoc Exchange of India[52]. In Figure 3.7, I show the ZCYC estimated by NSE at the beginning May 2012.

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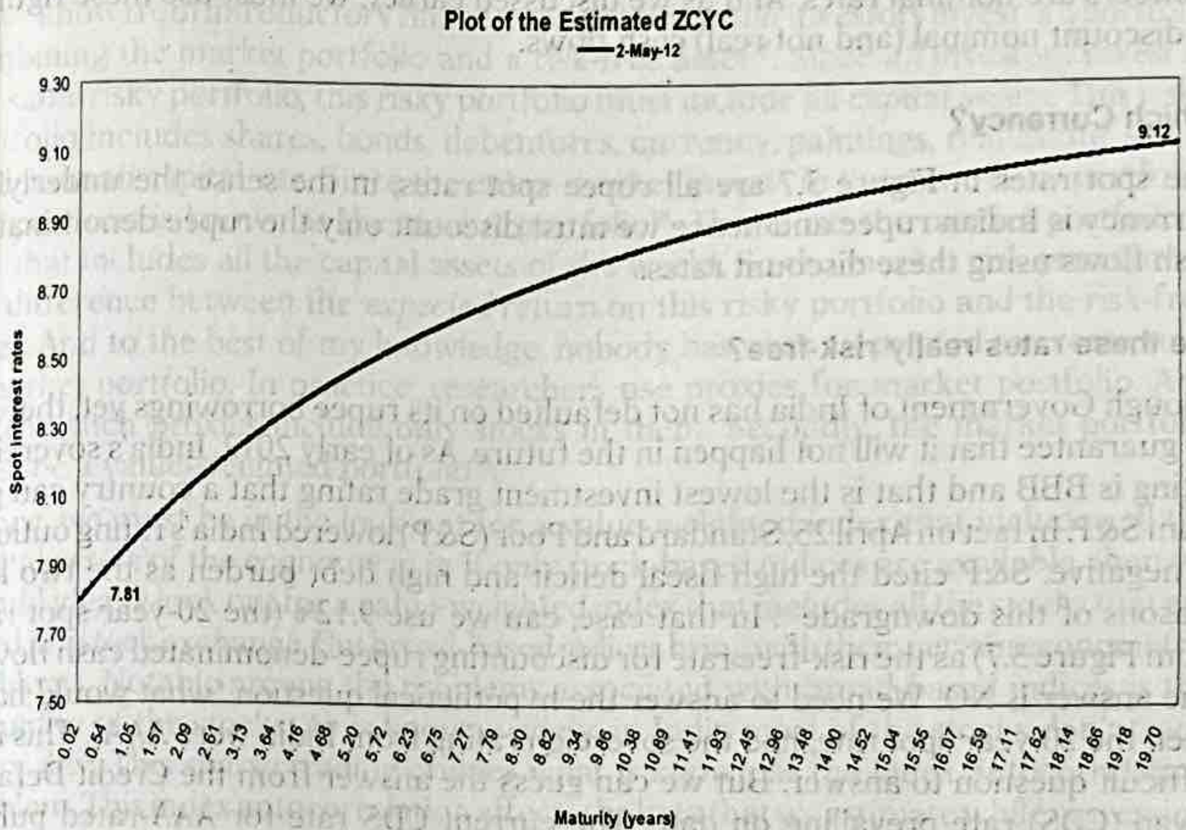
50. The cash flows for the three years change to 200, 100, and 100.
51. And that is what almost everyone in the business does.
52. http://www.nseindia.com/p. oducts/content/debt/wdm/archieve_debt.htm

FIGURE 3.7: ZERO COUPON YIELD CURVE ESTIMATED BY THE NSE



We can see from Figure 3.7 that the yield curve was upward sloping on 2nd May, 2012. The 20-year spot rate is higher than the 1-year spot rate. You will also notice that the yield curve is getting almost flat towards the extreme right. Thus for example, on 2 May, 2012, the 20-year spot rate was 9.12% and if we assume the spot rates to remain at 9.12% for longer than 20-year periods, then we can use 9.12% as an approximate yield for all the cash flows while valuing the company. This is not an exact solution to the problem we have, but will come pretty closer to the number we are looking for as we shall be discounting cash flows for the next infinite number of years.[53] Instead of using the spot rates, if you prefer to use the yield on coupon paying securities issued by the RBI, then you will get a different answer. The yield on the 10-year G-Sec was 8.69%[54], whereas the average yield is 9.12%[55]. However, never use the one-year spot rates or yield figures while valuing a company. These numbers may be used for discounting cash flows from a project that has a one-year life.

In addition, you also must be aware of the following technical issues while estimating the risk-free rate.

Real or Nominal Rate?:

The spot rates we showed in Figure 3.7 are all nominal rates. These numbers have been derived from yield data of Government Securities (G-Secs). When we invest in

53. I find the yield to be 9.06% when I considered 100 years of cash flows with cash flows growing at a constant rate of 5%. The yield was 9.05% when I considered only 50 years' cash flows.
54. The 10-year spot rate on this date was 8.81%.
55. Not everybody uses the ZCYC prepared by the NSE. But you must use the longest term spot rate available from your ZCYC.

G-Sec, we do not get any additional compensation for the inflation. So all the abov numbers are nominal rates. And as we discussed earlier, we must use these figur to discount nominal (and not real) cash flows.

Which Currency?

The spot rates in Figure 3.7 are all rupee spot rates, in the sense the underlyin currency is Indian rupee and hence we must discount only the rupee denominate cash flows using these discount rates.

Are these rates really risk-free?

Though Government of India has not defaulted on its rupee borrowings yet, there i no guarantee that it will not happen in the future. As of early 2012, India's sovereig rating is BBB and that is the lowest investment grade rating that a country can ge from S&P. In fact on April 25, Standard and Poor (S&P) lowered India's rating outloo to negative. S&P cited the high fiscal deficit and high debt burden as the two ke reasons of this downgrade[56]. In that case, can we use 9.12% (the 20-year spot ra from Figure 3.7) as the risk-free rate for discounting rupee-denominated cash flows The answer is NO. We need to answer the hypothetical question "what would hav been the 20-year spot rate, had the sovereign rating from India were AAA". This is difficult question to answer. But we can guess the answer from the Credit Defau Swap (CDS) rate prevailing on date. The current CDS rate for AAA-rated publi sector undertakings in India is around 100 bps. The Indian rupee risk-free rate the will probably equal 9.12% - 1% that is 8.12%.

However, in this book, I use 9.12% as the risk-free rate. If one uses 8.12% as the risk free rate, then one has to increase the market risk premium by around 1%. I do no do this adjustment.

Estimation of the Market Risk Premium

We need an estimate of the market risk premium while estimating the cost of equit using the Capital Assets Pricing Model. We will first try to understand the role marke risk premium plays in CAPM (from a theoretical point of view) and then we will se how the market risk premium can be estimated.

Market risk premium (MRP) is simply the difference between the return on the marke portfolio and the return on the risk-free asset. In order to estimate the market ris premium, we must have answers to certain theoretical issues.

[56] http://www.tribuneindia.com/2012/20120426/main1.htm

What is the definition of the market portfolio in CAPM?

As we know from introductory finance, in equilibrium, *all* investors invest in a portfolio combining the market portfolio and a risk-free asset[57]. Since *all* investors invest in the *same* risky portfolio, this risky portfolio must include all capital assets. This risky portfolio includes shares, bonds, debentures, currency, paintings, real estate, silver, gold, human capital, etc. Since the entire market invests in the same risky portfolio, this portfolio is known as the market portfolio[58]. Therefore, the market portfolio is one that includes all the capital assets of the world. So the market risk premium is the difference between the *expected* return on this risky portfolio and the risk-free asset. And to the best of my knowledge, nobody has ever attempted to create such a market portfolio. In practice, researchers use proxies for market portfolio. And usually such proxies include only stocks in them. Secondly, the market portfolio must be a value-weighted portfolio[59].

Hence, we must be in the look-out for a value-weighted index that includes all the capital assets of the economy in it. If only stock-based indices are available, then we should ideally look out for a value-weighted index that includes all the stocks that are listed in a stock exchange. But broad-based indices bring with them certain econometric problems. Notable among the problems associated with broad-based indices is the illiquidity of the stocks. As is known widely, in India most of the stocks don't trade every day. This sparse trading creates what is known as the index autocorrelation problem. This index autocorrelation affects the beta that we estimate while regressing the stock returns on the index returns.

Secondly, the return on an index should include both the dividend yield and the capital gains. Most of the indices, however, are not adjusted for dividend. Let's take an extreme example to understand the effect of dividend non-adjustment on the index. Assume that the dividend yield of all the stocks included in an index is 1per cent. If all the stocks go ex-dividend on the same day, then the index will fall by 1per cent. The estimated return on the index will be lower by 1% compared to the true return on the index. So, if you use an index that is not adjusted for dividend, then remember to add back the dividend yield to estimate the market risk premium.

In India, the Bombay Stock Exchange and the National Stock Exchange use what is known as free-float adjusted weights to estimate the value of most of the indices. I explain this methodology by using Sensex as an example in Table 3.4.

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57. In the presence of the risk-free asset, the capital market line (CML) acts as the efficient frontier. A linear combination of the risky market portfolio and the risk-free asset dominates all other risky portfolios. See Tobin, James, 1958, Liquidity Preference as Behaviour Toward Risk, The Review of Economics Studies, February.
58. Of course, in practice, not everyone invests in the same risky portfolio. But this is what is assumed when we derive the CAPM equation.
59. This directly follows from the fact that under the assumption of homogeneous expectations, all the shareholders will hold the market portfolio in equilibrium. It can be easily proved that if every investor invests in the same risky portfolio, that portfolio must be a value-weighted portfolio.

TABLE 3.4: FREE FLOAT ADJUSTED SENSEX (DATA AS ON 10 MAY, 2012)

Company	Close Price	No.of Shares (normal)	Full Mkt. Cap.	Free-Float Adj. Factor	Free-Float Mkt. Cap	Weight in In
I T C LTD	240.05	7,818,424,300	187,681.28	0.7	131,376.89	
RELIANCE	695.1	3,274,550,412	227,614.00	0.55	125,187.70	
INFOSYS LTD	2,367.70	574,230,001	135,960.44	0.85	115,566.37	
HDFC BANK LT	512.45	2,343,641,695	120,099.92	0.8	96,079.93	
HOUSING DEVE	650.5	1,473,847,551	95,873.78	1	95,873.78	
ICICI BANK L	821.8	1,152,718,462	94,730.40	1	94,730.40	
TCS LTD	1,225.90	1,957,220,996	239,935.72	0.3	71,980.72	
LARSEN & TOU	1,158.90	612,398,899	70,970.91	0.9	63,873.82	
ONG CORP LTD	256.3	8,555,490,120	219,277.21	0.25	54,819.30	
STATE BANK O	1,887.60	634,999,595	119,862.52	0.45	53,938.14	
TATA MOTORS	293	2,691,486,150	78,860.54	0.65	51,259.35	
HIND UNI LT	432.2	2,160,879,033	93,393.19	0.5	46,696.60	
BHARTI ARTL	309.8	3,797,530,096	117,647.48	0.35	41,176.62	
MAHINDRA & M	655.15	613,974,839	40,224.56	0.75	30,168.42	
TATA STL	417.35	971,214,779	40,533.65	0.7	28,373.55	
WIPRO LTD	406.7	2,458,687,839	99,994.83	0.25	24,998.71	
NTPC LTD	151.45	8,245,464,400	124,877.56	0.2	24,975.51	
SUN PHARMACE	599.8	1,035,550,385	62,112.31	0.4	24,844.92	
BAJAJ AUTO	1,511.75	289,367,020	43,745.06	0.5	21,872.53	
COAL INDIA	326.45	6,316,364,400	206,197.72	0.1	20,619.77	
JINDAL STEEL	470.65	934,833,818	43,997.95	0.45	19,799.08	
MARUTISUZUK	1,301.95	288,910,060	37,614.65	0.5	18,807.32	
HEROMOTOCO	1,871.40	199,687,500	37,369.52	0.5	18,684.76	
BHEL	216.15	2,447,600,000	52,904.87	0.35	18,516.71	
CIPLA LTD.	322.6	802,921,357	25,902.24	0.65	16,838.46	
GAIL INDIA	322.55	1,268,477,400	40,914.74	0.4	16,365.90	
TATA POWER	97.7	2,373,072,360	23,184.92	0.7	16,229.44	
HINDALCO IN	119	1,918,607,775	22,831.43	0.7	15,982.00	
STERLITE IN	96.55	3,360,704,478	32,447.60	0.45	14,601.42	
DLF LIMITED	182.1	1,698,282,733	30,925.73	0.25	7,731.43	

Source: http://www.bseindia.com/about/abindices/bse30.asp

Depending on the promoter's holding in the company, BSE assigns a free-fl adjustment factor to each stock. This is shown in the fifth column on Table 3.4.1 actual market capitalization is then multiplied with this free-float factor to obtain free-float market capitalization. The weight of each stock is subsequently derived dividing the stock's free-float market capitalization with the total free-float ma capitalization of all the 30 stocks. There are undoubtedly some merits in compu the value of the index by looking only at stocks that are not with the promo However, the market portfolio in the CAPM equation is simply a market-cap w weighted index.

From the above discussion, we can conclude that we need to estimate the ma risk premium by using an index as a proxy for the true (but unobservable) m portfolio. The market proxy must have the following properties:

1. It should be a broad-based index. An index consisting of smaller numb stocks will not capture the mood of the overall capital market.
2. It should be a value-based index. The weight of each stock in the index be proportionate to its market capitalization. This represents the actual tr behaviour of the entire market.

3. It must include liquid stocks. If some of the stocks are illiquid, then the index does not fully capture the effect of any macro-economic event on the given date. It also affects the beta computations (as we will see shortly).
4. The return on the index must include dividend as well as capital gains. Since most indices do not include dividend, one must add the average dividend yield to the estimated market risk premium.

In Table 3.5, I compare three measures of liquidity of three indices in India, namely, BSE 500, Sensex, and Nifty. You can immediately notice that stocks belonging to Sensex and Nifty are much more liquid as compared to the remaining stocks. An average Sensex stock is 8 times larger than an average BSE 500 stock. It also trades seven times more frequently. Let's also look at the frequency distribution of number of transactions of the BSE 500 stocks shown in Figure 3.8.

TABLE 3.5: LIQUIDITY AND MARKET CAPITALIZATION OF AN AVERAGE STOCK IN THE THREE INDICES

	Total Mkt Cap (Rs. Crores)	Avg. Mkt Cap (Rs. Crores)	Turnover (Rs. Crores)	Shares Traded (No.)	Number of Transactions
BSE 500	50,471,540	100,943	26	195,337	1,866
Sensex	26,304,926	876,831	174	381,633	8,146
Nifty	31,200,087	624,002	129	370,227	6,547

Source: Prowess database.

Around 25% of the stocks report less than 250 transactions per day. This clearly shows that stocks belonging to BSE 500 are illiquid stocks. If we decide to use Sensex or Nifty as our market index, then we will almost completely ignore the small-cap stocks in India. So can we use Sensex or Nifty as the market index?

In Table 3.6, I show the correlation coefficients in the returns of five stock indices. I compute the correlations between the indices using monthly returns on these five indices. You can see that the indices are highly correlated with each other with the minimum correlation being 0.97.

These data show that we can use either Nifty or Sensex as our market proxy. These stocks are highly liquid and capture the overall mood of the market. These indices are highly correlated with the other broad-based indices as well[60].

60. The average monthly return was 1.3% for Sensex and 1.46% for BSE 500 over the sample period 2001-2012. The higher return of BSE 500 reflects the fact that small stocks, on an average generate higher returns because of their exposure to size risk of Fama and French. Instead of using Sensex, if we use BSE 500, for example, we should get a higher market risk premium.

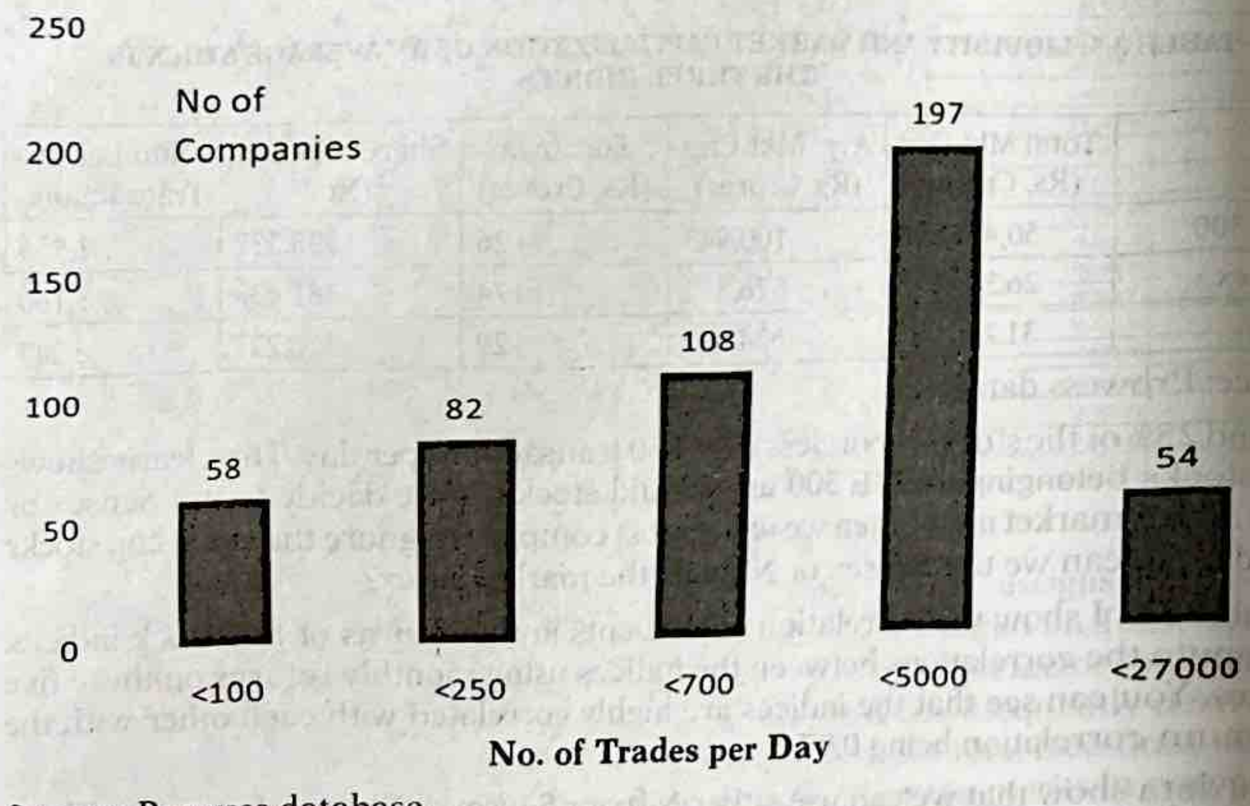
TABLE 3.6: PAIR-WISE CORRELATION BETWEEN MONTHLY INDEX RETURNS (2001-2012)*

	Sensex	Nifty	BSE 100	BSE 200	BSE 500
Sensex	1.00	0.99	0.99	0.98	0.
Nifty	0.99	1.00	0.99	0.98	0.9
BSE 100	0.99	0.99	1.00	1.00	0.
BSE 200	0.98	0.98	1.00	1.00	1.
BSE 500	0.97	0.97	0.99	1.00	1.

*Figures rounded off to two decimal places.

Source: Prowess database.

FIGURE 3.8: NO. OF TRADES PER DAY OF THE BSE 500 STOCKS



Source: Prowess database.

There is another advantage of using Sensex as the market index for estimatin market risk premium. We can get historical returns data on Sensex from the yea 1979. No other index boasts of such a large period of data availability in India. If w use Sensex as the market proxy in our study, we can get statistically better estima of historical market risk premium in India.

In CAPM we require the expected market risk premium. What is required in th context of CAPM is the expected return on the market portfolio and not the actu return. We estimate the expected market risk premium by using one of the followin two methods. In the first method, we estimate the historical market risk premiu by using the historical average of the market risk premium figures and then assum that the future risk premium will be similar to the historically realized market ris premium. In the second method, we estimate the expected market risk premium by computing the implied market risk premium in the current market valuation.

Estimation of Historical Risk Premium

In order to estimate the historical market risk premium, for each historical time period, we need to estimate the monthly realized index return and the monthly risk-free rate prevailing at that time. Since, we use the long term risk-free rates for company valuation, we should subtract the long term risk-free rate from the index return to estimate the excess index return for each time period. Then we can take the average of these excess returns to estimate the monthly risk premium. We can then annualize this figure.

What should be the estimation period?

Though we have data on the index from 1979, should we use the entire sample period, or should we exclude the pre-liberalization period? There are no clear answers to these questions. There is always some merit in using a larger sample period. The period must be large enough to include as many business cycles in it as possible. If we use a very short time period, then we will definitely get a biased answer. Thus for example, if you compute the historical risk premium immediately after a stock market boom, you will get a very high market risk premium figure. A larger sample period will average out all the ups and downs in the stock market and we get a better estimate of the market risk premium. However, if the characteristic of the entire stock market has changed (as has happened in India after 1991), then you may like to use a shorter time period. However, ensure that the estimation period includes a few stock market boom and bear phases. I show the historical market risk premium figure (with Sensex as the market index) in Table 3.7.

TABLE 3.7: HISTORICAL MARKET RISK PREMIUM IN INDIA (INDEX RETURNS INCLUDE THE DIVIDEND YIELD)

Time Period	*Sensex return*	*Average G-Sec Yield*	*Market Risk Premium*
1979-2011	22.89%	9.29%	13.61%
1991-2011	20.37%	9.82%	10.55%

Source: Estimated with data collected from the RBI website

Table 3.7 suggests 10.55% as the market risk premium for India (excluding the pre-1991 time period). I am not suggesting that in the next year, Sensex will actually give an excess return of 10.55%. This was the average excess return that Sensex generated in the past 20 odd years.

Survivorship Bias in Historical Premium:

Bombay Stock Exchange has made changes in Sensex composition more than 50 times between 1992 and 2012. Stocks that do not perform well or that go out of existence because of mergers or bankruptcy are removed from the index. Thus for example, on 12 January, 2009, BSE replaced Satyam Computers with Sun Pharmaceuticals.

Prior to January 2009, the market participants probably did not know of this change. But the post 2009 period monthly returns of Sensex include the actual returns of Sun Pharma and not that of Satyam Computers. That shows that the above figure of 10.55% has an upward bias. It reflects the actual performance of stocks that survived and performed well during the historical period. It does not correctly reflect the return expected by the market participants in the past.

How should we compute the average:

We can compute the average of the historical excess returns of the index by using either the arithmetic or the geometric average. If we use arithmetic average, then the average annualized market risk premium in India is 10.55%, whereas it is 7.13% if one uses geometric average. Since the difference is high, we must clearly understand the difference between the two and understand which of the two methods is better for our purpose.

Example 3.13: Let's take a simple numerical example to understand the difference between arithmetic and geometric average of stock returns. The following information is available about a stock on 3 actual dates.

Date	*Closing price*
1.1.2011	₹100
1.1.2012	₹50
1.1.2013	₹100

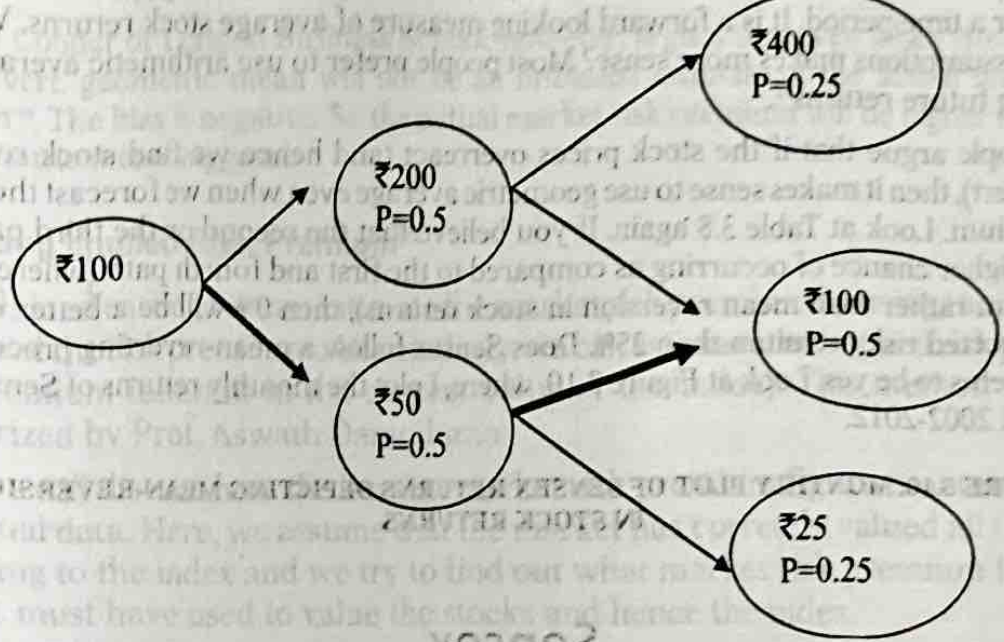
What is the annualized return here? Here, the return for the year ending 1.1.2012 is -50%, and 100% for the year ending 1.1.2013. If we use arithmetic averaging, then the average return will be (-50%+100%)/2 that is 25% per annum. If we use geometric averaging, on the other hand, then the average return will be 0% as shown here:

$$\sqrt{(1-0.5)\times(1+1)}-1=0\%.$$

Which number should we use? On the face of it, the geometric average seems to be a reasonable number because the stock has not appreciated at all over the two years' time period, and therefore has actually generated a 0% return in the sample period. This is also known as the **Buy and Hold Return**. Because if somebody buys the share on 1.1.2011 and holds it for two years, then he actually gets a return of 0%.

What is the significance of the 25% figure (which we obtain by using Arithmetic averaging) then? Based on the realized returns in the last 2 years (-50% and +100%), if we have to forecast the one-year return for next year then what will be our answer? Since we just have two data points, we can answer by saying: "...the stock will either give -50% return of +100% return over a one-year holding period..." It is as if the stock price follows a binomial return generating process (see Figure 3.9). At the end of one year, the stock either goes up by 100% or goes down by 50%. Therefore, as on 1.1.2014, the stock can take a value equal to either ₹200 or ₹50. Similarly as on 1.1.2015, the stock can take a value of either ₹400 or ₹100 or ₹25.

FIGURE 3.9: POSSIBLE PRICES OF OUR STOCK IN EXAMPLE 3.14



If we assume that there is an equal probability of the stock of either going up or down, then the probability that the stock will take a value equal to ₹400, as on 1.1.2014 is 0.25. The probabilities of the other figures are also given in the above figure.

When somebody makes the investment on 1.1.2013, what is the expected value of his investment as on 1.1.2015? The expected return for the 2-year period is 0.25(300%) +0.5(0%) +0.25(-75%) =56.25%. The implied annualized rate is, therefore 25%[61].

This is exactly the arithmetic rate of return in this example. In this example, the stock price could have taken any of the following four paths over the last 2-year period:

TABLE 3.8: POSSIBLE PRICE PATHS OF THE STOCK BETWEEN 2011 AND 2013

Path	*Probability*
100 - 200 - 400	0.25
100 - 200 - 100	0.25
100 - 50 - 100	**0.25**
100 - 50 - 25	0.25

However, we actually observe the third path, namely 100 - 50 - 100. In Figure 3.9 I have shown this path using bold arrow strips. Given that this is the path we observed in the last two years, which path we expect to see in the future?

If your answer is 'the third path', then you should use geometric average. That is if you believe that the future is going to be very much like the past, then use geometric average. So geometric average is a backward looking measure of average stock returns.

61. The annualized return can be obtained as follows:

$$(1+r)^2 = 1.5625$$

$$\Rightarrow r = \sqrt{1.5625} - 1 = 0.25 \quad or \quad 25\%\ pa.$$

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$\sqrt{(1-0.5)\times(1+1)}-1=0\%.$

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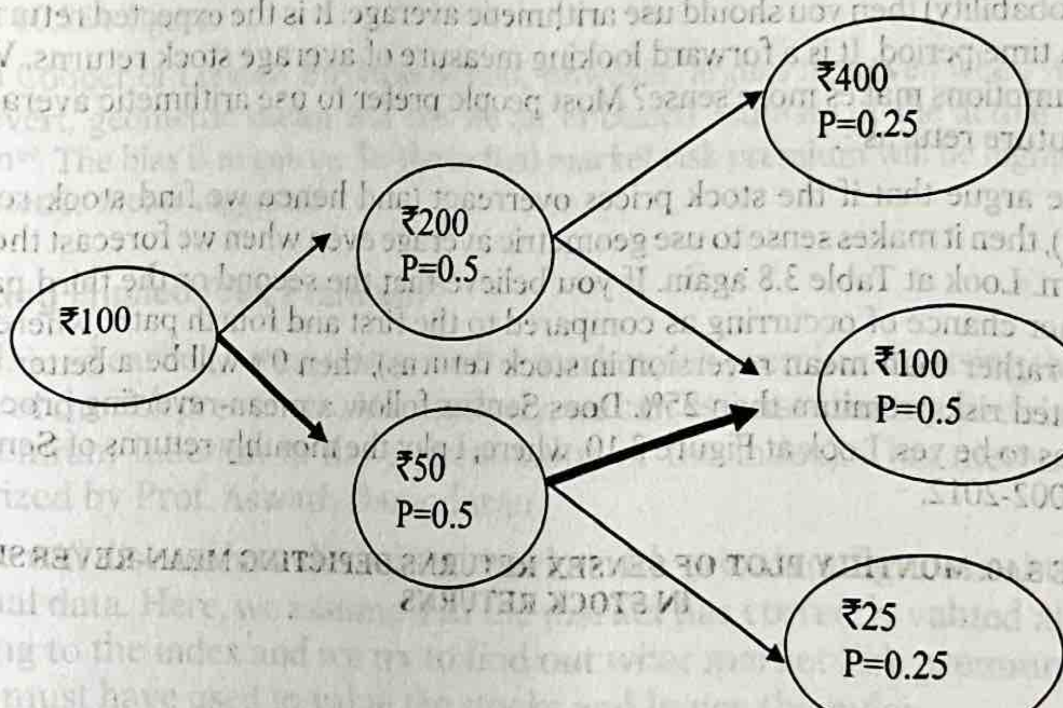

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61. The annualized return can be obtained as follows:

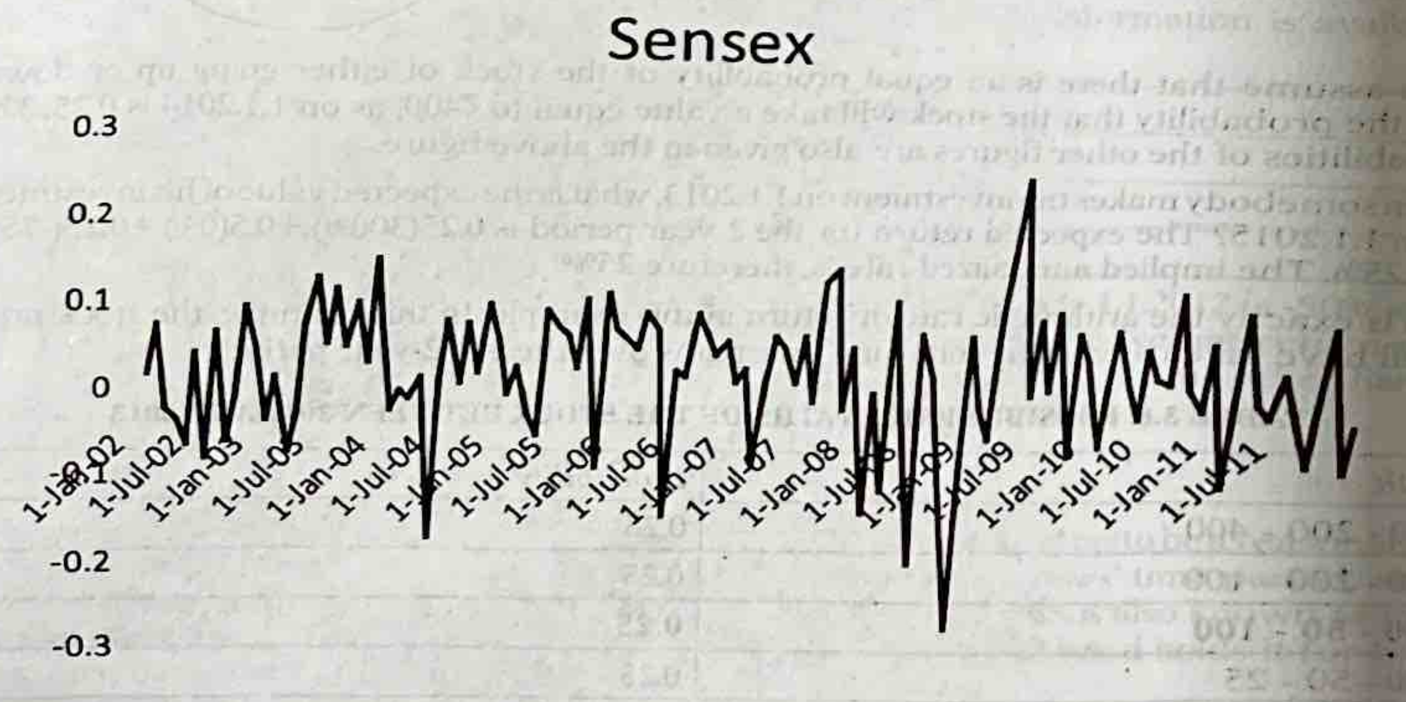
$(1+r)^2 = 1.5625$

$\Rightarrow r = \sqrt{1.5625} - 1 = 0.25$

On the other hand, if you believe that the stock price can take any of the above four path the given probability) then you should use arithmetic average. It is the expected returns stock over a time period. It is a forward looking measure of average stock returns. Wh the two assumptions makes more sense? Most people prefer to use arithmetic average predicting future returns[62].

Some people argue that if the stock prices overreact (and hence we find stock retur mean revert), then it makes sense to use geometric average even when we forecast the m risk premium. Look at Table 3.8 again. If you believe that the second or the third path a much higher chance of occurring as compared to the first and fourth path (where th momentum rather than mean reversion in stock returns), then 0% will be a better esti of the expected risk premium than 25%. Does Sensex follow a mean-reverting process answer seems to be yes. Look at Figure 3.10, where, I plot the monthly returns of Sensex the period 2002-2012.

FIGURE 3.10: MONTHLY PLOT OF SENSEX RETURNS DEPICTING MEAN-REVERSION IN STOCK RETURNS



Financial economists compare the long run volatility of stock returns with short run volatility find out if stock returns mean revert. If stock returns follow a random path, then the 2-year variance in Sensex returns should approximately be equal to 2 times their yearly variance find that 2-year variance is about half of 2-times the 1-year variance of Sensex returns. Th clearly shows that Sensex returns mean-revert[63]. If we expect this mean reverting proces

62. Blume (1974) first noted this property of Arithmetic mean. It is an unbiased estimate of the true mark risk premium. However, the discount factors for the second year onwards are biased estimators o the true discount factors, no matter whether we use arithmetic mean or geometric mean (because th discount factor is a non-linear function of the true market risk premium). See Blume, Marshall, E., 1974 Unbiased Estimators of Long Run Expected Rates of Return, Journal of American Statistical Association Vol. 69, 634 – 638.

63. While some economists interpret this as a sign of market inefficiency, others interpret it as time varying market risk premium.

to continue in the future, then 7.13% is a better proxy for the market risk premium in India than the 10.55% figure.

Prof. Ian Cooper of London Business School, however, argues that even when stock returns mean revert, geometric mean will not be an unbiased estimate of the actual market risk premium[64]. The bias is negative. So the actual market risk premium will be higher than what the geometric mean suggests.

Estimating Implied Risk Premium[65]

As I explained earlier, we can estimate the market risk premium by using the historical average of the risk premiums. Alternatively, we can estimate the implied risk premium (risk premium inherent in the current value of the index). This method has been popularized by Prof. Aswath Damodaran.

Now, we will discuss how the implied market risk premium figure is obtained using the actual data. Here, we assume that the market has correctly valued all the stocks belonging to the index and we try to find out what market risk premium figure the market must have used to value the stocks and hence the index.

In Table 3.9, I have shown the weights of the 30 stocks in Sensex on 10 May, 2012. Sensex closed at 16,420 points on 10 May, 2012. I first estimate how much dividend (and FCFE) an investor will get from each share if he invests ₹16,420 in the portfolio mimicking Sensex.

To illustrate the process, the weight of Bajaj Auto in Sensex was 1.58%. That means, 1.58% of ₹16,420, that is, ₹259.44 would be invested in the shares of Bajaj Auto in the mimicking portfolio. Since the closing price of Bajaj Auto on 10 May, 2012 was ₹1,677.9, one can buy 0.155 shares of Bajaj Auto with ₹259.44. This will entitle the investor to receive a total dividend of ₹6.17 in this mimicking portfolio. Similarly the FCFE per share comes to ₹10.96 from the investment made in Bajaj Auto.

We can similarly find out how much dividend one will receive from the remaining shares of Sensex. The total dividend from this investment comes to ₹248.68. The total FCFE comes to ₹410.23. You can notice from Figure 3.9 that the free cash flow to equity as negative for a few companies. However, I assume that the aggregate FCFE from Sensex is a better proxy for the overall FCFE of all these 30 companies.

TABLE 3.9: TOTAL DIVIDENDS AND FCFE FROM SENSEX (10 MAY, 2012)

Company Name	Closing Price	DPS	FCFE	Weights (%)	Amount Invested	No. of shares	Dividend	FCFE
Bajaj Auto Ltd.	1,677.90	39.93	70.90	1.58	259.44	0.15	6.17	10.96
Bharat Heavy Electricals Ltd.	256.95	6.30	5.64	1.34	220.03	0.86	5.39	4.83
Bharti Airtel Ltd.	336.75	1.01	-10.66	2.98	489.32	1.45	1.47	-15.49
Cipla Ltd.	304.55	2.01	0.54	1.22	200.32	0.66	1.32	0.35

64. Cooper, I., 1996, Arithmetic versus Geometric Mean Estimators: Setting Discount Rates for Capital Budgeting, European Financial Management, Vol 2, No. 2, 157 – 167.

65. I estimate the implied equity risk premium for Indian market every quarter and make the number available in my site: www.sites.google.com/a/xlri.ac.in/profmohanty. Visit my site to get the most recent estimate of the market risk premium.

Company Name	Closing Price	DPS	FCFE	Weights (%)	Amount Invested	No. of shares	Dividend	FCFE
Coal India Ltd.	343.10	9.92	7.41	1.49	244.66	0.71	7.07	5.2[illegible]
D L F Ltd.	201.50	1.99	3.46	0.56	91.95	0.46	0.91	15[illegible]
G A I L (India) Ltd.	374.95	8.51	-17.10	1.18	193.76	0.52	4.40	-8.8[illegible]
H D F C Bank Ltd.	520.05	3.28	-7.04	6.95	1,141.19	2.19	7.19	-15.4[illegible]
Hero Motocorp Ltd.	2,054.85	105.00	59.49	1.35	221.67	0.11	11.33	6.4[illegible]
Hindalco Industries Ltd.	129.45	1.50	0.57	1.16	190.47	1.47	2.21	0.5[illegible]
Hindustan Unilever Ltd.	409.90	7.01	9.94	3.38	555.00	1.35	9.49	13.4[illegible]
Housing Development Finance Corpn. Ltd.	673.60	9.03	23.09	6.94	1,139.55	1.69	15.27	39.0[illegible]
I C I C I Bank Ltd.	887.25	14.02	-35.15	6.85	1,124.77	1.27	17.77	-44.56
I T C Ltd.	226.85	4.45	5.99	9.51	1,561.54	6.88	30.61	41.25
Infosys Ltd.	2,864.95	34.95	137.47	8.36	1,372.71	0.48	16.75	65.87
Jindal Steel & Power Ltd.	545.00	1.53	-38.99	1.43	234.81	0.43	10.66	-16.80
Larsen & Toubro Ltd.	1,306.85	14.51	32.54	4.62	758.60	0.58	8.42	18.89
Mahindra & Mahindra Ltd.	696.90	11.50	-12.20	2.18	357.96	0.51	5.91	-6.27
Maruti Suzuki India Ltd.	1,349.10	7.55	118.83	1.36	223.31	0.17	1.25	19.67
N T P C Ltd.	162.70	4.30	0.80	1.81	297.20	1.83	7.85	1.47
Oil & Natural Gas Corpn. Ltd.	267.30	8.50	23.11	3.97	651.87	2.44	20.73	56.35
Reliance Industries Ltd.	748.25	8.01	34.55	9.06	1,487.65	1.99	15.92	68.69
State Bank of India	2,095.00	29.96	447.62	3.90	640.38	0.31	9.16	136.83
Sterlite Industries (India) Ltd.	111.10	2.10	17.56	1.06	174.05	1.57	3.29	27.51
Sun Pharmaceutical Inds. Ltd.	569.50	3.47	14.48	1.80	295.56	0.52	1.80	7.51
Tata Consultancy Services Ltd.	1,167.85	17.05	37.72	5.21	855.48	0.73	12.49	27.63
Tata Motors Ltd.	275.70	4.00	-8.25	3.71	609.18	2.21	8.83	-18.24
Tata Power Co. Ltd.	100.85	1.25	-4.74	1.17	192.11	1.90	2.38	-9.03
Tata Steel Ltd.	470.40	12.00	-16.97	2.05	336.61	0.72	8.58	-12.14
Wipro Ltd.	439.00	6.01	3.84	1.81	297.20	0.68	4.07	2.60
						Total	248.68	410.23

Source: Data collected from the website of BSE and Prowess database.

The next step is to estimate the projected growth rate the market expects from these stocks. There are a number of options available here. You can use the historical growth rate in FCFE for these 30 stocks. You can use the growth rate projected by the analysts. You can alternatively use the projected GDP growth rate (adjusted for inflation) as a proxy for the growth rate in FCFE.

The Central Statistical Organization has estimated that the GDP growth rate for 2011-12 is going to be around 6.9%[66]. Though the RBI, and the Economic Advisor to the Prime Minister initially projected a growth rate of around 8% for the year, the revised growth rate is lower by more than 1%. The March 2012 IIP figures are also lower than the March 2011 IIP figures by around 3.5%.

66. Source: The Economic Times, February 12, 2012.

It is difficult to make economic projections. Let's assume that the GDP growth rate will increase to 7% in 2013 and then will remain at 7% till 2015, and then will decline to around 5% by 2022. Let's also assume that the GDP growth rate will remain constant at 5% after 2022. As the size of Indian economy becomes larger, the projected GDP growth rate must come down because of the larger base.

The GDP growth rate is a real growth rate. Our free cash flows to equity are nominal cash flows. The current inflation rate in India is around 7%. However, the long-term inflation rate is expected to come down to about 5%. I also assume that the FCFE will grow at the same rate as the GDP growth rate (adjusted for inflation)[67]. In Table 3.10, I show the projected FCFE figures.

TABLE 3.10: PROJECTED FREE CASH FLOW TO EQUITY (FCFE) FROM SENSEX

	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Growth Rate		12%	12%	12%	11.50%	11.50%	11%	11%	10.50%	10.50%	10%	10%
FCFE	410.23	459.46	514.59	576.34	642.62	716.52	795.34	882.83	975.53	1,077.96	1,185.75	1,304.33

The present value of all the projected FCFE must equal 16,420. The risk-free rate is 2%. The beta of Sensex is 1, as it is a diversified portfolio. Let the implied market risk premium be m. So the required return from investing in Sensex is 9.12% + m. Since 'm' is the only unknown here, we can solve for 'm':

$$16{,}420 = \frac{459.46}{(1+9.12\%+m)} + \frac{514.59}{(1+9.12\%+m)^2} + \dots + \frac{1{,}077.96 + [(1{,}185.75)/(9.12\%+m-10\%)]}{(1+9.12\%+m)^9} \quad \dots 4(7)$$

Solving for 'm', we get:

Market risk premium = m = 3.92%[68].

This figure is extremely low and is much lower compared to what we obtained using the historical method. This means the market probably used much higher growth rates than what we have assumed here. Notice that the implied growth rate figure is extremely sensitive to the projected growth rate.

You will realize from the above equation that the implied premium will vary inversely with the level of Sensex. When the market goes up, the implied risk premium will come down. That also makes intuitive sense. Investors generally demand higher risk premium when the market is down. When the market is in a bullish phase, the perceived risk is low and hence one demands lower risk premium.

We got three estimates of the market risk premium so far: If we use the arithmetic mean, then the market risk premium is 10.55%. If we use geometric mean, then the market risk premium is 7.13%. The implied market risk premium is about 3.92%. Which one should we use? Going by the evidence presented in Figure 3.10, I will have more trust on the geometric mean figure. So I shall use 7.13% as the market risk premium in the rest of this book. If you believe that the stock prices will not

67. This assumption may not make sense for an individual company. However, this is a sensible assumption to make for a diversified portfolio.
68. Visit sites.securities.com/a/xlri.ac.in/profmohanty for getting an up-to-date estimate of the implied market risk premium in India.

mean revert in the future and that they will follow a random walk process, then by all means use the 10.55% figure.

Estimation of Beta[69]

We now need to know how beta is estimated so that we can estimate the cost of equity of a company. Beta measures the sensitivity of the stock to the market index. It is computed by dividing the covariance of the stock return with the market return ($\sigma_{i,m}$) and the variance of the market return (σ_m^2)[70].

$$\beta = \frac{Covariance(R_i, R_m)}{Variance(R_m)} = \frac{\sigma_{i,m}}{\sigma_m^2} \quad \text{... 3(8)}$$

Before discussing about the different estimation issues, let's understand two important theoretical issues regarding beta. Firstly, beta is the right measure of risk only if the investor is diversified. Secondly, beta is affected by both the business risk and the financing risk of a company.

While deriving the CAPM equation, we assume that the market does not price diversifiable risk. The well-diversified investors invest in a portfolio combining the risky market portfolio and the risk-free asset. If an investor is not diversified, then beta of the stock is not a measure of risk for such an investor and we cannot use the CAPM equation to estimate the required return for such an investor. This is not a serious limitation when we value listed companies as it is safe to assume the marginal investor to be well-diversified here. However, when we value private companies, we can no more assume the investor (the owner of the company) to be well-diversified. We will discuss this issue in more detail while discussing about valuation of private companies in Chapter 9 of the book.

Secondly, the beta of a stock is affected by both its operating and financing risk. We will discuss this issue in detail here as understanding this helps us finding beta of stocks which are not listed.

Beta and Operating Risk

Beta measures the systematic risk of a company. We essentially try to find out how sensitive the stock returns are to the overall movements of a market index. This depends on the nature of the business (or operations) of a company. This depends on the products that the company sells. This depends on what happens to the cash flow of a company (when the company sells such products) when the economy goes through ups and downs.

The business risk of a toothpaste manufacturing company is lower than that of an automobile company. The total revenue (and hence the total profit) of a toothpaste manufacturing company will not go down by 6%, say, when the GDP growth is down

69. In Appendix B, I discuss some of the advanced issues relating estimation of beta.
70. We will obtain this value of beta by regressing the stock returns of company 'I' on that of the market. We never regress stock returns on a sector index. Thus for example, ***never*** find the beta of Infosys by regressing Infosys stock returns on an IT index returns.

by 6% or when the stock market index goes down by 6%. Nor does the total revenue of the toothpaste company increase by 6%, when the stock index goes up by 6%. The business of the automobile company, however, depends a lot on the state of the economy. Automobile companies generate substantial amount of revenue in bullish economies. The stock index also remains high during such time period. However, during down times, the automobile companies' sales come down as the inventories pile up. The beta of the toothpaste manufacturing company will be low. The beta will be high for the automobile company.

The cost structure in the company also affects the business risk of the company. The total costs of a company can be loosely divided into fixed costs and variable costs. When the operations of a company remain confined to certain pre-specified level, the fixed costs do not move with time. But the variable costs do. Therefore, if the proportion of fixed costs to the total costs of a company is very high, then the company must sell a larger number of its output to breakeven (and make profit). If however, the ratio of fixed costs to total costs is low, then the breakeven output level is low. If the fixed costs of a company are high, then the company faces the risk of not earning profit when there is recession in the economy. Therefore, high fixed costs add to the riskiness of a company.

We can write the market value of assets as equal to the present value of expected cash flows that is[71]:

PV (Assets) = PV (Cash Flows). This can approximately be written as:

PV (Cash Flows) = PV (Revenue) - PV (Fixed costs) - PV (Variable Costs).

Here we can look at the present value of the cash flows of a company as a portfolio of a long position in the PV of revenues, and a simultaneous short position in the PV of both fixed and variable costs. The beta of the PV of the cash flows will therefore be equal to the weighted average beta of the three components :

$$\beta_{Assets} = \beta_{Revenue} \times \frac{PV(Revenue)}{PV(Assets)} - \beta_{Fixed\_costs} \times \frac{PV(Fixed\_Costs)}{PV(Assets)} - \beta_{Variable\_costs} \times \frac{PV(Variable\_Costs)}{PV(Assets)}$$

The beta of fixed costs can be assumed to be zero. If we can pass on the increase in input prices in the form of higher price, then beta of revenue and beta of variable costs will be approximately equal. Therefore, we can rewrite the above expression as:

$$\beta_{Assets} = \beta_{Revenue} \times \left(\frac{PV(Revenue) - PV(Variable\_Costs)}{PV(Assets)} \right)$$

$$= \beta_{Revenue} \times \left(\frac{PV(Assets) + PV(Fixed\_Costs)}{PV(Assets)} \right) \quad \text{... 3(9)}$$

$$= \beta_{Revenue} \times \left(1 + \frac{PV(Fixed\_Costs)}{PV(Assets)} \right)$$

71. I borrowed this idea from Brealey, Myers, Allen and Mohanty (2014). See, Brealey, R., S C Myers, F Allen and P Mohanty, 2014, Principles of Corporate Finance, 11th Edition, Chapter 9, McGraw Hill Publishers, New Delhi.

Therefore, higher the proportion of fixed costs in the total costs, higher will be beta of the assets, and hence higher the equity-beta.

So you should expect high operating risk and hence higher beta from the followi types of companies[72]:

1. **Cyclical companies**: The cyclicality of a business affects the beta of a sto Companies operating in cyclical industries will usually have a beta greater th 1. The revenue of cyclical companies are highly volatile and hence for a giv cost structure, it is possible that the company will not earn enough profit.
2. **Young Companies**: The fixed costs of young companies will be higher (as of total costs) as compared to established companies in the industry. You companies face the daunting task of maintaining decent amount of pro when the economy goes through a bad time. Of course, these companies d much better compared to the established companies in the industry when th economy does well.
3. **Companies that sell discretionary products:** If a company sells products tha customers can go without, then they will avoid such products during recessionar times. Such companies will find it difficult to generate enough profit durin recessionary times.
4. **Companies having low brand loyalty:** If a company does not have enoug consumers who are loyal to its brands, then it will face problems during dow times.

In order to find out whether the operating risk of a company is high or not, just ask yourself a simple question: "How will this company perform during downtimes? During boom times?" If you believe that the company is not affected much by the state of the economy, then the operating risk (and hence the beta) is low. Else it is high.

Beta and Financial Leverage

The presence of debt adds to the overall riskiness of the equity and hence the beta of the equity increases with debt. A levered company has to pay interest on its debt and hence one can treat the interest expense as a fixed expense. It is possible that the company does not make enough profit to pay interest on its debt capital. The net income of a company will be positive only if the EBIT is greater than its interest expense. Since the EBIT remains low when the economy is down, chances of earning positive net income come down drastically with high level of debt. Let's quantify the relationship.

We saw in Chapter 2 that we can write the value of a levered company (under MM assumptions) as:

$$V_L = V_U + D \times t \quad \text{... 3(10)}$$

See http://people.stern.nyu.edu/adamodar/ for a more detailed analysis.

We can write V_L as $D_L + E_L$ or simply as $D + E_L$. We can also write V_U as E_U.[73] Therefore, Equation 3(10) reduces to :

$$D + E_L = E_U + D \times t$$
$$\Rightarrow E_L = E_U - D \times (1 - t)$$

Therefore, one can look at the equity in a levered company as a portfolio consisting of long position in the equity in an otherwise similar unlevered company and a short position in the Debt × (1-t) of the same levered company. The beta of E_L will also be equal to the beta of E_U - D×(1-t)[74].

$$\beta_L = \beta_U \times \left(\frac{E_U}{E_L}\right) - \beta_D \times \frac{D(1-t)}{E_L}$$
$$= \beta_U \times \left(\frac{E_L + D \times (1-t)}{E_L}\right) - \beta_D \times \frac{D}{E_L} \times (1-t)$$
$$= \beta_U \times \left(1 + \frac{D}{E_L} \times (1-t)\right) - \beta_D \times \frac{D}{E_L} \times (1-t)$$
$$= \beta_U \times \left(1 + \frac{D}{E} \times (1-t)\right) - \beta_D \times \frac{D}{E} \times (1-t) \quad \text{... 3(11)}$$

This is the famous Hamada equation often used in corporate finance literature. Some analysts usually assume that the beta of debt is equal to zero[75]. In that case, Equation 3(11) reduces to:

$$\beta_L = \beta_U \times \left(1 + \frac{D}{E} \times (1-t)\right) \quad \text{... 3(12)}$$

Here, we assumed the rupee value of debt to remain constant over the life of the company. In the presence of growth, the relationship between the levered and unlevered beta is slightly different (as we saw in Appendix to Chapter 2):

$$\beta_L = \beta_U \times \left(1 + \frac{D}{E}\right) \quad \text{... 3(12a)}$$

If we look at the Hamada equation [Equation 3(12) above], we will notice that the following condition must be satisfied to ensure a positive relationship between levered beta and financial leverage[76].

$$\beta_U \times (1-t) - \beta_D \times (1-t) \succ 0$$
$$\Rightarrow \beta_U \succ \beta_D$$

73. Here, E_U stands for the equity of the unlevered company. Other terms have similar interpretations.
74. In equation 3(11), I used β_L to represent the beta of equity. Instead, one can use β_E here.
75. It is important to understand the implications of the assumption that the beta of debt is zero. If the beta of debt is zero, then using CAPM, we can prove that the required return on the debt is equal to the risk free rate of return. In many cases, the beta of debt will be too small to have any impact on the beta of equity. Hence in practice it is ignored.
76. This can be derived by differentiating levered beta with respect to the debt-to-equity ratio.

The beta of assets will always be greater than the beta of debt. Therefore, β_A-β_D will always be positive. Therefore, beta will move positively with the leverage of the company.

Levered Beta in the Presence of Preference Capital

While deriving Equation 3(11), we assumed that debt and equity are the only two sources of capital. What if preference capital is also present? I will first show how the beta of equity is found out when equity and preference capital are present. Then I will show you how to find levered beta when equity, debt and preference capital are present[77].

Case 1: Only Equity and Preference Capital Are Present

Since payment of dividend on preference capital does not give any tax benefits to the company, the levered and unlevered values will be equal.

$$V_L = V_U = P + E_L$$

$$\Rightarrow \beta_U = \beta_P \frac{P}{V} + \beta_E \frac{E}{V} \text{...4(13)}$$

$$\Rightarrow \beta_E = \beta_U + (\beta_U - \beta_P) \times \frac{P}{E} \text{...3(14)}$$

Case 2: Debt, Equity, and Preference Capital Are Present

We get the benefits of tax shield only from payment of interest and not from dividend payment on the preference shares. Again, using MM assumptions (rupee value of debt remaining constant over the life of the company)

$$V_U + D \times t = E_L + D + P$$

$$\Rightarrow V_U = E_L + D(1-t) + P$$

$$\Rightarrow \beta_U = \beta_L \frac{E}{E + D \times (1-t) + P} + \beta_D \frac{D \times (1-t)}{E + D \times (1-t) + P} + \beta_P \frac{P}{E + D(1-t) + P} \text{...3(15)}$$

Rearranging terms, we obtain:

$$\beta_L = \beta_U + (\beta_U - \beta_D) \times \frac{D}{E} \times (1-t) + (\beta_U - \beta_P) \times \frac{P}{E} \text{...3(16)}$$

What if the companies rebalance their debt continuously to keep the debt ratio constant? In that case, we cannot use Equation 3(16). Of course, Equation 3(14) will remain unchanged. However, in place of Equation 3(16), we should use Equation 3(16a) to find the unlevered beta:

$$\beta_L = \beta_U + (\beta_U - \beta_D) \times \frac{D}{E} + (\beta_U - \beta_P) \times \frac{P}{E} \quad \text{... 3(16a)}$$

Now that we understood the theoretical properties of beta, let's see how beta is estimated. Beta of a stock can be estimated by either using the historical stock returns

77. I derived these two equations in my website. You can check them at https://sites.google.com/a/xlri.ac.in/profmohanty/understanding-valuation/findingunleveredbetawhenpreferencecapitalispresent

data of the company, or by using industry beta (after appropriate adjustment for financial leverage), or by using our commonsense. We will now discuss about these methods in detail.

Estimating Beta using historical stock returns data

We saw earlier that the beta is estimated using the following equation[78]:

$$\beta = \frac{Covariance(R_i, R_m)}{Variance(R_m)} = \frac{\sigma_{i,m}}{\sigma_m^2} \quad ...3(17)$$

We require historical returns of the stock and the index to use this equation. In order to estimate beta, we must have answers to the following two important questions[79]:

(i) What should be the estimation period for the above equation? Should we use the last one year's data, 2 years' data or 5 years' of data?

(ii) How frequently should we compute returns? That is, should we use daily returns or weekly returns or monthly returns?

Keep the following two principles in mind while answering these two questions:

- We must ensure that neither the operating risk nor the financing risk has changed significantly during the estimation period. Thus for example, we should ensure that the company's capital structure (a measure of the financing risk of a company) has not changed significantly during the estimation period. We should also ensure that the company has not launched any new product, hived off any division, gone for mergers or acquisitions, etc. during the beta estimation period.
- A larger sample size always gives better estimate of the sample statistic. This means use daily returns data (you will get about 250 returns in a year). This also means use a larger sample period. You will get more data points in a five-year estimation period than in a one-year estimation period. Of course, sample size is not the only consideration, when beta is estimated.

The operating risk and financing risk must remain the same during the estimation period:

Tata Motors borrowed heavily during 1998 to finance the Indica car project. If we are estimating the beta of Tata Motors, then we should not use the pre-1998 period. We should similarly ensure that the operating risk of the company has not changed significantly during the estimation period. Thus for example, Tata Steel was also producing cement till December 1998 when the cement plants were hived off to Lafarge S A of France for a consideration of ₹550 crores. Now it predominantly produces steel. Again we should not take the pre- Dec-98 period while estimating the beta of Tata Steel.

78. Equivalently, you can estimate beta by regressing the stock returns on that of the market index.

79. In Appendix A to this chapter, I have discussed the different practical issues one encounters while estimating the stock returns.

The beta of assets will always be greater than the beta of debt. Therefore, β_A-β_D always be positive. Therefore, beta will move positively with the leverage of company.

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Rearranging terms, we obtain:

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Estimating Beta using historical stock returns data

We saw earlier that the beta is estimated using the following equation[78]:

$$\beta = \frac{Covariance(R_i, R_m)}{Variance(R_m)} = \frac{\sigma_{i,m}}{\sigma_m^2} \quad ...3(17)$$

We require historical returns of the stock and the index to use this equation. In order to estimate beta, we must have answers to the following two important questions[79]:

(*i*) What should be the estimation period for the above equation? Should we use the last one year's data, 2 years' data or 5 years' of data?

(*ii*) How frequently should we compute returns? That is, should we use daily returns or weekly returns or monthly returns?

Keep the following two principles in mind while answering these two questions:

- We must ensure that neither the operating risk nor the financing risk has changed significantly during the estimation period. Thus for example, we should ensure that the company's capital structure (a measure of the financing risk of a company) has not changed significantly during the estimation period. We should also ensure that the company has not launched any new product, hived off any division, gone for mergers or acquisitions, etc. during the beta estimation period.
- A larger sample size always gives better estimate of the sample statistic. This means use daily returns data (you will get about 250 returns in a year). This also means use a larger sample period. You will get more data points in a five-year estimation period than in a one-year estimation period. Of course, sample size is not the only consideration, when beta is estimated.

The operating risk and financing risk must remain the same during the estimation period:

Tata Motors borrowed heavily during 1998 to finance the Indica car project. If we are estimating the beta of Tata Motors, then we should not use the pre-1998 period. We should similarly ensure that the operating risk of the company has not changed significantly during the estimation period. Thus for example, Tata Steel was also producing cement till December 1998 when the cement plants were hived off to Lafarge S A of France for a consideration of ₹550 crores. Now it predominantly produces steel. Again we should not take the pre- Dec-98 period while estimating the beta of Tata Steel.

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78. Equivalently, you can estimate beta by regressing the stock returns on that of the market index.
79. In Appendix A to this chapter, I have discussed the different practical issues one encounters while estimating the stock returns.

Longer estimation period is preferred:

It is ideal to have a longer time period because the stock returns are so volatile th a shorter time period may not really help us in finding out how sensitive the sto return is to that of the market. It is possible that when the stock is doing badly, t market index is doing well. This happened during the pre-2000 time period. Becau there was a boom in the software sector, both Infosys and Wipro were doing we This pushed up the returns of Sensex during that time period. However, the rest the economy was not doing well at that time. Thus for example, we would get 0. as the beta of Tata Steel if we use daily stock returns data of Tata Steel and Sens for the period 1998-99. Steel is a cyclical product. The operating risk of Tata Steel high. And so is its financing risk. Its beta will be much higher than 0.22.

If we take a much longer time period, then there is every chance that the operatin and financing risk of the company have changed during the estimation period. S a judicious balance needs to be kept between the above two requirements whi deciding the return estimation period. In Table 3.11, I show the beta of 10 companie in the energy sector in India.

TABLE 3.11: BETA OF ENERGY COMPANIES OVER DIFFERENT ESTIMATION PERIODS

	BPCL	Cairn India	GAIL	Gujarat State Petronet Ltd.	HPCL	IOL	ONGC	OIL	Petronet LNG	RIL
1-Year	0.73	0.83	0.29	0.40	0.57	0.30	0.47	0.06	0.15	1.12
2-Years	0.55	0.59	0.40	0.51	0.57	0.25	0.53	0.30	0.09	1.06
3-Years	0.63	0.70	0.56	0.91	0.75	0.57	0.72	0.46	0.69	1.06
4-Years	0.66	0.74	0.69	1.09	0.63	0.55	0.79	0.46	1.07	1.07
5-Years	0.68	0.77	0.75	1.07	0.65	0.67	0.86	0.46	1.12	1.09

Source: Stock returns data collected from prowess database.

For Reliance Industries Limited, the beta remains close to 1.1, no matter what th estimation period is. However, for Petronet LNG, the beta figure is 1.12 if we use a 5-year estimation period (2007 - 12) and only 0.06 if we use a 1-year estimation perio (2011-12). You will also notice that the betas for GAIL, Gujarat State Petronet, ONGC and Petronet LNG increase along with the estimation period. In such cases, which beta figure should we use? If we assume that the operating and financing risk of these energy companies have not changed much during the 5-year estimation period then the beta estimated over the five-year period is more reliable.

Since the business risk for these 10 companies should be similar, the difference in beta can be largely attributed to their different financing risk. I use Equation 3(18) to find out the unlevered beta of the ten energy companies. Remember that, I am assuming that these companies will keep their debt constant at the current level Instead, if I assume the debt ratio to remain constant, then I need to take the (1-t) term out of the equation.

$$\beta_L = \beta_U \times \left(1 + \frac{D}{E} \times (1 - t)\right) \quad ...3(18)$$

TABLE 3.12: UNLEVERED BETA OF ENERGY COMPANIES

Company Name	DE*	NDE**	Beta-L	Beta-U#	Beta-U##
Bharat Petroleum Corpn. Ltd.	0.75	0.74	0.68	0.45	0.46
Cairn India Ltd.	0.02	0.01	0.77	0.76	0.77
G A I L (India) Ltd.	0.05	0.00	0.75	0.72	0.75
Gujarat State Petronet Ltd.	0.34	0.29	1.07	0.87	0.90
Hindustan Petroleum Corpn. Ltd.	2.44	2.43	0.65	0.25	0.25
Indian Oil Corpn. Ltd.	0.83	0.81	0.67	0.44	0.44
Oil & Natural Gas Corpn. Ltd.	0.08	-0.02	0.86	0.82	0.88
Oil India Ltd.	0.03	-0.35	0.46	0.45	0.60
Petronet L N G Ltd.	0.25	0.24	1.12	0.96	0.97
Reliance Industries Ltd.	0.28	0.16	1.09	0.92	0.98
			Average	0.66	0.70
			Median	0.74	0.76

Source: Author's own analysis based on stock returns data collected from Prowess database

* DE is computed as the ratio of interest bearing liability and the market capitalization as on 1 April, 2012

** NDE is computed as the ratio of (debt - cash and marketable securities) and market capitalization.

Beta-U is computed using the DE ratio

Beta-U is computed using the NDE ratio.

I compute the unlevered beta by using both the debt-equity ratio and the net debt-equity ratio as both are accepted as acceptable practices[80]. I have used 33.99% as the tax rate for these 10 companies. Once the unlevered betas are computed, one can compute the median of these unlevered betas to find out the unlevered beta for the sector. As I show in the last column, we can use 0.76 as the unlevered beta for the energy sector in India.

Once you find the unlevered beta, you can find the levered beta of a company by using Equation 3(18) again. Thus for example, for Petronet LNG, the levered beta will be 0.88. There is less noise associated with the estimation of the sector beta and hence this is a much more reliable figure.

80. See Appendix D at the end of the chapter for a detailed discussion on the difference between net debt and gross debt. In the appendix, I show that the net-debt method is the preferred method.

What if the market value of equity figure is not available for the company? We ma be interested in finding the beta of a company whose shares are not listed. We ma be valuing a private company. We may be interested in valuing the stock for an IPO We may also be valuing the shares of a listed company whose shares do not trad frequently and hence we cannot use the market capitalization figure as a proxy fo its true market value of equity. The E in Equation 3(18) expects us to put the marke value of equity and not the book value of equity. In such cases, instead of putting th book value of equity as a proxy for the market value of equity, we should always pu an imputed value for the market value of equity. The imputed value can be derive by using the multiplier method of valuation. This imputed value can work as a prox for the market value of equity.

Let's take an example to understand the process. Suppose, Orissa Oil is an oil an natural gas company based in Balasore, the coastal town in Orissa. It has generate net sales of ₹6.5 billion in the previous fiscal year ending March 2012. The media enterprise value to sales ratio for the energy companies is 1.11 (I show the detail in Table 3.13).

TABLE 3.13: ENTERPRISE VALUE TO SALES RATIO FOR ENERGY STOCKS (31 MARCH, 2012)

Company Name	Sales	Enterprise value	EV/Sales
Bharat Petroleum Corpn. Ltd.	1,632,182	360,911	0.22
G A I L (India) Ltd.	329,071	469,258	1.43
Gujarat State Petronet Ltd.	10,391	55,353	5.33
Hindustan Petroleum Corpn. Ltd.	1,437,768	266,642	0.19
Indian Oil Corpn. Ltd.	3,574,220	1,113,691	0.31
Oil & Natural Gas Corpn. Ltd.	702,535	2,033,352	2.89
Oil India Ltd.	119,143	199,530	1.67
Petronet L N G Ltd.	131,973	146,359	1.11
Reliance Industries Ltd.	2,586,512	2,697,042	1.04
		Median	1.11

Source: Prowess database.

So the imputed enterprise value of Orissa Oil is given by:

$$₹6.5 \text{ billion} \times 1.11 = ₹7.215 \text{ billion}$$

If the total debt of Orissa Oil is ₹3.5 billion as on 31 March, 2012, then the imputed market value of equity comes to:

$$₹7.215 \text{ billion} - ₹3.5 \text{ billion} = ₹3.715 \text{ billion}$$

We should use ₹3.715 billion as the proxy for the market value of equity of Orissa Oi to estimate its levered beta. Since the unlevered beta for the energy sector is 0.76, the levered beta of Orissa Oil is given by:

$$0.76 \times \left(1 + (1 - 0.3399) \times \frac{3.5}{3.715}\right) = 1.23$$

Frequency of Return Estimation

The next issue that we must address is about the frequency of return computation. Should we use daily returns or weekly returns or monthly returns? The only advantage in using daily returns is the availability of a large number of data points. If we use data for the last five years (say), then we can use approximately 1,300 data points to estimate the beta of the stock. If we decide to use weekly data, then we will be left with around 260 data points. Use of daily returns data is particularly useful when we do not have time series data extending to (say) last five years. Thus for example, if we have only two years' of data with us, then using weekly or monthly returns data will really reduce the statistical power of any regression analysis[81]. But there are certain estimation problems that we must keep in mind while using daily returns data.

Infrequent trading: If the stock is subject to infrequent trading, then we will not be able to observe the true daily returns. Thus for example, if the stock trades on Monday and not on Tuesday, then we will not be able to observe the true daily returns on Tuesday and Wednesday. In such cases using weekly return will be more appropriate.

Non-synchronous trading: The stock may suffer from what is called non-synchronous trading. A particular stock may be trading every day, but it may not trade as frequently as the other stocks in the index. If a stock suffers from non-synchronous trading, then the stock will exhibit positive serial auto-correlation.

Therefore, the question of whether to use daily or weekly return boils down to a trade off between having more data points, and not being able to observe the true data. If the stock is an actively traded stock, and if we have five years of data available with us, then the choice between daily and monthly data is merely one of convenience. On the other hand, if the stock suffers from infrequent trading problem, then we should use weekly or monthly return data. I computed the beta of the ten energy companies using both daily and month returns data. I show the output in Table 3.14.

TABLE 3.14: BETA ESTIMATION OF ENERGY COMPANIES USING DAILY DATA

	BPCL	Cairn India	GAIL	Gujarat State Petronet Ltd.	HPCL	IOL	ONGC	OIL	Petronet L N G	RIL
Daily data (1-Year)	0.47	0.90	0.55	0.67	0.52	0.34	0.58	0.39	0.52	1.26
Daily Data (5 Years)	0.51	0.91	0.73	0.83	0.55	0.49	0.82	0.36	0.90	1.17
Monthly Data (5 Years)	0.68	0.77	0.75	1.07	0.65	0.67	0.86	0.46	1.12	1.09

Source: Estimated by the author using returns data from Prowess, computed using daily returns data for the 2011-12 period.

81. In case, you are trying to find the beta of a company with less than 2 years of data, it will be better to use the sector beta.

For most of the companies, the difference between daily beta and monthly beta significant (both statistically and economically). Look at Petronet LNG, for example, get 0.52 as its levered beta, if we use daily returns figure for just one year. However we use monthly returns data, we get a beta of 0.15 (with 12 monthly returns figure or 1.12 (with 60 monthly returns figures). One of the reasons for this discrepan is the illiquid nature of the stock. During the 2007-12 period, on 17 occasions, t stock did not trade at all in the market. Daily returns will be less reliable for su stocks. We should go by the beta we get using monthly returns figure than by da returns figure. And whenever we use monthly returns figure, we should have arou 48-60 data points with us. If the stock is highly liquid, then the frequency of retu estimation will not make much of a difference. Look at Reliance, for example. I beta is 1.17, if we use daily returns data for five years. If we use monthly retur data, we get 1.09 as the beta.

There are a few websites that also report the beta of Indian stocks. Most of the seem to be using daily returns data over the last 1 year to estimate the beta. If you a looking at a relatively illiquid stock, do not use such beta figures. Prowess databa also reports the beta figure for around 10,000 Indian companies. However, it us daily returns data to estimate the beta and hence these figures are less reliable.

Many authors recommend regressing excess stock return on excess market retur Thus instead of regressing actual return of the stock on that of the market, the advocate running the following regression equation:

$r_{i,t} = \alpha_i + \beta_i \times r_{m,t} + \varepsilon_{i,t}$....3(19)

Here, $r_{i,t}$ is the difference in return for stock i for period t and the risk free rate c return for period t. Similarly, $r_{m,t}$ is the difference between the actual market retur for period t and the risk free rate of return for period t. One can show that if th risk free rate of return is constant across the estimation period, then both the abov regression equations will give identical beta estimates. If we are using daily return then we can run either of the two regression equations. But if we are using monthl return (say), then it is better to use the excess-return based regression equatio We can substitute the 1- month T-bill yield as $r_{f,t}$ in the same equation. However, i practice, no one uses excess return while estimating the beta.

Here, I showed how to estimate the beta of a stock by regressing its stock return on the returns of an index. Using the example of the energy sector, I showed ho to compute the beta of the sector and then use that to compute the beta of a stock In Appendix B, I have discussed how we can use our commonsense to estimate th beta of stocks.

Once we estimate the beta of a stock, we can estimate the cost of equity by using the famous CAPM equation.

$E(R_i) = R_f + \beta \times (E(R_m) - R_f)$...3(20)

Example 3.14: Estimating Cost of Equity of Petronet LNG

Now that we know how to estimate the three inputs required under CAPM to estimate the cost of equity of a company, let's take the example of Petronet LNG as an example here to see how you should estimate the cost of equity for a real life company.

The risk-free rate of return (r_f) equals 9.12%.

The market risk premium ($r_m - r_f$) equals 7.13%.

Beta (using sector beta approach) equals 0.88.

So the cost of equity of Petronet LNG as on the date would equal:

$Ke = 9.12\% + 0.88 \times 7.13\% = 15.39\%$

Other Adjustments Usually Done While Computing Cost of Equity

Some analysts prefer to make other adjustments like adding (or subtracting) premiums (or discounts) for risks like country risk, illiquidity risk, etc. Thus for example, it is customary to add 1 to 3 per cent points to the cost of equity while valuing illiquid stocks. Similarly, when investors from the developed countries invest in emerging markets, they add a few percentage points to the cost of equity before valuing the stocks. Let's briefly discuss about these two adjustments.

Adjustment for illiquidity

Investors in illiquid stocks take the additional risk of not being able to sell the stock whenever they want. This risk cannot be diversified away by investing in a diversified portfolio because the portfolio becomes less liquid with illiquid stocks included in it. This is obvious when we compare market indices like Sensex and Nifty with BSE 500 or NSE's CNX 500 indices.

However, instead of adding an arbitrary illiquidity premium to the cost of equity, it is better to use the multi-Factor model approach of Fama and French.[82]. This model can adjust for illiquidity risk by adding the liquidity risk premium that is determined by multiplying the liquidity beta of the stock (that measures the sensitivity of the stock return to liquidity risk) with the liquidity risk premium. Here, illiquidity works as the fourth risk factor in the traditional Fama-French model.

Adjustment for country risk

Investors in the developed countries take an additional risk when they invest in stocks belonging to the emerging markets. However, while finding the cost of equity, the question we should ask is "Is the country risk diversifiable". If this is a diversifiable risk, then we should not worry about it at all if we have invested in stocks around the world. If this risk is not diversifiable, then most likely it is already priced in all the stocks around the world. In order to understand this issue in detail, let's take a simple example.

82. You can find a brief introduction to the 3-Factor model in Appendix C given at the end of this chapter.

Example 3.15: An American investor wants to invest in Greek stocks. He is worried about th economic scenario in Greece and hence wants to add an additional country risk premiu to the required rate of return figure that CAPM gives. Let's assume that the required retu from a particular Greek company is 11%. Should our American investor add a premium t this 11% figure just because he is taking some additional risk?

International markets are highly integrated these days with poor stock performance in on country affecting the stocks in other countries as well. If the economic situation in Greece i a matter of concern for everybody in the world, then even Indian investors become cautiou when they invest in Indian stocks. Poor performance of the European countries is one of th key reasons why stock markets around the world remained suppressed in early 2012. Th risk is priced anyway as part of the market risk. Even if the American investor does not inves in the Greek stocks, he is still taking this non-diversifiable country risk associated with th poor performance of Greece.

When analysts talk about adding a country risk premium, they usually do not refer to th risk (that is priced in the markets all over the world anyway). They refer to that addition risk which an investor takes only when he invests in stocks of that particular country. I our American investor has so far invested in stocks belong to the U.S. and as part of hi international diversification has started investing in Greek stocks, then he expects an additiona risk premium from his Greek investments. However, that part of the risk which is Greece specific is a diversifiable risk. International investors invest in all the countries of the world an hence do not care about such country-specific risk. So even if our American investor woul require an added premium for investing in Greek stocks, other investors will not oblige a they are not taking this country risk in the margin. Such risks will not be priced in a globall competitive capital market.

Therefore, never add a premium for the country-specific risk. If the country risk is reall non-diversifiable, then that risk is already priced in the market anyway. The market ris premium includes this. When we estimate the market risk premium using historical marke returns data, we include all those risks that are priced anyway. So do not inflate this figur by adding arbitrary premiums.[83]

Can we take CAPM seriously? CAPM is based on some very intuitive principles an all that one needs to know about a company is its beta to estimate the cost of equity of a stock. But does CAPM work in real life? That is, can we use beta of stocks to explain the cross-sectional variation in stock returns?

Richard Roll in a classic paper once commented that if the true market portfoli is mean-variance efficient, then the relationship between returns and beta will be exact[84]. If the market portfolio is not mean-variance efficient, then the relationshi between expected return and beta is not exact. Since it is difficult to observe (and construct) the true market portfolio, we usually use market proxies like Sensex or nifty to estimate beta. If these indices are inefficient, then no sense can be made of either the beta that we get from the regression or the expected return that we get by plugging the beta into the CAPM equation.

Roll and Ross (1994) similarly argue that the true cross-sectional expected return-beta relation is exact when the index is efficient, so no variable other than beta can explain

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83. For a different view point, refer to Damodarn's Damadaran on Valuation.
84. See Roll, R., 1977, A Critique of Asset Pricing Theory's Tests: On the Past and Potential Testability of the Theory, Journal of Financial Economics, Vol. 4, 129 – 176.

any part of the true cross section of expected returns[85]. Conversely, if the index is not efficient, any variable that happens to be cross-sectionally related to expected returns can have discernible power when the index proxy is ex ante inefficient. Similarly, Kandel and Stambaugh (1995) show that expected returns can display essentially no correlation with betas computed against an index portfolio that has an expected return arbitrarily close to that of the efficient portfolio with the same variance[86]. Alternatively, expected returns can display a nearly perfect linear relation to betas computed against an index portfolio that is grossly inefficient.

All the above imply that if the market portfolio that we use to estimate the beta is mean-variance inefficient, then we cannot take the results from CAPM seriously. In a related paper Eugene Fama and Kenneth French found that the relationship between realized stock returns and beta is flat in the USA[87]. This implies that there is no relationship between stock returns and beta. That is stocks with both large and small beta are expected to give more or less similar returns[88].

Financial economists use two other models that are used as substitutes for CAPM. These two models are :

- Arbitrage Pricing Theory, and
- Multi-factor Models

These two models are however, rarely used in practice. I briefly discuss about these models in Appendix C.

3.4 Cost of Other Instruments

So far, we discussed about how one can find the costs of plain-vanilla debt, preferred equity and common equity. For most companies in the world, debt and equity will be the two primary sources of fund. Companies sometimes also use instruments that have option-type features. We will briefly discuss about how you can find cost of such instruments.

3.4.1 Cost of Warrant

Many companies issue warrants to the promoters, bondholders and the shareholders. There are many instruments like convertible debentures that have features very similar to warrants. In order to find the cost of convertible debentures, we must know how the cost of warrant is found out.

85. See Roll, Richard, and Stephen A. Ross, 1994, On the Cross-sectional Relation between Expected Returns and Betas, Journal of Finance, Vol. 49, 101–121.
86. See Kandel, Shmuel, and Robert F. Stambaugh, 1995, Portfolio Inefficiency and the Cross-section of Expected Returns, The Journal of Finance, Vol. 50, 157 – 184.
87. See Fama., E., and K French, 1992, "On the Cross Section of Stock Returns", Journal of Finance.
88. Da, Guo, and Jagannathan (2012) however argue that we can use the CAPM to estimate the cost of equity for projects even when there are evidence against CAPM in cross-sectional studies. See Da Zhi., Re-Jin Guo, Ravi Jagannathan, 2012, CAPM for Estimating Cost of Equity Capital: Interpreting the Empirical Evidence, Journal of Financial Economics, Vol. 103, 204 – 220.

Since warrants are issued by the companies, dilution happens when warrants get exercised[89]. Therefore, while valuing a warrant, we must explicitly adjust for the dilution that takes place when the warrants get exercised. The value of a warrant can be obtained by dividing an otherwise equivalent call option with 1+p, where p is the ratio of total No. of warrants and total No. of shares outstanding before the warrants get exercised. That is :

$$warrant = \frac{call}{1+p} \quad ...3(21)$$

where

$$p = \frac{\text{number of warrants outstanding}}{\text{number of shares outstanding before exercise of warrants}}$$

We can find the cost of warrant by using the CAPM. For this we require an estimate of the beta of the warrant. It can be shown that the beta of a call option is given by

$$\beta_{Call} = N(d1) \times \frac{Stock\_Price}{Call\_Value} \times \beta_{Stock} \quad ...3(22)$$

Here,

$$d1 = \frac{\ln(S/K) + (r + \sigma^2/2) \times T}{\sigma\sqrt{T}} \quad ...3(23)$$

Equation 3(22) can be derived from the Black-Scholes option pricing model:

$$\text{Call} = S \times N(d_1) - K \times e^{-r \times t} \times N(d_2) \quad ...3(24)$$

By taking beta of both the sides and keeping in mind that the second term is a risk-free investment, we can derive the beta of the call option. Beta of the warrant can be found out by dividing beta of the call option with (1+p)[90].

Example: 3.16: Let's find out the cost of warrants issued by SMA Limited.

The Current stock price of SMA = ₹43.9

The strike rate of SMA = ₹45

Assume that the risk free rate of return is 10%. We should convert the discrete interest rate into a continuously compounded rate of return. This rate is given by ln(1+10%) = 0.09531. Suppose, the daily standard deviation of continuously compounded returns of SMA is 0.0404. Then the annualized volatility comes to about 59.9%.[91]

89. That happens because warrants will be exercised only when the exercise price is lower than the current market price of the share. The company has to issue shares at a price lower than the current market price and that causes the dilution.

90. This can be easily proved: Suppose: $Y = Z/\alpha$. If $Z = a + bX$, then $Y = a/\alpha + (b/\alpha)X$. Now replace Y with warrant, Z with call, α with (1+p), and b with the stock beta to get the derivation. Some people argue that since warrants are perfectly correlated with the call options, their betas should be the same. As you can see here, this is not correct.

91. The annualized volatility can be computed by multiplying the daily volatility with the square-root of the number of trading days.

Assume that the warrants are getting exercised after exactly one year. So the value of T is equal to 1.Using Equation 3(23), we can compute the value of d1 as equal to 0.417469.

From the standard normal table, we can find out that N(d1) = 0.661832.

We can compute *d2* by using the equation: $d2 = d1 - \sigma \times \sqrt{t}$.

In this problem, *d2* is equal to -0.18209.

Hence, N(*d2*) = 0.427756.

Plugging the above numbers in Equation 3(24), we can now compute the value of the call option. The value of the call option comes to ₹11.55.

To compute the value of the warrant, we must adjust for potential dilution. Suppose, the value of p = 0.01. Then the value of the warrant will be ₹11.55/(1.01) = ₹11.43.

What is the cost of this warrant?

The beta of warrant is given by N(d1) × S/C × beta of stock/(1+p)

Here, N(d1) = 0.661832.

S/C = 3.799

Beta of SMAI Limited is 1.12.

Hence, beta of the warrant is 2.79. Once the beta of the warrant is found out, we can find the cost of warrant by using the capital assets pricing model (CAPM) equation.

3.4.2 Convertible Debentures

Now that we know how to find the cost of a warrant, we can now find the cost of convertible debentures because convertible debentures have in-built warrants within them. Though one can find varieties of convertible debentures in India, optionally (fully) convertibles debentures are the most popular amongst them. Here, the entire face value of the debenture gets converted into equity after a certain time period. Thus for example, a company can issue debentures with face value of ₹100 each. At the end of one year, the bondholders will exchange the entire face value of ₹100 for certain pre-specified number of shares. This conversion may take place at the discretion of the bondholder. Here the bondholder will decide at the end of one year whether to convert the bond into certain pre-specified shares or take the money back. Such debentures are called **Optionally Convertible Debentures.** Alternatively, a company can issue convertible debentures where the entire face value gets compulsorily converted into shares after certain pre-specified time period. These debentures are called **Compulsorily Convertible Debentures.** There can also be **Partially Convertible Debentures** where, only a part of the face value of the debenture gets converted into equity. The remaining part of the face value of the debenture comes back to the debenture holder after certain pre-specified time period. As in the case of fully convertible debentures, here also we can have compulsorily (and optionally) convertible debentures.

3.4.2.1 Cost of Optionally Convertible Debentures (OCD)

Let's first discuss about optionally convertible debentures. Optionally convertible

Since warrants are issued by the companies, dilution happens when warrants a exercised[89]. Therefore, while valuing a warrant, we must explicitly adjust for th dilution that takes place when the warrants get exercised. The value of a warra can be obtained by dividing an otherwise equivalent call option with 1+p, where is the ratio of total No. of warrants and total No. of shares outstanding before th warrants get exercised. That is :

$$warrant = \frac{call}{1+p} \quad ...3(21)$$

where

$$p = \frac{\text{number of warrants outstanding}}{\text{number of shares outstanding before exercise of warrants}}$$

We can find the cost of warrant by using the CAPM. For this we require an estima of the beta of the warrant. It can be shown that the beta of a call option is given by

$$\beta_{Call} = N(d1) \times \frac{Stock\_Price}{Call\_Value} \times \beta_{Stock} \quad ...3(22)$$

Here,

$$d1 = \frac{\ln(S/K) + (r + \sigma^2/2) \times T}{\sigma\sqrt{T}} \quad ...3(23)$$

Equation 3(22) can be derived from the Black-Scholes option pricing model:

$$\text{Call} = S \times N(d_1) - K \times e^{-r \times t} \times N(d_2) \quad ...3(24)$$

By taking beta of both the sides and keeping in mind that the second term is a risk free investment, we can derive the beta of the call option. Beta of the warrant ca be found out by dividing beta of the call option with (1+p)[90].

Example: 3.16: Let's find out the cost of warrants issued by SMA Limited.

The Current stock price of SMA = ₹43.9

The strike rate of SMA = ₹45

Assume that the risk free rate of return is 10%. We should convert the discrete interest ra into a continuously compounded rate of return. This rate is given by ln(1+10%) = 0.0953 Suppose, the daily standard deviation of continuously compounded returns of SMA is 0.040 Then the annualized volatility comes to about 59.9%.[91]

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89. That happens because warrants will be exercised only when the exercise price is lower than the curr market price of the share. The company has to issue shares at a price lower than the current mark price and that causes the dilution.
90. This can be easily proved: Suppose: Y = Z/α. If Z = a + bX, then Y = a/α+(b/α)X. Now replace Y wi warrant, Z with call, α with (1+p), and b with the stock beta to get the derivation. Some people arg that since warrants are perfectly correlated with the call options, their betas should be the same. As yo can see here, this is not correct.
91. The annualized volatility can be computed by multiplying the daily volatility with the square-root of th number of trading days.

Assume that the warrants are getting exercised after exactly one year. So the value of T is equal to 1.Using Equation 3(23), we can compute the value of d1 as equal to 0.417469.

From the standard normal table, we can find out that N(d1) = 0.661832.

We can compute $d2$ by using the equation: $d2 = d1 - \sigma \times \sqrt{t}$.

In this problem, $d2$ is equal to -0.18209.

Hence, N($d2$) = 0.427756.

Plugging the above numbers in Equation 3(24), we can now compute the value of the call option. The value of the call option comes to ₹11.55.

To compute the value of the warrant, we must adjust for potential dilution. Suppose, the value of p = 0.01. Then the value of the warrant will be ₹11.55/(1.01) = ₹11.43.

What is the cost of this warrant?

The beta of warrant is given by N(d1) × S/C × beta of stock/(1+p)

Here, N(d1) = 0.661832.

S/C = 3.799

Beta of SMAI Limited is 1.12.

Hence, beta of the warrant is 2.79. Once the beta of the warrant is found out, we can find the cost of warrant by using the capital assets pricing model (CAPM) equation.

3.4.2 Convertible Debentures

Now that we know how to find the cost of a warrant, we can now find the cost of convertible debentures because convertible debentures have in-built warrants within them. Though one can find varieties of convertible debentures in India, optionally (fully) convertibles debentures are the most popular amongst them. Here, the entire face value of the debenture gets converted into equity after a certain time period. Thus for example, a company can issue debentures with face value of ₹100 each. At the end of one year, the bondholders will exchange the entire face value of ₹100 for certain pre-specified number of shares. This conversion may take place at the discretion of the bondholder. Here the bondholder will decide at the end of one year whether to convert the bond into certain pre-specified shares or take the money back. Such debentures are called **Optionally C**[illegible]**ebentures**. Alternatively, a company can issue convertible [illegible] tire face value gets compulsorily converted in[illegible] ne period. These debentures are called C[illegible] lso be **Partially Convertible Deben**[illegible] enture gets converted into e[illegible] back to the d[illegible] of fully co[illegible] convert[illegible]

3[illegible]

debentures can be actually looked as a portfolio of a straight bond (a plain vanilla bond) and a warrant.

Example 3.17: To illustrate, assume that a company sells a convertible debenture with the following features :

Face Value = ₹100

Coupon Rate = 5%

Yield on a straight bond = 12%

Conversion will take place after 2 years.

Here, if an investor buys the convertible debenture, he is actually buying a straight bond and a warrant with that. Keep in mind that the coupon rate is much lower than the yield on a straight bond. In fact in optionally convertible debentures, the coupon rate will always be lower than the yield on a straight bond. Otherwise, the debenture holders will get a free warrant from the company.

In the above example, the value of the straight bond is given by

$$Straight\_Bond\quad Value = \frac{5}{1.12} + \frac{105}{(1.12)^2} = 88.17$$

If the investor buys a straight bond issued by the company (with a coupon of 5% and YTM of 12%), he will pay ₹88.17 for the same. However, he is willing to pay ₹100 for the OCD. Therefore he is paying the difference of ₹11.83 for the warrant.

Once these numbers are obtained, we can treat the ₹88.17 as a straight bond and ₹11.83 as a warrant. We can estimate the cost of the warrant the way we did it in the previous section. We can subsequently find the cost of the convertible bond as a weighted average of cost of straight bond and cost of warrant:

$$K_{OCD} = \frac{86.17}{100} \times 12\% \times (1-t) + \frac{11.83}{100} \times K_{warrant}$$

You can find the cost of a partially convertible debenture in the same manner. Look at the chapter-end exercises for this.

3.4.2.2 Cost of a Compulsorily Convertible Debenture (CCD)

In the above problem, we assumed that the right of conversion lies with the investor. Now let's see how one can find the cost of a compulsorily convertible debenture.

Example 3.18: Let's consider Example 3.17 once again.

Face Value = ₹100.

Coupon Rate = 5%

Yield on a straight bond = 12%

However, let's now assume that the debenture will be compulsorily converted into 1 share after 2 years.

Here, we can look at the CCD as a portfolio consisting of a long position in a debenture with 5% coupon and 12% yield, and a long position in a forward contract on the stock. If the investor buys this CCD he will receive ₹5 at the end of the first year, and ₹5 plus a stock at the end of the second year.

How will we find the cost of the CCD here? Here, it is important to understand that the company gets tax shield on only the interest component. As far as the company is concerned, its net cash outflow is (₹5 - tax shield) in year 1 and 2, and it issues a share with certain expected value at the end of year 2. Therefore, we can directly find out the after-tax cost of the CCD as follows. The after tax-cost of CCD is the solution to 'r' given by the following equation :

$$PV = \frac{5 \times (1-t)}{(1+r)} + \frac{5 \times (1-t)}{(1+r)^2} + \frac{E(P)}{(1+r)^2}$$

Here, E(P) stands for the expected price of the stock at the time of conversion of the convertible debenture. Here, we have discounted the after-tax interest expenses and the expected stock price after two years at the same rate. The reason is that we want to obtain one figure for the cost of CCD rather than two separate numbers. Though this method does not give an exact answer, it is usually used because it saves us the trouble of doing separate adjustment for tax for the coupon components.

In order to find 'r' in the above equation, we need to find the expected stock price at the end of year 2. One can find the expected value of the stock price in the following manner :

Suppose, the current market price is P_0 and the cost of equity is K_e.

Then the expected cum-dividend price of the stock after one year is $P_0 \times (1+K_e)$.

If the expected dividend after year 1 is D_1, the expected ex-dividend price to prevail after one year is given by $P_0 \times (1+K_e) - D_1$. Let's denote this term by X.

Then the expected cum-dividend price to prevail after two years is given by $X \times (1+K_e)$. Usually, the CCD holders do not receive the dividends given in the second year. Therefore the expected price to prevail after two years will be given by: $X \times (1+K_e) - D_2$.

The expected price to prevail after two years can be written as:

$(P_0 \times (1+K_e) - D_1) \times (1+K_e) - D_2$.

Let's assume that the required rate of return in Example 3.18 as 20%. Let's also assume that the company pays ₹1 as dividend each year. Since its current stock price is ₹100, its expected cum-dividend price in year 1 (next year) will be ₹120. So the expected ex-dividend stock price is ₹119.

The expected cum-dividend stock price at the end of year 2 will be 119×1.2, that is, ₹142.8. Since the company is expected to pay only ₹1 as dividend even in the second year, the expected ex-dividend stock price is ₹141.8.

Assuming a marginal tax rate of 30%, we can now find the value of 'r':

$$100 = \frac{5 \times (1-0.3)}{(1+r)} + \frac{5 \times (1-0.3)}{(1+r)^2} + \frac{141.8}{(1+r)^2}$$

$$\Rightarrow r = 15.7\%$$

You can notice that the cost of compulsorily convertible bond is much higher than the coupon rate on the bond. Why is this difference so high? The reason is, when one is buying a compulsorily convertible bond, one is actually buying a plain vanilla bond and a forward contract on the stock at the conversion value. Therefore, a CCB has the feature of both debt and equity in it, and hence its cost must be higher than that of a plain vanilla bond. Here since we have multiplied the interest expenses with (1- tax rate), we do not need to multiply the 'r' we get with (1- tax rate) again.

One can similarly derive the cost of a partially compulsorily convertible debenture.

Conclusion: In this chapter, we discussed at length all that one needs to know to estimate the cost of capital of a company in practice. We discussed about both the theoretical concepts and the practical issues that one needs to keep in mind while finding the cost of capital. Always make sure that the cost of capital should be consistent with the cash flows being discounted. You should ensure that real (nominal) cash flows get discounted using the real (nominal) cost of capital. You should similarly ensure that rupee (dollar) cash flows get discounted using rupee (dollar) cost of capital.

Secondly, in practice people do a lot of adjustments while computing the cost of capital. They increase the cost of capital if the company belongs to an emerging market or if the stock is illiquid. Always understand the theoretical concepts before making any such adjustment.

QUESTIONS FOR DISCUSSION

1. Balaji Hotels and Enterprises Limited issued fully convertible debentures of ₹135 each in March 1995. Each FCD will be compulsorily converted into one share at a premium of ₹125 after 18 months. The debentures carry an interest rate of 9%. What is the cost of this FCD? Make necessary assumptions.
2. While discussing about NIBL, we mentioned that one does not need any adjustment, as it does not affect the value of equity. One can always make the same argument about interest-bearing liabilities also. Even if we do not make any adjustment for interest-bearing liabilities, we will get the same value of equity anyway. Comment on this argument.
3. Suppose an American investor is investing in Infosys. How will he find the beta of Infosys? Will he use regress Infosys stock returns on an Indian market index (like Nifty) or the US market index (like S&P 500)? If he decides to use the American index, how should he compute Infosys returns?
4. In Section 3.1, we valued JLT by assuming the growth rate to be 0. Show that the value will get the same value of JLT irrespective of whether you use nominal or real cash flows.
5. The cost of equity of Kavita Cosmetics Limited is 20%. The dividend yield of the company is 5%. The market has valued the company assuming no growth rate. The company has no debt outstanding. The expected net income of Kavita Cosmetics is ₹250 million. A project is available with an IRR of 10%. Show that the stock prices of Kavita Cosmetics Limited will fall if it invests in this project. How should Kavita Cosmetics define its cost of equity?
6. Go back to the example on beta in mergers and acquisitions given in Appendix B (Section B4). Assume that the market has valued these companies assuming the companies will keep the debt ratio constant. Find the levered beta of A+B assuming A is acquiring B by issuing debt. Make reasonable assumptions about cost of debt and growth rate (if required) to answer this question.
7. Look back at Example 3.17 again. Assume that the investor will get back ₹40 of the face value in cash after 2 years. The investor has the right to get one

share or the remaining ₹60 in cash after 2 years. Find its cost of debt in that case.

7. Assume the following spot rates that prevail at the end of the year. Suppose, you are valuing two bonds, namely, H Bond and the L Bond. The cash flows from these two bonds are also given here. Find the yield of these two bonds.

Year (i)	*i-year Spot Rates*	*Bond H (cash flow at the end of year i)*	*Bond L (Cash flow at the end of year L)*
1	7%	0	50
2	7.4%	0	40
3	8%	0	30
4	8.6%	10	20
5	9.5%	20	20
6	10.2%	30	20
7	11%	100	100

Mini Case 1 (Manthan Car Show Room)

Giridhar, the CFO of Manthan Car Show Room, got an offer from Kitty Real Estate to buy the property on which the car show room is located. Mr. Jilani, the owner of Kitty Real Estate asked for an up-front payment of Rs.5 crores to sell the property. Taxes, stamp duties and other expenses would come to Rs.30 lakhs. Manthan currently pays an annual rental of Rs. 18 lakhs per annum. Historically, the rentals have increased at 8% per annum.

Mr. Baby Thomas started Manthan Car Show Room about 10 years back. During the initial years, he thought, it would be prudent to take the land under lease as it saves immediate cash outflow. The value of the property was Rs.30 lakhs those days. The property prices in Kerala have increased by 15% p.a. in the past 5 years itself. Mr. Joseph believes that the company can now afford to buy the property and save the increasing lease rentals. The car show room is well known in Cochin. Cash is not a constraint anymore for Manthan.

The capital structure of Manthan is given in the following table:

	Market Value	*Before-tax Cost*
Debt	Rs.7 Crore	12%
Equity	Rs.18 crore	18%
Total	Rs.25 Crore	

The marginal tax rate of Manthan is 30%.

Mr. Giridhar has to take a final decision and inform Mr. Joseph about it. In particular, Mr. Giridhar is wondering what cost of capital to use to evaluate the proposal. Manthan has fixed deposits of Rs.8 crores with Allahabad Bank. Mr. Giridhar wonders if it would be prudent to break the fixed deposits and finance the land with that.

of debt. In addition, she could obtain the estimates of CDS (based on an in-hou[se] model of Bloomberg) for these steel companies. Table 4 presents the debt adjustme[nt] factor and the CDS for these large steel companies.

Table 4: Debt adjustment factor and CDS data for the steel companies

	Debt adjustment factor	*Credit Default Spread*[95]
Tata Steel	1.57	187 bps
JSW Steel	1.38	205 bps
SAIL	1.38	164 bps
Jindal Steel	1.38	268 bps

(Source: Bloomberg)

She was also not sure whether to use the effective tax rate or the statutory tax ra[te] to compute the after-tax cost of debt. Indian companies pay tax at the rate of 33.99[%] on their taxable income.

Required:

1. Download the stock price data for the four steel companies (and that of Sense[x]) for the relevant time period from Yahoo Finance and compute the beta of th[e] four companies. Compare your answers with that given in Table 1.
2. Compute the cost of equity of Tata Steel.
3. What options are available with Sunita to compute the cost of debt of Tat[a] Steel?
4. Compute the WACC for Tata Steel.

95. After selecting a company, use DRSK<GO> to obtain the CDS data in Bloomberg.

APPENDIX

APPENDIX A

In this appendix, I briefly discuss certain technical issues that you may find useful while estimating stock returns.

A1: Computing Return for Beta Estimation

There are two ways of computing the return from a stock. If the stock price at the time the investment is made is P_0 and the stock price at the time the stock is sold is P_1, then the return for the holding period can be computed in either of the following two ways :

$$\text{Return} = \ln\left(\frac{P_1}{P_0}\right)$$

or

$$\text{Return} = \frac{P_1 - P_0}{P_0}$$

Most people use the term 'returns' for the second definition of return and 'continuously compounded return' for the first term. Which of these two return definitions should we use? There is a small technical problem with the second definition. The return (as defined in the second method) does not follow normal distribution. The minimum value that return can take during a particular time period is *minus 100%* (i.e., -100%) because of the limited liability effect[1]. Normal distribution, on the other hand assumes that there is some (however small) probability of getting a return that is less than -100%. When we use daily stock returns, we do not need to worry about this limitation as it is very unlikely that the company will go bankrupt within one single day. As the return horizon increases, however, the possibility of bankruptcy increases and hence the return may not follow a normal distribution. Fama and Miller (1971)[2] argue that if the variances of stock returns are finite then under certain restricted assumptions, the return using the second definition will follow normal distribution.

We will not face this problem if we use the first definition of return. If the price relative, P_1/P_0 follows a log-normal distribution, then the return (following the first definition) will follow a normal distribution. It is advisable to follow the first definition while using long-term returns like monthly returns[3].

The reason we are so much bothered about normal distribution is that if the utility function of the investor is not quadratic, then the asset returns must follow normal distribution so that CAPM equation will hold. If asset returns do not follow normal distribution, then the

1. The stock price will never go below zero since the shareholders benefit from what is called the limited liability clause. Hence the result.
2. Fama, E.F., and M. H. Miller, 1971, The Theory of Finance, Chapter 6, Holt, Rinehart and Winston, New York.
3. Refer to Chapter 4 of Quantitative Methods in Finance by Terry Watsham and Keith Parramore (1997) for a detailed discussion. This is an International Thomson Business Press Publication.

relationship between systematic risk and required return may be different from what CAPM predicts. Since we use the CAPM to estimate the cost of equity, we must define the stock return in such a way that it follows normal distribution. That is why, it is always preferable to compute the stock returns (particularly weekly and monthly returns) using the first definition.[4]

A2: Accounting Adjustments

While computing returns, we make certain accounting adjustments. These accounting adjustments are necessary because sometimes the companies issue bonus shares, split their stocks, etc., and this makes the stock prices before and after the above capital change incomparable. In this section, we will see what adjustments one has to make while computing the returns. In particular, we will discuss adjustments to be made in the following cases:

- Adjust for dividends.
- Adjust for stock split and bonus issue.
- Adjust for rights issue.

Doing the above adjustments is not difficult once we understand that the rate of return is the actual return an investor will get if he buys the stock in the beginning of the holding period and sells it at the end of the holding period. Thus for example, suppose we buy an asset at time 't' at a cost of X_0 and sell the asset at time 't+1' and get X_1. Then our return is computed by comparing X_1 with X_0 using either of the two definitions I mention earlier.

Adjustment for Dividend

This is the simplest of all the adjustments to be made. Suppose, an investor gets a dividend of D_1 and then sells the stock immediately at P_1. Then his return can be computed as:

$$\text{Return} = \frac{P_1 + D_1 - P_0}{P_0} \quad or \quad \ln\left(\frac{P_1 + D_1}{P_0}\right)$$

This is a very straightforward adjustment to make when we compute daily return.

But we need to make certain adjustments when we compute weekly (or monthly or yearly) returns. To see this clearly, let's take the following example :

Price of stock on 1.1.2013:	₹50
Dividend received on 15.1.2013:	₹2
Price of stock on 1.2.2013:	₹52

What return an investor will get here if he buys the stock on 1.1.2013 and sells it on 1.2.2015? In short, how will we compute the monthly return here? Here, you can notice that, the dividend is not paid at the end of the month. Thus, the dividend paid by the company remained with the investor for half of the month. How will we adjust for this? There are 3 methods available to adjust for this dividend.

Method 1: Ignore the 15-days period, and assume that the dividend is paid at the end of the month. In this case, the return will be (using the second definition)

$$Return = \frac{52 + 2 - 50}{50} = 8\%.$$

4. As pointed out by Campbell, Lo, and Mackinlay (2007), the first definition of return has one disadvantage. Since log(X+Y) is not equal to Log(X) + log(Y), portfolio return is not a weighted average of the individual stock returns. See Campbell, John Y., Andrew W. Lo, and A. Craig MacKinlay, The Econometrics of Financial markets, Princeton University Press, New Jersey.

Here we have assumed that though the investor has received the dividend in the middle of the month, he has not invested that money anywhere. Another point that you must remember is that the return of 8% is not an annualized return. It is merely the holding period return.

Method 2: Assume that the dividend is reinvested in the stock itself. Suppose, the ex-dividend price of the stock is ₹50. If we reinvest ₹2 in the stock itself, then we can buy 1/25th of a stock. We can sell it on 1.2.2013 at ₹52/25. Therefore, the return (using the second definition) will be

$$Return = \frac{52 + 52/25 - 50}{50} = 8.16\%.$$

This is actually the holding period return. All that the above equation says is that if you invest ₹50 in the stock, and reinvest the dividend in the stock itself, then after the holding period (1-month), you will get a return of 8.16% on your initial investment.

Method 3: In this method, we assume that the dividend is reinvested in a risk free asset. Suppose, the fifteen-day risk-free rate of return is 0.38%. Then at the end of the month, we would get 2×(1.038)= ₹2.076. Therefore, our return (using the second definition) would be given by:

$$Return = \frac{52 + 2.076 - 50}{50} = 8.152\%.$$

This return however is a mixture of a pure stock return and a risk free rate of return. Method 2, however, gives us the pure stock return and hence must be used while computing stock returns.

Adjustment for Bonus and Stock Split

Though there are some accounting differences between a bonus issue and stock split, from a purely financial angle, there is no difference between them. The number of shares that a shareholder owns increases after the bonus issue (or split) and the share price reduces almost proportionately[5]. Suppose, the company announces a bonus issue in the ratio of m:n. that is, for every n-shares that a shareholder has, he gets m-shares free. If the cum-bonus price is P, then the theoretical ex-bonus price should be

$$Theoretical \quad Ex-Bonus \quad Price = \frac{p \times n}{m+n}.$$

Thus for example, if the stock price is ₹120, and the company announces a 2:3 bonus (that is for every three shares that the shareholder has, he gets another two shares free), then the theoretical ex-bonus price is given by :

$$\frac{120 \times 3}{2+3} = 72.$$

The actual ex-bonus price will be usually somewhat higher than the theoretic ex-bonus price. In the above example, suppose somebody buys the share at ₹120 and sells the very next day at ₹73.[6] Then how much return he gets? We certainly cannot compute the return as :

5. Usually, the stock price does not reduce exactly proportionately because bonus issues and stock splits increase the liquidity of the stocks. The rupee dividend paid to the shareholders also increase after the bonus issue (not after the split) and therefore stock price usually increases after the bonus issue is announced.

6. Here the actual price is ₹1 higher than the theoretical ex-bonus price. Other than the liquidity effect and the dividend effect, the actual stock price may be higher just because the market is in a bullish mode on that day.

$$\text{Return} = \frac{73-120}{120} = -39.17\%.$$

What is wrong with the above definition? The above definition actually compares apples wi oranges. One cannot compare the ex-bonus price of ₹73 with the cum-bonus price of ₹12 One must adjust for the fact that, the number of shares that the shareholder has, has actua increased. Therefore, we can compare 3 shares with a price of ₹120 with five shares with price of ₹73. Therefore the true return (using the second definition) is given by:

$$\text{Return} = \frac{73\times5-120\times3}{120\times3} = \frac{365-360}{360} = 1.39\%.$$

The adjustment for stock split is similar. When there is a stock-split, of m:n, then the investo actually receives m-n extra shares for every n shares that he has. Thus, for example, whe a company announces 10:1 stock split, then the shareholder gets 10 shares of face value ₹ each by exchanging 1 share of face value ₹10x. Therefore, the shareholder actually receive 9 extra shares free. Therefore, one can use 9 (rather than 10) for 'm' in the above equatio to estimate the expected return.

Adjustment for Rights issue

Rights issues are made to the existing shareholders. As per section 62 of the Companies Ac 2013, if a company wants to issue equity, then it must give the existing shareholders th right to buy the shares first. Hence the term rights issues. The rights issue must be made a discount to the current market price for otherwise the issue devolves.

Since the rights issue is made at a discount, it can be looked as a portfolio of a public issue a the market price and a bonus issue. To see this more clearly, let's take a numerical exampl

Issue Price:	₹50
Current Market Price:	₹200
Rights Ratio:	1:4

The theoretical ex-rights price will be :

$$Ex\_Rights\_Price = \frac{200\times4+50}{5} = 170.$$

After the rights issue, the price of the share is expected to fall to ₹ 170 per share. Assum that an investor has ₹ 850 with him to invest in the stock. Without the rights issue, he ca buy 4.25 shares with ₹ 850[7]. However, if the company makes a rights issue, then he can bu 5 shares with that money since the ex-rights price falls to ₹ 170 per share. Therefore, he i able to buy 0.75 shares more because of the rights issue. It is as if the company is giving 0.7 shares free for every 4.25 shares. This is equivalent to a bonus ratio of 0.75: 4.25 that is 3:17 Thus we can look at the above rights issue as a portfolio of a public issue at ₹ 200 per shar and a simultaneous bonus issue in the ratio of 3:17.

There is a simpler way of finding the bonus ratio in the rights issue. The bonus ratio can al be obtained by using the following formula :

7. Do not worry about fractional shares here.

8. Keep in mind that this bonus is not actually making the shareholders any better. The value of 4.25 shar before the rights issue is exactly the same as the value of 5 shares after the rights issue. This is true f all bonus issues.

$$Bonus\_Ratio = \frac{Stock\_Price}{Theoretical \quad Ex-Rights \quad Stock \quad rice} - 1$$

Thus for example, in the above example, we can obtain the bonus ratio by dividing ₹200 with ₹170 and then subtracting 1 from the ratio. Once you find out the bonus component in the rights issue, treat it like any bonus issue while estimating the return.

A3: Certain other market micro-structure issues

We need to be aware about certain market micro-structure issues while estimating stock returns (particularly daily returns). Often we observe what is known as discontinuous trading. Discontinuous trading can occur if the stock market remains closed on certain days. It can also happen if a particular stock does not trade on a particular day because there is no trading interest in that stock. In this section, we will briefly discuss the problems that are encountered when the entire market remains closed on certain days. In Appendix B, we will discuss about the problems one encounters when there is discontinuous trading for the individual stocks.

Most of the stock exchanges in the world remain open from Monday to Friday. No trading takes place on Saturdays and Sundays. Other than these two holidays, there may be other holidays like Diwali and Christmas. In such cases how will we estimate the daily return on Monday? The last trading for a stock has taken place on Friday.

Suppose, the closing price on a particular Friday is P_f and that of the next Monday is P_m. In this case, the return on Monday is computed in either of the following two ways:

$$Return = \ln\left(\frac{P_m}{P_f}\right) \quad or \quad \frac{P_m - P_f}{P_f}$$

But is it a daily return or a 3-days return? If no new information about the stock gets released to the market on Saturday and Sunday, then for all practical purposes, one can treat the above return as a daily return. If however, the market gets extra value-relevant information about the stock during the weekends, then the above return can no longer be called a daily return.

To understand the above problem better, let's take the following hypothetical example. The price of a stock as on 31 March, 2014 is ₹100 and the price of the same stock is ₹120 as on 31 March, 2015. Assume that the stock has traded on all the trading days during the year. In this case, we can say that the annual return of the stock was 20% or 18.23% if one uses the logarithmic price difference.

To create some twist in the example, try to answer the following hypothetical but interesting question. "What would have been the price of the stock as on 31 March, 2015 had there been no trading in the stock in the entire year?" In an efficient market, one can argue that the price would have been ₹120[9]. In that case will you call the 20%-return a daily return?

So when we use daily returns, we actually combine a number of 1-day returns with a few 3-day returns (or 2-day returns when the market remains closed for one day). It is like estimating beta (or standard deviation) by using returns of all possible horizons. The beta will be less reliable. The 3-day returns will usually be larger in magnitude compared to the

9. Because the market will value the stock based on the available information with it as on 31 March, 2015 irrespective of whether the stock traded in the last one year or not. But market microstructure theory will not agree with this view.

1-day returns. Therefore, when we use daily returns, we encounter what is known as 'jumps' in the stock returns[10].

Many people completely ignore this problem while computing returns. However, one may decide not to compute the daily return on all the days following holidays. Thus, for example, we will not compute returns on Mondays, etc. The major problem with this approach is loss of data. If we don't have stock price figures available for a longer time period, then this may not be the preferred approach.

APPENDIX B :Some Advanced Issues Regarding Estimation of Beta

In this appendix, we will discuss some of the advanced issues regarding the estimation of beta. In particular, we will look at the following four issues :

- Estimation of beta in the presence of infrequent trading or non-synchronous trading
- Estimation of Beta for divisions of companies or for private companies.
- Estimation of beta when beta mean-reverts.
- Estimation of Beta in mergers and acquisitions.

B.1: Estimation of beta in the presence of infrequent trading or non-synchronous trading

For illiquid stocks (with infrequent trading or non-synchronous trading), covariance of the stock returns with the market return remains downward biased and this creates a downward bias in the estimated beta. When the stock does not trade every day, the computed returns will be zero on some of the days and this leads to underestimation of beta[11]. We have the following options when we find the cost of equity in case of illiquid stocks.

(i) Use adjustments suggested by Dimson or William and Scholes.

(ii) Use beta of comparable companies.

Dimson Adjustment

Prof. Elroy Dimson of London Business School suggested a simple way to solve this problem[12]. Here, we regress $R_{i,t}$ on $R_{m,t-1}$, $R_{m,t}$ and $R_{m,t+1}$ in a multiple regression and then find the beta by summing up the three betas thus obtained.[13]

$$R_{i,t} = \alpha_i + \beta_t \times R_{m,t} + \beta_{t-1} \times R_{m,t-1} + \beta_{t+1} \times R_{m,t+1} + \varepsilon_{i,t}.$$

$$\textit{Dimson Adjusted Beta} = \beta_{t-1} + \beta_t + \beta_{t+1}$$

While doing Dimson adjustment, remember to put a return equal to zero on all the days when the stock has not traded. Thus for example, assume that the stock has not traded on Tuesday. Then the return on Tuesday will be equal to zero. We will assume that the closing

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10. In fact this is one of the reasons why stock returns do not follow a normal distribution. We observe kurtosis when we look at daily returns largely because of such jumps.
11. One of the components of covariance is the correlation coefficient. When some of the return components are zero, the correlation will be downward biased.
12. See, Dimson, E., 1979, Risk Measurement when Shares are Subject to Infrequent Trading, The Journal of Financial Economics, Vol. 7, pp 197 - 226.
13. Sometimes researchers regress the stock return on the market return with two (or more) lags and leads. In this case, the Dimson adjusted beta will be equal to the sum of the five (or more) betas.

price on Tuesday is equal to that of Monday. The return on Wednesday will be computed by comparing Wednesday's price with Monday's return.[14]

The following table gives the beta values and Dimson adjusted beta values for BPL and Mirc Electronics :

Name of the Company	*Simple beta*	*Adjusted beta*
BPL	0.9842	1.405
Mirc Electronics	0.924	1.07

The betas of illiquid stocks increase after the Dimson adjustment is done. In the above regression, we used only 1 lead and lag term for the market returns. If the stock does not trade for longer time period, then one needs to use longer lead and lag periods.

William and Scholes (WS) Adjustment[15]

William and Scholes adjustment differs from Dimson adjustment in two fundamental ways:

- First, we delete all the days when the actual observed daily returns cannot be computed. Thus for example, if a stock has not traded on Wednesday, we delete both Wednesday's return and Thursday's returns from our sample as we cannot compute the actual returns on either of these two days. In Dimson adjustment, one has to put a return equal to zero for Wednesday, and a two-day return computed over the period Thursday to Tuesday for Thursday's return.
- The second major difference between Dimson and WS adjustment is that in WS adjustment, instead of running multiple regressions, we run three (or more if necessary) two-variable OLS regressions.

We run the following three regressions while doing WS adjustment.

$$R_{i,t} = \alpha_1 + \beta_t \times R_{m,t} + \varepsilon_1$$

$$R_{i,t-1} = \alpha_2 + \beta_{t-1} \times R_{m,t-1} + \varepsilon_2$$

$$R_{i,t+1} = \alpha_3 + \beta_{t+1} \times R_{m,t+1} + \varepsilon_3$$

$$WS \quad Adjusted \quad Beta = \frac{\beta_t + \beta_{t-1} + \beta_{t+1}}{1 + 2 \times \text{correlation}(R_{m,t}, R_{m,t-1})}$$

Instead of doing Dimson adjustment or WS adjustment, one can also use the sector beta as the company's beta. Since we have discussed this in this chapter, I will not discuss about it here.

B2: Finding Cost of Capital for a division

When we value a division of a company, we need an estimate of the cost of capital of the division rather than the whole company. This creates its own problem because it is very difficult to obtain data at the division level to obtain the cost of capital. Thus for example, to obtain the beta of the stock, we require data about the past stock prices. But we cannot get stock price data for the divisions.

Here, we will first discuss methods to find the cost of debt of divisions, and then discuss how the cost of equity of divisions is found out.

14. This is consistent with our assumption that the closing price on Tuesday is the same as that for Monday.
15. See Scholes, M., and J William, 1977, Estimating Betas from Nonsynchronous Data, Journal of Financial Economics, Vol. 5, pp 309 - 327

Finding Cost of Debt of Divisions

If the economic viability of all the divisions is the same, then we can use the same cost debt for each of the divisions. Thus for example, if the cash flow generated from each of t divisions of a company are more than sufficient to pay the interests and the company has a good rating, then one can use the same rating (and hence cost of debt) for each division for the whole company. But if the cash flow generating abilities of the different divisions a different, then one can no longer use the same cost of debt for each division.

One of the methods that we can use is to look at the rating of the instruments the compa has issued at the time of the initiation of the project (and its subsequent expansion) and fi the cost of debt based on the existing rating of the same bond instrument. Thus for exampl at the time of initiation of project A, suppose Debenture-series A was issued to partly finan the project. Let's assume that the rest of the project was financed by equity. Suppose, t current rating of debenture-series A is BBB+. And companies with similar rating are able borrow at 10.5% per annum. Then we can use 10.5% as the cost of debt of the division.

The above method is based on two assumptions, however and one needs to understand th implications of these two assumptions before using this method.

1. We are assuming that the rating of debenture series A reflects the interest payin capabilities of division A only. When credit rating agencies rate different bon instruments, they also look at the overall financial situation of the company. Thus fo example, assume that a cash-rich FMCG company decides to enter into a new proje with lot of capital expenditure requirement, and not much expected cash inflow in th near future. The company may still issue bonds at a much cheaper rate. Credit ratin agencies will assign a better rating based on the prospects of its existing businesse Therefore, using the rating of a particular instrument may not always give us a corre picture.
2. Secondly, we are assuming that the debt financing of the project has come only fro the proceeds of the issue of debenture series A. This may not always hold. It is possibl that the company has borrowed money earlier and has used that money along wi the proceeds of debenture series A to finance division A.

We can use a second method where we first obtain division specific data from the company an do a rating directly based on this data. Reporting segment-specific data became mandatory i India only in 2008. Now companies do report certain key accounting figures for the differen divisions. We can find the imputed rating of the divisions by using accounting ratios using thes reported accounting figures. Alternatively we can ask "what if" questions to the managemen to get some approximate idea about the debt-to-equity ratio of divisions. Questions like "I this division were not there... would your capital structure be any different?" Can help us i this regard.

Cost of Equity of Divisions

Since we do not get division-specific stock price data, we cannot estimate beta for the divisions This is the major obstacle we face while estimating the cost of equity of divisions. In practic one can use an imputed figure for beta of the particular division to find its cost of equity.

We can use the following method to find the imputed beta of the division. Firstly, we need to identify listed companies which are in the same line of business as the division we are valuing Such companies are called ***pure play companies***, and this approach is known as ***pure pla approach.***

The next step is to estimate the betas of these pure play companies. Here we need to estimate the unlevered beta of the pure play companies. Once we estimate the beta (levered) of the pure-plays using OLS regression, we can unlever the beta by using the Hamada equation. After obtaining the unlevered betas of all the pure play companies, we should estimate the median beta to find the industry unlevered beta. We can use the industry unlevered beta as the unlevered beta of the division. Using the method explained in this chapter, we can find the imputed market debt-equity ratio for the division and then find the levered beta for the division.

Finding the cost of capital of private limited Companies

We can use the above method to find the cost of capital of private limited companies. If the private company is diversified, we have to find out the cost of capital of each division and then find the cost of capital for the entire company[16].

Empirical research work shows that portfolio beta is always more stable that individual stock beta. The unsystematic risk of the different stocks is uncorrelated with each other. Therefore, beta in a portfolio is measured much more accurately than individual stocks. Therefore, it is always advisable to estimate the beta of the stock and compare the same with the industry beta. If there is a significant difference between individual stock beta and industry beta, then it is always safer to use the industry beta.

Estimating Betas using Commonsense

It becomes easier to find the beta if a comparable company trades in the market. We can estimate the levered beta of the comparable company and after suitable adjustment for the leverage of the company, we can estimate the beta of another company in the same industry. But what if the comparable company does not exist or its shares do not trade in the market? In that case, we can use our common sense while finding the beta. The business risk and the financing risk of the company should give us an idea about what the beta value should be for a company. Let's take some examples here.

Beta of an Airport

Let's suppose that we are interested in finding the beta of Mumbai Airport. Stocks of no airport trade in Indian stock exchanges. There are a few listed airports in the world. For example, stocks of Auckland International Airport are listed in Australian Securities Exchange. The stocks of airport of Malaysia, Malaysia Airports Holdings Berhad, are listed in Kuala Lumpur Stock Exchange. The asset beta of New Zealand airport has been estimated as 0.65[17]. If you believe that the asset beta (unlevered beta) of Mumbai airport is similar to that of New Zealand airport, then you can use 0.65 as the unlevered beta of Mumbai airport.

Alternatively, you can look at the risk of the cash flow of Mumbai airport. It gets its revenue primarily from the airline companies. If the airline companies face a downtime in India, that will substantially affect the cash flows (and returns) of Mumbai airport. Therefore, you can also use the asset beta (unlevered beta) of airline companies as a proxy for the asset beta of Mumbai airport.

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16. We can, for example, find the beta of each division (by first finding the unlevered betas of comparable listed companies and then leaving the median beta back) and then find the beta of the private company. We do not need to find the cost of debt of each division separately here. Once we find the cost of equity and cost of debt of the private company, we can then find its cost of capital.
17. Source: www.comcom.govt.nz/.

Beta of a Travel Agent

Let's assume that we are interested in finding the beta of Yatra.com. Stocks of no other tra portal trade in India. If you study the revenue model of a travel agent, you will find that th get most of their revenue by booking airline tickets and hotel tickets. So the beta of a tra agent will be related to the beta of airlines and hotels. If airlines face a downturn in th business, the travel agents will also face a downturn and it affects their business. So a tra agent beta will be a weighted average of the airline beta and hotel beta[18].

Beta of an IPL team

Suppose, we want to estimate the beta of Kolkata Knight Riders (KKR). This is owned Knight Riders Sports Private Limited. The shares of no other IPL team trade in the mark The closest competitor that you can find are the professional sports clubs like Sport Lisb e Benfica and Sporting Clube de Portugal, Società Sportiva Lazio S.p.A.[19] Alternatively, can look at the revenue model of KKR and try to understand the nature and magnitude the business risk of KKR. It gets most of its revenue from sponsorship tie-ups. As of M 2012, KKR had 22 sponsorship tie-ups. For the fiscal 2012, the company posted profit of ₹ crores.[20] Another source of revenue for KKR will be the trading of key players. As of now th company has not explored this option, but in future this may also be a source of revenue fo the company. KKR also gets money from sale of tickets and other accessories like t-shir caps, etc[21].

How sensitive are the above revenue items to the overall market movements. The reven from sale of tickets and other accessories will not be as dependent on the economy, giv that Indians love cricket a lot. However, during down times, the sponsors may not contin to have the tie-up or may be willing to pay lower sponsorship fees to KKR. Since it is a priva company, details of the revenue break-down is not known. However, assuming more th 75% of the revenue comes from sponsorship fees, we can estimate the beta of KKR by taki a weighted average of these different lines of business.[22]

B3: Time-Varying Beta

When we estimate beta by regressing past stock returns on that of the market portfolio, implicitly assume that the beta remains stationery over time. However, empirical resear shows that beta varies over time. If beta varies over time, then we cannot use the beta bas on past data to estimate the required stock returns. In this section, we will briefly discu why beta varies over time, and then we will discuss what adjustments one needs to make adjust for this time varying property of beta.

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18. Of course the beta of five-start hotels will be higher than that of, say, 3-star hotels. If our travel age primarily books hotels in 3-star hotels, then we can use the beta of 3-star hotels.
19. Source: http://en.wikipedia.org/wiki/Sports_club.
20. Source:http://articles.economictimes.indiatimes.com/2012-05-18/news/31765468_1_kkr-kolkata-knigh riders-venky-mysore.
21. Source: http://kkr.in/store.aspx.
22. We also need to look at the business risk of all the sponsors. Assuming they are all sports companies, can use the beta of sports companies as a proxy for the beta of sponsors.

Why Beta Varies Over Time?

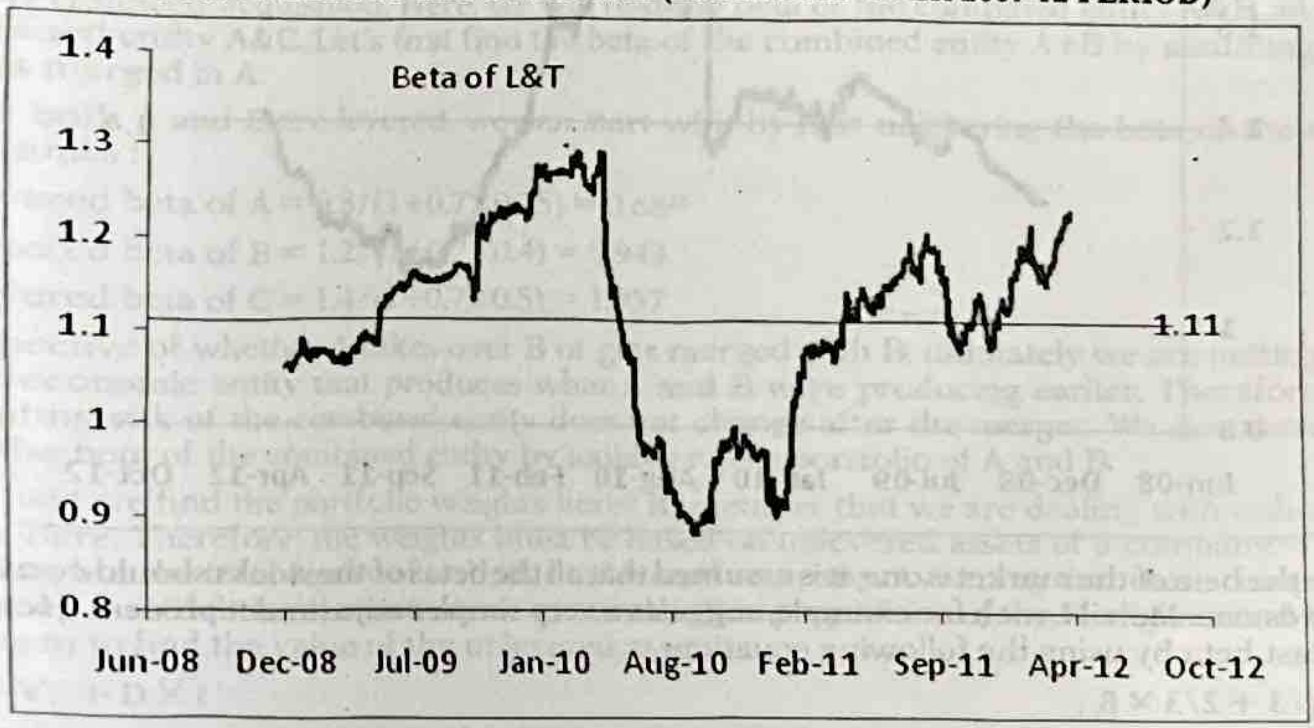
Hamda shows that beta is linked to the leverage of the firm[23]. Therefore when the leverage of a firm changes, its beta also changes. Fischer Black similarly argues that when the stock price changes, the market leverage also changes leading to a change in beta[24].

Beta is a measure of risk of an asset vis-à-vis the market. Any news that does not affect the stock market and the stock uniformly will change the beta of a stock. Changes in the risk free interest rate also can cause change in beta.

Stock returns and index returns exhibit time varying volatility. Since beta is defined as the ratio of the covariance between the stock and the market and the variance of the market, beta exhibits time varying properties.

Irrespective of what factors cause the time variation in beta, this is certainly a cause for concern as far as estimation of cost of equity is concerned. Let's plot the time series of beta of L&T Limited. Here we estimate the beta of L&T over a four-year period. Each beta is estimated using the previous 240 days of stock returns data. I show a chart depicting the time-series beta figures for L&T in Figure 1.

FIGURE 1: BETA OF L&T OVER TIME (ESTIMATED OVER 2007-12 PERIOD)



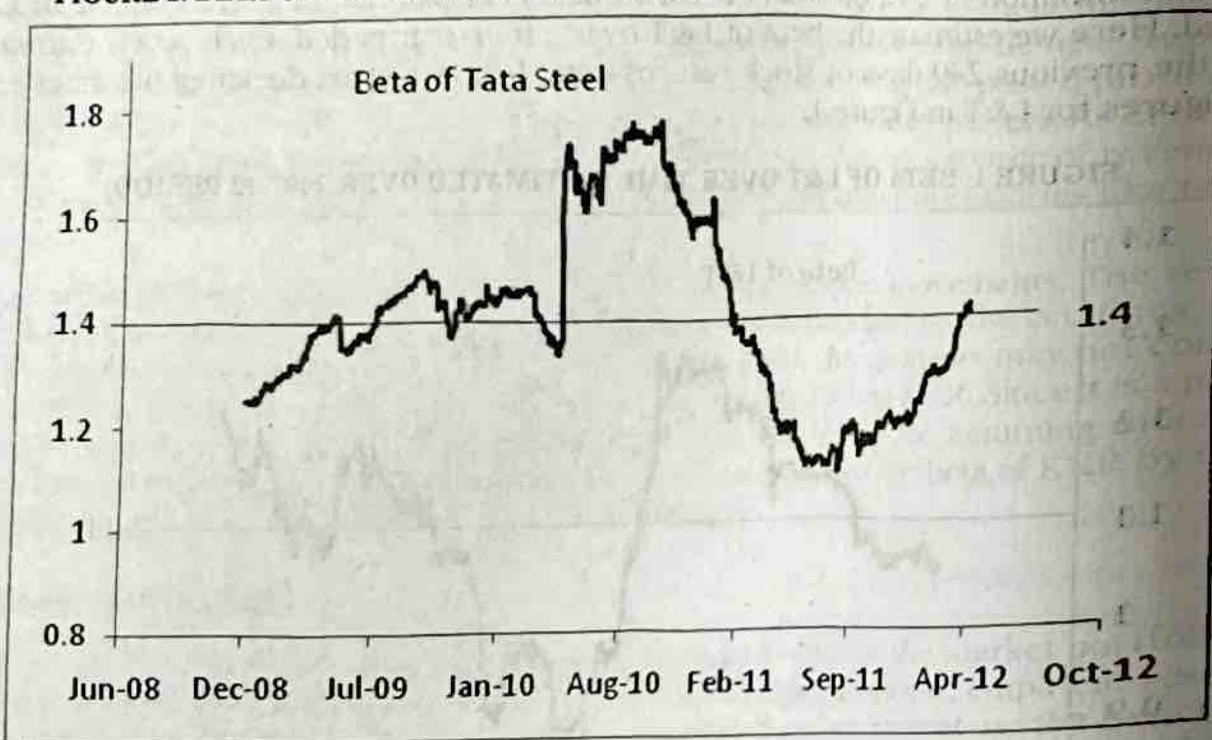
The average beta of L&T is 1.11. The most recent beta value is 1.23. Given the time-varying nature of beta of L&T, if we estimate the beta of L&T now (in March 2012), what value should we use? 1.23 or 1.11 or some other number?

23. See Hamada, Robert 1972, The Effect of the Firm's Capital Structure on the Systematic Risk of Common Stock, Journal of Finance, Vol. 27, pp. 435 - 452.
24. See Black, Fischer, 1976, Studies of Stock Price Volatility Changes, Proceedings of the 1976 Meetings of the Business and Economics Statistics Section, American Statistical Association, 177-181.

Since we are projecting cash flows over the next infinite number of years, we should us beta figure that is appropriate for all the next infinite number of cash flows and not just n year's cash flow. We can use 1.23 as the beta for L&T only if we believe that L&T's avera beta over the next infinite number of years will remain at 1.23. That seems unlikely given evidence presented in Figure 1. It will be better to estimate L&T's cost of equity using 1 as the beta.

Let's take one more example. In Figure 2, I show the time series of beta figures for Tata St over the period 2007-12. The average beta of Tata Steel during this period was 1.4. The mo recent beta figure is 1.42 (a notch above the historical average). Here also, we can use historical average of the time series beta figures. You can estimate the cost of equity of Ta Steel using 1.4 as the beta value.

FIGURE 2: BETA OF TATA STEEL OVER TIME (ESTIMATED OVER 2007-12 PERIOD)



Since the beta of the market is one, it is assumed that all the betas of the stocks should conver towards one. Merril Lynch for example, suggests a very simpler adjustment process. Here w forecast beta by using the following equation:

$\beta_t = 1/3 + 2/3 \times \beta_{t-1}$

Let's assume that there are two stocks with historical beta as equal to 0.8, and 1.2 respectively Since the betas have a tendency to revert towards 1, we should expect the projected beta o the first stock to increase, and that of the second stock to decrease.

The projected betas of the two stocks will be

$0.33 + 0.67 \times 0.8 = 0.866$

$0.33 + 0.67 \times 1.2 = 1.134$

As you can see the betas of both the stocks moved closer to 1 after the adjustment.

Blume (1971) suggested a method similar to the one suggested by Merril Lynch[25]. Blume estimates beta by using the following formulae.

$\beta_t = 0.343 + 0.677 \times \beta_{t-1}$

B4: Beta in Mergers and Acquisitions

In this section, I will discuss how to find the beta of a merged company after the merger.[26] I explain it using a simple numerical example. Assume the following for three companies, which are operating, in the same line of business.

The market value of Debt and Equity are in ₹ million.

Company	*Levered Beta*[27]	*Market Value of Debt*	*Market Value of Equity*
A	0.8	25	100
B	1.2	40	100
C	1.4	50	100

Assume that the corporate tax rate is 30%.

Assume that A and B are planning for a merger. Also assume that C wants to take over A in a cash-financed acquisition. Here, we will find the beta of the combined entity A&B and the combined entity A&C. Let's first find the beta of the combined entity A+B by assuming that B gets merged in A.

Since both A and B are levered, we can start with by first unlevering the beta of the three companies :

Unlevered beta of A = 0.8/(1+0.7×0.25) = 0.68[28]

Unlevered beta of B = 1.2/(1+0.7×0.4) = 0.943

Unlevered beta of C = 1.4/(1+0.7×0.5) = 1.037

Irrespective of whether A takes over B or gets merged with B, ultimately we are getting one large economic entity that produces what A and B were producing earlier. Therefore, the operating risk of the combined entity does not change after the merger. We can therefore find the beta of the combined entity by looking it as a portfolio of A and B.

How will we find the portfolio weights here? Remember that we are dealing with unlevered betas here. Therefore, the weights must be based on unlevered assets of a company. This is nothing but the total equity of a similar unlevered company. Assuming that these companies will keep their debt level constant, we can use the first proposition of the Modigliani and Miller theorem to find the value of the unlevered assets.

$V_L = V_U + D \times t$

Here, V_U is the value of the unlevered assets. Keep in mind here that while deriving the unlevered assets we have kept the financing decision separate from the investment decision. That is when we remove the debt from the company's balance sheet we have assumed that the company is financing the debt redemption through an issue of stock. Therefore, one can estimate the unlevered assets by using the following equation:

25. See Blume, M. E., 1971, On Assessment of Risk, The Journal of Finance, Vol. 24, 1 - 10.
26. In the merger parlance, a merged company is one that retains its identity after the merger, and a merging company is one that loses its identity (it gets merged in the merged company) after the merger.
27. It is not necessary to explicitly mention that the beta is levered. In Finance, beta by default refers to levered beta.
28. Keep in mind that, here I am assuming that the market has valued these companies by keeping their current level of rupee debt constant.

$$Assets_U = Assets_L - D \times t$$

In the following we have estimated the unlevered assets of all the three companies using t above formulae. The unlevered asset of A is 125-25×0.3 = 117.5. The unlevered assets of and C can be similarly found out :

Unlevered Assets of A	117.5
Unlevered Assets of B	128
Unlevered Assets of C	135

Now that we have estimated the unlevered betas and the assets of the two companies, we ca find the unlevered beta of A+B. The unlevered beta of A+B is 0.81. This is a simple weighte average of the unlevered beta of A and B:

$$\beta_{U,A+B} = \frac{117.5}{117.5+128} \times 0.68 + \frac{128}{117.5+128} \times 0.943 = 0.81$$

The levered beta of A+B will depend on the actual leverage of the combined company A+B If A and B are merging, then the actual debt and equity of the combined company will be jus the sum of the two individual companies.

The total debt of A+B	65
Total Equity of A+B	200

The levered beta of A+B can be computed using the Hamada Equation:

$$\beta_{L,A+B} = 0.81 \times \left(1 + (1-0.3) \times \frac{65}{200}\right) = 1$$

Here, you will notice a very interesting thing. We can obtain this levered beta by taking th weighted average of the levered betas of A and B directly. Thus for example, the beta of A and B are 0.8, and 1.2 respectively. The market values of the equity of the two companies ar 100 each. Therefore, the simple arithmetic average of the two betas will give us the figure w want. **We can always find the levered beta of the combined entity by taking a simple weighte average of the individual companies if the merger is financed with stock.** However, if th merger is financed with cash or with a combination of stock and cash, then we need to fin out the unlevered beta of the combined company before finding its levered beta.

What are the adjustments that one has to make while computing the beta in case of acquisitions This actually depends on whether the acquisition is being financed with debt issue or equit issue[29]. This also depends on whether we are buying the entire company or part of the compan Suppose, C wants to take over A by issue of debt. That is C will issue debt to buy the share of A. Let's look at this transaction independent of the previous transaction. That is C take over A, *which has not merged with B.*

At a first step we must find out the unlevered betas and the unlevered assets of the tw companies to find out the unlevered beta of A+C. The following table shows the calculations

29. Acquisitions are usually financed with debt or with a mix of debt and equity and existing cash.

Unlevered Beta of A+C	0.87
Total Debt of A+C	175
Total Equity of A+C	100+30[30]
Levered Beta of A+C	1.943

Will our answer be any different if A takes over C? The unlevered beta will remain the same irrespective whether A takes over C or C takes over A. But there will be a major difference to the levered beta if the acquisition is being financed through debt.

If A takes over C by the issue of debt, then A has to issue debt worth Rs.100. As we know debt issue will increase the value of the equity of the company by Debt times the tax rate. Therefore, the market value of A will increase to $100 + 100 \times 0.3$ that is 130. Then the debt to-equity ratio of A+C will be $(25+50+100)/130 = 1.346$.

The combined beta of A+C is given by $0.68 \times (117.5/(117.5+135)) + 1.037 \times (135/(117.5+135)) = 0.87$.

The levered beta of A+C will be $0.87 \times (1 + 1.346 \times 0.7) = 1.69$. This is different from the beta (1.943) we obtain when C takes over A.

APPENDIX C : Asset Pricing Models

In Section 3.3, we discussed in detail how one can estimate the cost of equity by using the Capital Assets Pricing Model (CAPM). There are two other asset pricing models that can also be used while computing the cost of equity of a company. In this appendix, I briefly discuss about them.

C1: Arbitrage Pricing Theory (APT)

APT is a different approach to determining equilibrium returns of assets. The APT description of equilibrium is more general than that provided by CAPM. One just requires the homogeneous expectation assumption (among the assumptions required in CAPM). The central core of the APT as outlined by Stephen Ross is that the return generating process for a population of N assets is a linear function of only a few, say k ($k < N$), systematic factors or components. The return generating process can be written in equation form as :

$$r_{i} = E(r_{i,t}) + \beta_1 F_{1,t} + \beta_2 F_{2,t} + \dots + \beta_k F_{k,t} + e_{i,t} \quad ...(1)$$

Here

r(i) is the observed return of stock i,

E(i) is the expected return of stock i,

β(i,j) is the factor loading of stock i with respect to factor j,

F(j) is the jth factor score.

We assume that the expected values of the factor scores as well as the idiosyncratic error terms are zero. So if there are enough securities in the market so that a portfolio can be formed with the following portfolio weights:

30. The market value of C will increase because of the debt issue and the tax shield.

$x_1 + x_2 + ... + x_N = 0.$[31]

$x_1 \times \beta(1,i) + x_2 \times \beta(2,i) + ... + x_N \times \beta(N,i) = 0$, i= 1,...,k.

$x_1 \times e(1) + ... + x_N \times e(N) = 0,$

then E(i) can be written as a linear combination of the β(i,j)'s and a constant vector. So

$$E(r_{i,t}) = \lambda_0 + \beta_1 \times \lambda_1 + \beta_2 \times \lambda_2 + ... + \beta_k \times \lambda_k \quad ... (2)$$

Equation 2 can be used to estimate the cost of equity of a company. As can be seen, we need estimates of the factor betas and the factor risk premiums to estimate the cost of equity of a stock. One can find details of the estimation procedure from Roll and Ross (1980)[32]. We explain the procedure used very briefly by giving the example of Titan Company Limited.

Estimation of the Cost of Equity of Titan Company Limited

When we use the APT to obtain the cost of equity of Titan Company Limited[33], we assume the following return generating model:

$$r_{Titan,t} = E(r_{Titan,t}) + \beta_1 F_{1,t} + \beta_2 F_{2,t}, + ... + \beta_k F_{k,t} + e_{Titan,t}$$

Under the assumption of APT, this leads to

$$E(r_{Titan,t}) = \lambda_0 + \beta_1 \lambda_1 + \beta_2 \lambda_2 + ... + \beta_k \lambda_k$$

It is the second equation above that is used to estimate the cost of equity for a company. In order to estimate the cost of equity for Titan, we need estimates for $\lambda_0, \lambda_1, ..., \lambda_k$, and $\beta_1, \beta_2, ..., \beta_k$. To estimate $\beta_1, \beta_2, ..., \beta_k$ we normally use factor analysis. However, to run factor analysis on stock returns, we need the returns of few more stocks. This is another advantage of APT because you can simultaneously estimate the cost of equity of a group of stocks. Here I use the stock returns of another 23 companies (apart from Titan). We ensure that no two companies belong to the same industry. This adjustment is necessary for otherwise, we may find a spurious industry factor in the factor analysis. Since we believe that industry risk can be diversified away in a well-diversified portfolio and hence does not get priced, we do the above adjustment.

Running maximum likelihood factor analysis gives us two factors[34]. This means that in India two factors (and not one as CAPM predicts) do explain the variation in stock returns. If we use one factor (market factor or some other factor), then we may not be able to explain the variation in stock returns in India. The beta of Titan for the first factor is 0.81, and that on the second factor is -0.11.

Our next step is to estimate λ_1 and λ_1. We run least squares regression of stock returns on β_1 and β_2 for the 24 companies we have in our sample. We obtain

$\lambda_1 = 0.098$[35], and

$\lambda_1 = 0.00857$

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31. This of course, is possible if we assume that short-sales are possible. But latter literature has shown that this assumption is not necessary.
32. See Roll, Richard, and S. R. Ross, 1980, An Empirical Investigation of the Arbitrage Pricing Theory, The Journal of Finance, Vol. 35, 1073-1103.
33. The name of the company has been changed to Titan Company Limited.
34. We do not need to know what the factors are. From a computational point of view this is definitely an advantage of APT. However, it becomes difficult to convince someone about the required rate of return figure when we do not know the identity of these two factors.
35. Notice that λ_1 comes very close to the market risk premium we got earlier. The first factor in APT, is probably the market factor.

Using the above we can estimate the cost of equity of Titan Company Limited. It is given by:

$E(r) = 0.0812 + 0.81 \times 0.098 - 0.11 \times 0.00857$

$= 15.96\%$.

C2: Multi-Factor Models

One of the main drawbacks of Arbitrage Pricing Theory is that we do not know the identity of the factors. Thus for example, we obtain 2 factors while running factor analysis on the stock returns of the 24 stocks. But we do not know the identity of these factors. Eugene Fama and Kenneth French found out that three factors namely market risk, size risk and book-to-market risk can explain the cross-section of stock returns in the US. In a similar paper I found that two factors namely market risk premium and size risk premium can explain the cross-section of stock returns in India[36]. In CAPM we consider only the market factor. The above statistical-model shows that the market factor alone does not adequately capture the entire variation in stock returns. We need additional explanatory variables.

Using the original model developed by Eugene Fama and Kenneth French in 1993, one can obtain the expected return (and hence the cost of equity) of a stock by using the following equation.

$$E(r) = r_f + \beta_1 \times (E(r_m) - r_f) + \beta_2 \times size \quad risk \quad premium$$
$$+ \beta_3 \times Book-to-Market \quad risk \quad premium$$

Fama and French observed that small stocks are fundamentally more riskier than large-cap stocks. Similarly stocks with high book-to-market are fundamentally more riskier than stocks with low book-to-market. A similar research done by me showed that in the Indian context, we do not need to include the book-to-market risk. Hence we can find the cost of equity of an Indian company by using the following equation:

$$E(r) = r_f + \beta_1 \times (E(r_m) - r_f) + \beta_2 \times size \quad risk \quad premium$$

Here, I will show how the above two-factor model can be used to find the cost of equity of Titan Company Limited. We need estimates of the risk free rate of return, β_1, β_2, market risk premium and size risk premium to estimate the cost of equity. The size risk premium is found to be equal to 43.3062%. The market risk premium (using this database, which I created specially to test this model) is found to be 22.0216%.

Titan's market beta is 0.6367, and size beta is -0.169. The cost of equity is found to be 11.4%. If we do not consider size risk premium at all, we will obtain Titan's cost of equity as equal to 21.69%. Titan has negative risk exposure to size risk. If one does not adjust for this, then one will overestimate the riskiness of the stock, and consequently overestimate the required return on the stock. If we include book-to-market risk too (on which Titan is found to have a positive loading), the cost of equity is equal to 12.66%.

How does one estimate the size risk premium?

To obtain the size risk premium, I created three portfolios by ranking the stocks based on their market capitalizations. In September of each year t, I ranked all the stocks based on size. Then the stocks were divided into three size groups based on the breakpoints for the bottom 30%

36. See http://www.nseindia.com/content/research/paper13.pdf

(small), middle 40% (Median) and top 30% (Big) of the ranked values of size. These portfoli are value-weighted portfolios.[37]

I created a portfolio called SMB (small minus big) by taking a long position in the small stock and a short position in the large stocks. This is the simple difference, each month, betwee the averages of the returns on the small portfolio and the large portfolio. I used this retur difference as a proxy for size risk premium here.

APPENDIX D : Maintaining consistency between cash and bet (risk consistency)

Analysts normally treat cash as a non-operating asset and ignore it when they value a enterprise. Cash does not generate any business income for a company. When a potenti buyer is valuing a company, he will look at those assets that will generate future busine income. If a company has ₹10 million of cash, its market value is also equal to ₹10 million. N one will pay ₹11 million to buy this asset. The owner of the company will not accept anythin less than ₹10 million for this asset. Operating assets generate business income in the futur and the potential buyer is interested only in this business income. That is why cash is usuall treated as a non-operating asset.

In this appendix, we will learn about the role 'cash' plays in influencing the beta of the stoc In this chapter, we saw that presence of debt increases the financing risk of the stock an hence the beta increases. In the same manner, cash can be seen to decrease the financing ris and hence it reduces the beta of a stock. If a company has lots of cash (invested in risk-fre money market instruments or T-bills), then this company will be less affected by a recessio in the economy as compared to another company that is into the same business, but has n cash in its balance sheet. I use a few examples to explain how cash affects the beta of a stoc Understanding this helps us maintaining the *risk consistency* in the valuation model.

Example 1: Assume the risk-free rate (after-tax) to be 5%. Also assume the market risk premiu to be 6% in an economy. The beta of the stock of KLM Limited (with no debt and no cash) i 0.5. The company is expected to earn constant EBIT of \$100 million in each year in the futur Assume that the company pays tax at the rate of 20% each year.

In this case, the cost of equity equals 8%. The projected free cash flow to equity is \$80 millio and hence the intrinsic value of equity should be \$1,000 million, as shown here.

$$K_e = r_f + \beta \times MRP = 5\% + 0.5 \times 6\% = 8\%$$

FCFE = Net Income - Net Investment = \$100 million × (1-0.2) - 0 = \$80 million[38]

$$Equity = \frac{FCFE}{Ke} = \frac{\$80m}{8\%} = \$1{,}000m$$

Example 2: Now, think of MJL Limited, another company that is similar to KLM in ever aspect expecting that it has got \$50 million of excess cash with it. This surplus cash has bee invested in government securities that generate 6.3% return (before tax) each year. This is risk-free investment and generates the same after-tax return (5%) that we used in Exampl 1. How much value should we put on the equity of MJL Limited?

37. Fama and French (1993) argue that using value weighted components results in mimicking portfolio that capture the return behaviours of small and large stocks.
38. Here, the company has no debt and hence the taxable income equals the EBIT. The projected ne investment equals 0 as we are valuing the company assuming the growth rate to be 0.

Common sense tells us that the intrinsic value of equity of MJL should be $1,050 million. As MJL is nothing but KLM Limited and an additional $50 million of cash that has been invested in risk-free assets. In order to find the value of equity of MJL, we need its beta and FCFE. Since cash makes a company relatively safer, the beta of MJL must be lower as compared to KLM. But how much low?

In order to compute the beta of MJL, look at it as a portfolio of $1,000 million of operating assets (invested in the regular operations of the company) and $50 million of excess cash. So, out of the total assets of $1,050 million, the risk of the $1,000 million of assets is captured by the beta of 0.5 (same as that of KLM) and the risk of the $50 million of cash is captured by the beta of cash, which is 0. So the beta of MJL must be 0.4762, as shown here :

$$\beta_{MJL} = \frac{1,000}{1,000+50} \times 0.5 + \frac{50}{1,000+50} \times 0 = 0.4762 \approx 0.48$$

Even if both KLM and MJL are into the same business, the beta of MJL is lower than that of KLM, as MJL has $50 million of risk-free assets in its balance sheet. Analysts normally look at cash as negative debt. So, one can also argue that MJL has net debt of − $50 million and this reduces the beta of MJL.

We can now compute the cost of equity of MJL using 0.48 as the levered beta. We also need to compute the FCFE of MJL after taking care of the additional (after-tax) interest income it earns on the $50 million of cash.

$$K_e = r_f + \beta \times MRP = 5\% + 0.48 \times 6\% = 7.86\%$$

FCFE = Net Income - Net Investment = ($100 million + $50 million × 6.3%) × (1-0.2) - 0 = $82.5 million

$$Equity = \frac{FCFE}{Ke} = \frac{\$82.5m}{7.86\%} = \$1,050m$$

This is exactly, what we expected. Look at the way, we maintained the **risk consistency**. The numerator included some risk-free cash flow (interest on cash). The denominator reflected the risk as well, as we used a proportionately lower beta to estimate the cost of equity.

Example 3: Now think of a third company, ASL Limited, that is similar to MJL Limited in every aspect, excepting that ASL Limited has borrowed $100 million (@7%) and that the book equity of ASL is exactly lower by $100 million as compared to that of MJL Limited. What is the beta of ASL and what is the intrinsic value of equity of ASL?

ASL Limited is riskier than MJL Limited as the former has an additional $100 million of debt, while the latter has no debt. So the beta of ASL should be higher than that of MJL. It would be instructive to compare the beta of ASL with that of KLM. ASL has $50 million of cash and $100 million of debt that KLM does not have, and hence the beta of ASL should be low (cash effect) and high (debt effect). How do we adjust for these effects?

Before, we learn about the computation of cost of equity, let's learn about the concept of **'regression beta[39]'**. This is the beta that we obtain by regressing stock returns on that of the market portfolio. If the company has invested cash in risk-free assets, then its return will exhibit lower volatility compared to a similar company without any such risk-free investment. Though both the companies will have similar business risk, one (with cash) will have lower regression beta compared to the other company. In Example 2, MJL has got $50 million cash

39. This also equals the levered beta of the stock. However, I use the term regression beta to show that we compute this number using stock returns data of the company. The levered beta of a company can be computed by using peer companies' data.

in its balance sheet that KLM does not. Since these two companies are otherwise similar, MJL will have lower regression beta compared to KLM. Stock returns of MJL will exhibit lower volatility.

If a company has more debt, then its stock returns will exhibit higher volatility and hence its regression beta will be higher. This is consistent with the concept of efficient market hypothesis. In an efficient market, the stock returns correctly reflect the riskness of the stock (higher cash reduces risk and higher debt increases risk) and hence regression beta reflects the actual risk of the company.

In order to adhere to the concept of risk consistency, we must ensure that the riskness of the cash flows being discounted matches with the discount rate. So, if we use regression beta to compute the cost of equity, we essentially assume that the company has no cash and its debt has been reduced by the amount of cash. The regression beta of ASL therefore, assumes the cash to be 0 and debt to be $50 million. Therefore, while computing the free cash flow of ASL, we must ignore the interest income that cash generates. This is the so-called net-debt method of computation of beta.

Net-debt method

In the net-debt method, the unlevered beta is 0.50. In the net debt method, when we talk about unlevered beta, we refer to the hypothetical situation, where both the debt and cash are equal to 0 (and hence the net debt equals 0). The levered beta equals 0.5017. Since the cost of borrowing is 7% and cash generates interest of 6.3%, we need to find the average interest expense (income) on the net debt. This equals 7.75%[40]. This will be higher than the cost of debt, as cash gets a negative weight. The implied debt beta is 0.4583[41].

The levered beta can be computed using Equation 3(11).

$$\beta_L = \beta_U \times \left(1 + (1-t) \times \frac{D}{E}\right) - \beta_D \times (1-t) \times \frac{D}{E}$$

$$= 0.5 \times \left(1 + (1-0.2) \times \frac{50}{960}\right) - 0.4583 \times (1-0.2) \times \frac{50}{960} = 0.5017$$

Note that, in order to compute the levered beta, we need the value of equity. I compute this number using the APV method that we learn in Chapter 2. The cost of equity comes to 8.01% and the cost of capital comes to 7.92%. In the net-debt method, we totally ignore the interest income from cash, while computing the free cash flow. So the free cash flow remains $80 million. The enterprise value comes to $1010 million and the equity value comes to $960 million. I show the key computations here.

$$K_e = 5\% + 0.5017 \times 6\% = 8.01\%$$

$$\text{Cost of capital} = \frac{50}{1{,}010} \times 7.75\% \times (1-20\%) + \frac{960}{1{,}010} \times 8.01\% = 7.92\%$$

Free Cash Flow = $100 million × (1 - 0.2) = $80 million

Enterprise Value = $80 million/7.92% = $1,010 million

Equity Value = Enterprise Value - Net Debt = $1,010 - $50 = $960 million

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40. This equals (100 × 7% - 50 × 6.3%)/(100 - 50).

41. This figure can be computed by back-computation. I use CAPM to derive this number. It equals (7.75% - 5%)/6%.

Gross-debt method

If we use the gross-debt method, we still obtain the same levered beta and hence the same cost of equity. It is important to understand that, no matter which method we use, we will obtain the same regression beta. Regression beta is obtained by regressing stock returns on the market returns. It has nothing to do with how we value a company.

You may wonder, how we obtain the same levered beta, when the debt-equity ratio that we use are different in the net-debt and the gross-debt method. The answer lies in the way we interpret the unlevered beta in the two methods. Unlevered beta in the net-debt method tries to estimate beta in the hypothetical case, when the net debt is 0 (that is both cash and debt are 0). In the gross-debt method, unlevered beta is the beta with the debt equal to 0 (cash is still positive). So the unlevered beta in the gross-debt method is lower compared to the unlevered beta in the net-debt method.

In this example, the unlevered beta in the net-debt method is 0.5. How do we obtain the unlevered beta in the gross-debt method? We must derive the answer from the regression beta. If we back-calculate using Equation 3(11), we obtain 0.4888 as the unlevered beta in the gross-debt method. Of course, we do not need to compute the unlevered beta in the gross-debt method, as the cost of equity is computed using the levered beta. This discussion is necessary to make us understand, why we use the same levered beta in both the net-debt and the gross-debt method.

For ASL, the cost of equity comes to 8.01% (as is the case in the net-debt method) and the cost of capital comes to 7.78%. In the gross-debt method, the effect of cash is explicitly considered while computing the cost of capital. So the interest income from cash must be considered while computing the free cash flow. For ASL, the free cash flow comes to \$82.5 million. The enterprise value comes to \$1,060 million. The intrinsic value of equity comes to \$960 million. I show the workings here:

$$K_e = 5\% + 0.5017 \times 6\% = 8.01\%$$

$$\text{Cost of capital} = \frac{100}{1{,}060} \times 7\% \times (1 - 20\%) + \frac{960}{1{,}060} \times 8.01\% = 7.78\%$$

Free Cash Flow = (\$100 million + \$50 million × 6.3%) × (1 - 0.2) = \$82.5 million

Enterprise Value = \$82.5 million/7.78% = \$1,060 million

Equity Value = Enterprise Value - Gross Debt = \$1,060 - \$100 = \$960 million

Both the net-debt and the gross-debt methods yield the same equity value. Of course, we obtained different cost of capital figures and different enterprise value figures. However, these are irrelevant numbers. In a transaction, we need only the equity value. That is what matters to us.

If both the methods yield the same equity value, which method should we use? Since we are getting the same equity value in both the cases, we must look at computational convenience to decide which method we should use. If you use the net-debt method, you must compute the average cost of debt, while computing the cost of capital. You do not have to do that in the gross-debt method. In the example of ASL, the cost of debt of 7% was readily available. We had to compute the average cost of debt of 7.75% figure because cash earns different interest than what the company pays on its borrowings. You do not need to do this in the gross-debt method.

However, in the gross debt method, you must find the extra income the company earns on its cash and add that to EBIT before computing the free cash flow. Interest income on cash is usually reported after EBIT is computed. So this is an extra adjustment we need to make while using the gross-debt method.

Computing Unlevered Beta from Comparable Companies

We sometimes, compute beta from the comparable companies. Since regression beta is always estimated with some statistical noise, it is always advisable to compute the beta from the other comparable companies. We first estimate the levered beta of the comparable companies. But how should we unlever them? In Chapter 3, I unlevered the betas by using both the net-debt and gross-debt (See Table 3.12). However, which method is better?

For ASL Limited, we find that the unlevered beta is 0.5 in the net debt method. This is the beta of KLM Limited (a company having no debt and no cash). This shows that we get theoretically consistent estimates of beta if we use the net debt to equity ratio to unlevered beta as it gives us the right estimate of beta. The problem with the gross debt-to-equity ratio is that we obtain regression beta of 0.4762 (for a company with $50 million of cash and no debt) and 0.4888 (when we unlever the beta from the regression beta using the gross debt method). This inconsistency does not arise in the net debt method. Of course, the difference is small and may be ignored in most practical applications.

The key point to remember is that we must maintain risk consistency while doing valuation. Once we ensure that the free cash flow is consistently computed, it really does not matter whether we use the net debt method or the gross debt method.

APPENDIX E: Steps Involved in Computing Cost of Capital (in a nutshell)

In this appendix, I list the steps involved in computing the cost of capital for a company. Use it as a checkbox to ensure that no step is left out in the process. If you find the theory involved in computing the cost of capital to be little involved, you can quickly go through this appendix to understand the steps involved. Refer to the main chapter to know the reason as to why something is treated in a particular manner.

1. Identify sources of capital

First identify all the sources of capital for the company. Equity will be one of the item for every company. Debt will appear for most other companies. Preference capital may be present in a few cases. As far as debt is concerned, divide them into interest bearing debt and other debt. Ignore the other debt while computing the cost of capital. Find out if the company uses 'operating lease' to finance any asset. If it is an insignificant number, then ignore it. Else consider it as a source of fund as well.

2. Find out the cost of debt

In order to find the cost of debt, follow the pecking order prescribed in this chapter. Most likely you will use the credit rating method. Find out the rating of the bonds issued by the company. Find out the default spread for the rating from the website of www.fimmda.org. Find out the risk-free rate from the website of www.nseindia.com. If it is a foreign currency loan, find the corresponding rupee-equivalent cost of debt (assuming your cash flows are denominated in Indian rupees). Compute the tax rate using Equation 3(4) of the book and then compute the after-tax cost of debt.

3. Find out the cost of equity

First find out the comparable companies and compute their unlevered beta using the methodology stated in this book. You can obtain the beta values from some websites (check the website of www.in.reuters.com). But it is advisable to compute it yourself so that you have more control over it.

Then find out the average unlevered beta for the sector and then lever it (using the debt ratio of the company). Find out the market risk premium (most recent quarter figure can be obtained from my website) and then compute the cost of equity. If the stock is not listed and if no comparable company is listed in the exchange, then find out beta of similar assets listed in other exchanges.

4. Find out the market value of assets

If the stocks of the company are listed, use the market capitalization as a proxy for the value of equity. If the stocks are not listed, try to compute the imputed value by using multiplier method. Thus for example, if it is a private company, then you can find out the average price-earnings multiple from the sector and multiply that with the net income of the private company to obtain an estimate of its market value of equity.

As far as debt is concerned, find out the book value of debt from the balance sheet. If you know the exact redemption schedule of all the debt, then you can compute the market value of debt by discounting the coupon on the bonds with the cost of debt (computed earlier). Otherwise, use the book value of debt as a proxy for the market value of debt. If you are treating operating lease as a source of fund, then find out the present value of operating lease rentals payable and add that to the book value of debt. If cash is being treated as a non-operating asset in the valuation, then compute the net debt.

Now you have all the numbers ready. Compute the weighted average cost of capital by using Equation 3(1).